

(12) **United States Patent**
Ikeda et al.

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(45) **Date of Patent:** **Aug. 12, 2025**

(54) **SEMICONDUCTOR DEVICE AND ELECTRONIC DEVICE**
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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 825 days.

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(Continued)

(51) **Int. Cl.**
H03F 1/22 (2006.01)
H01L 23/66 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 62/80** (2025.01); **H01L 23/66** (2013.01); **H03F 3/193** (2013.01); **H10D 84/01** (2025.01);
(Continued)

(58) **Field of Classification Search**
CPC H03F 1/22
(Continued)

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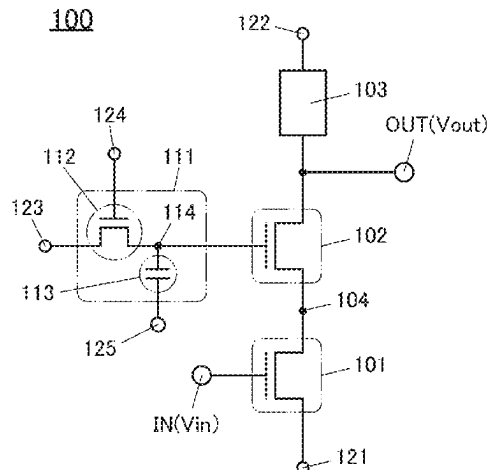
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Primary Examiner — Henry Choe
(74) *Attorney, Agent, or Firm* — Eric J. Robinson;
Robinson Intellectual Property Law Office, P.C.

(57) **ABSTRACT**
A semiconductor device with low power consumption is provided. In a cascode circuit including a first transistor provided on a low power supply potential side and a second transistor provided on a high power supply potential side, a source or a drain of a third transistor and a capacitor are connected to a gate of the second transistor. A gate of the first transistor is electrically connected to a back gate of the second transistor. An OS transistor is used as the third transistor.

20 Claims, 31 Drawing Sheets



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Jun. 21, 2019	(JP) 2019-115159	2017/0069596 A1 3/2017 Pillarisetty et al.
(51)	Int. Cl.	2022/0231644 A1 7/2022 Ito et al.
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<i>H10D 62/80</i>	(2025.01)	2022/0255579 A1 8/2022 Ikeda et al.
<i>H10D 84/01</i>	(2025.01)	FOREIGN PATENT DOCUMENTS
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(52)	U.S. Cl.	JP 2015-046592 A 3/2015
CPC <i>H10D 84/811</i> (2025.01); <i>H01L 2223/6677</i>	(2013.01)	JP 2015-047061 A 3/2015
(58)	Field of Classification Search	JP 2015-188209 A 10/2015
USPC 330/310, 311	See application file for complete search history.	JP 2017-510979 4/2017
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FIG. 1A

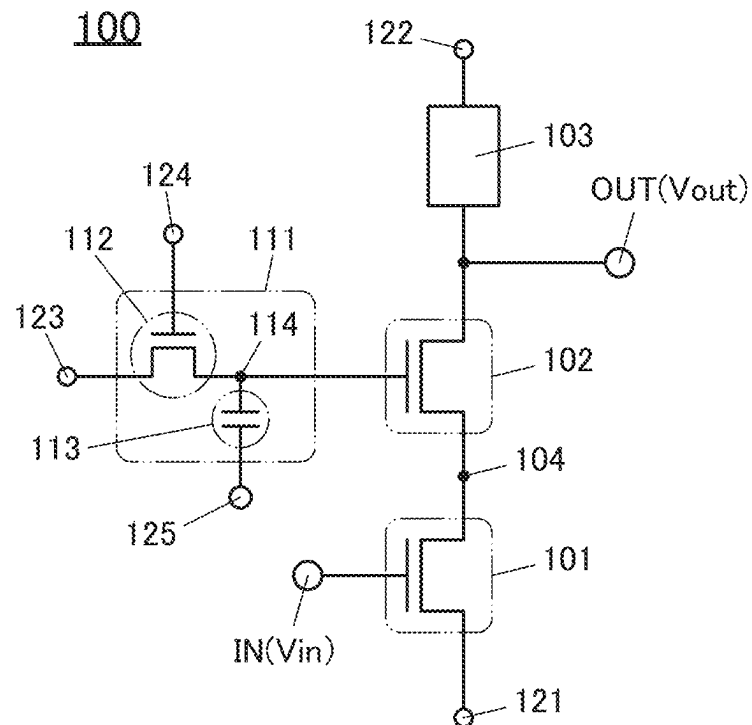


FIG. 1B

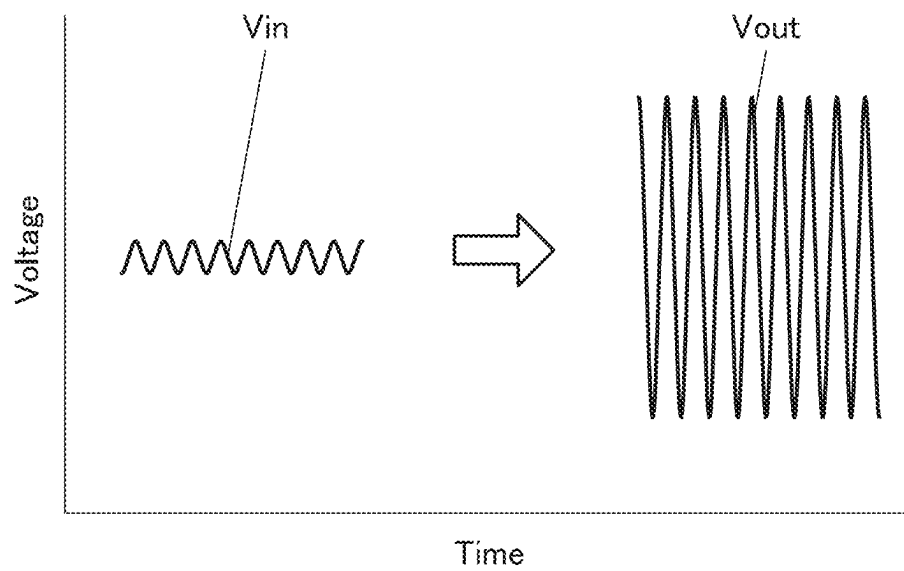


FIG. 2A

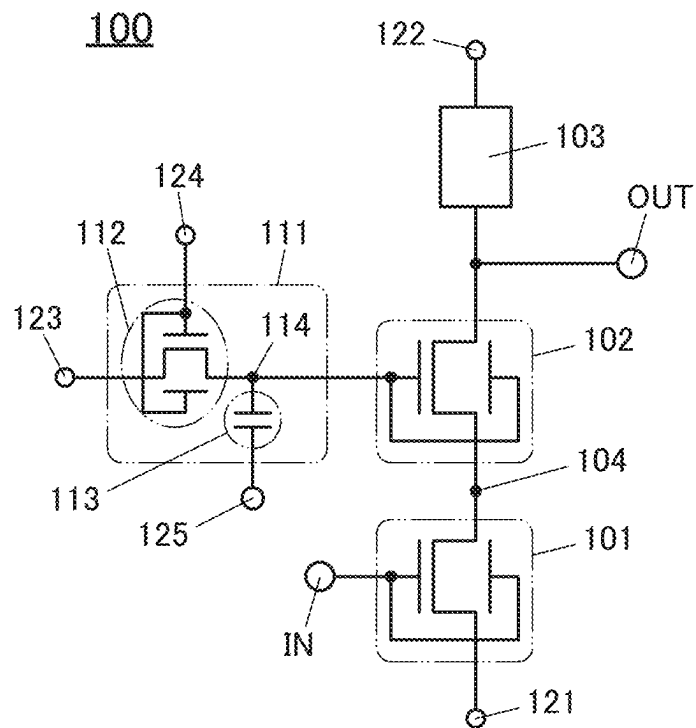


FIG. 2B

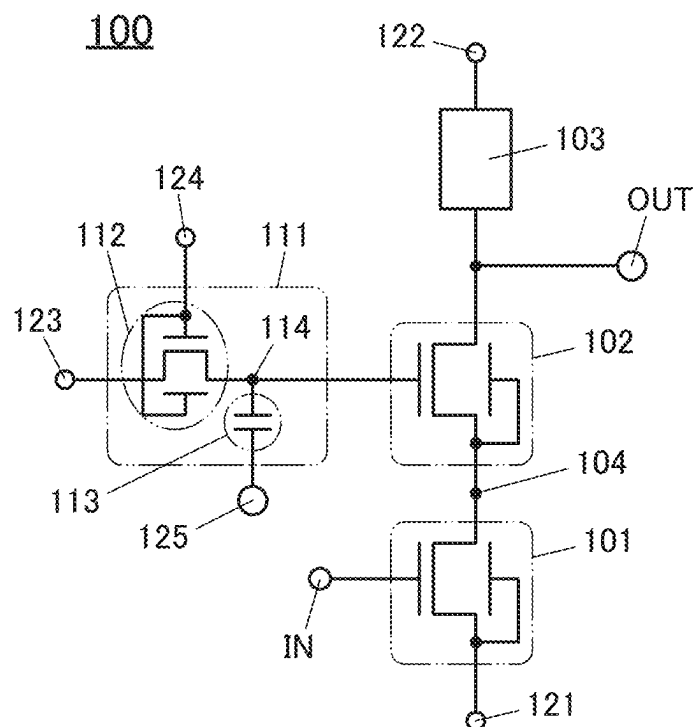


FIG. 3A

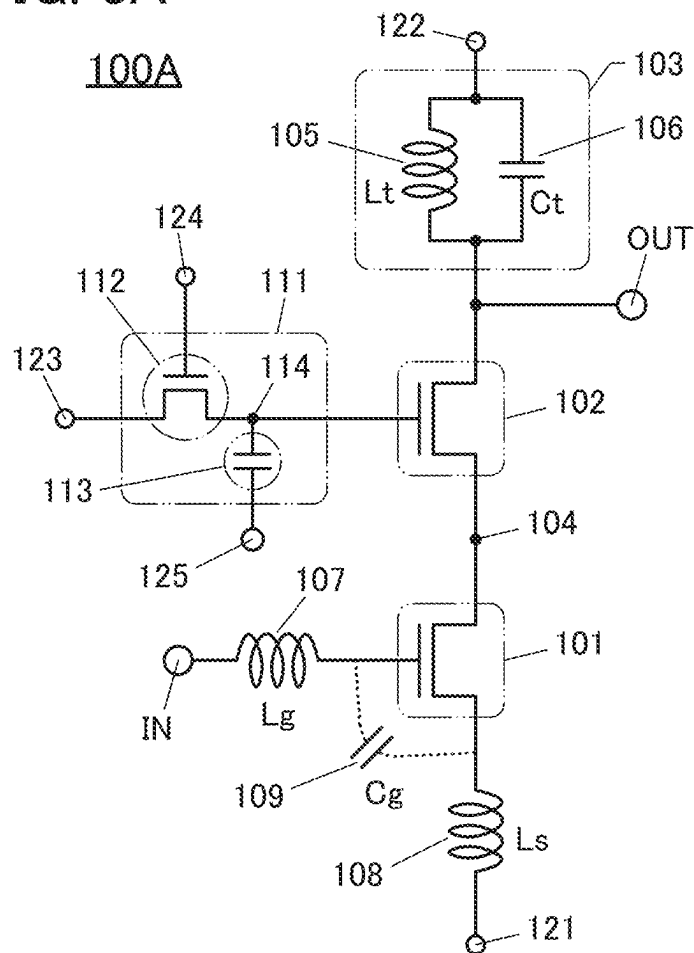


FIG. 3B

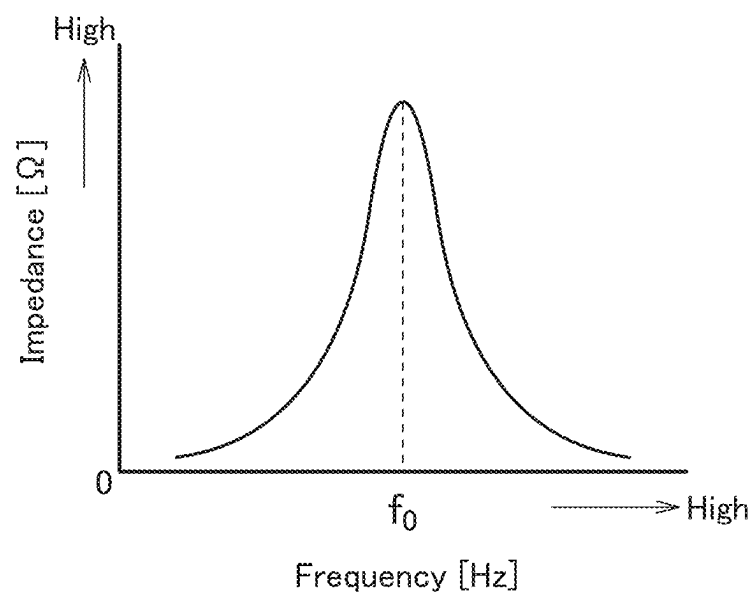


FIG. 4A

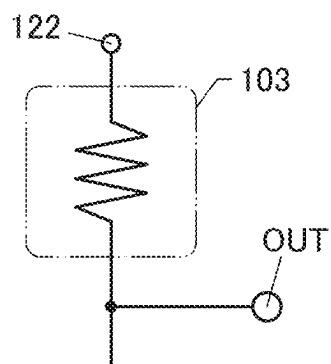


FIG. 4B

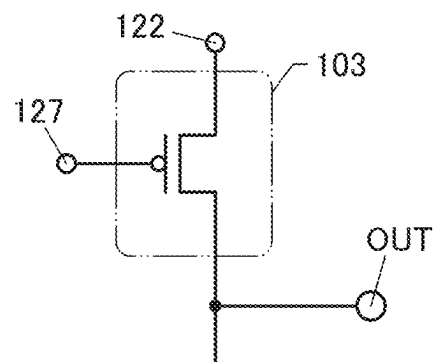


FIG. 4C

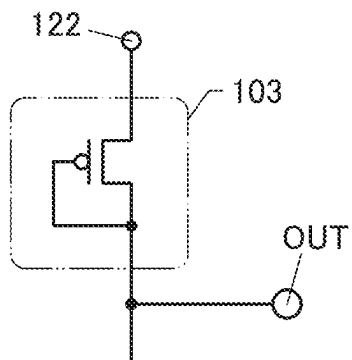


FIG. 4D

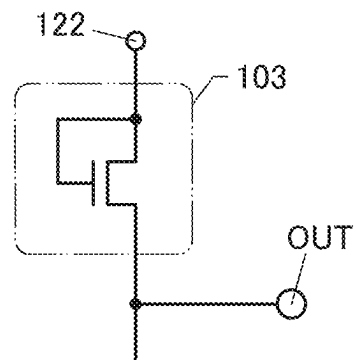


FIG. 5

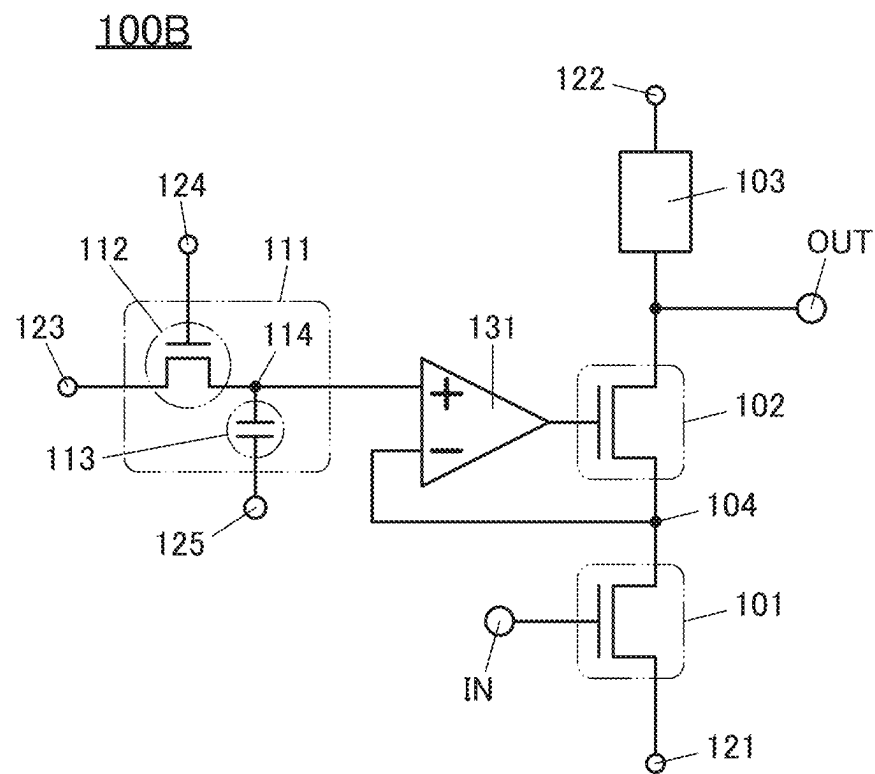


FIG. 6A

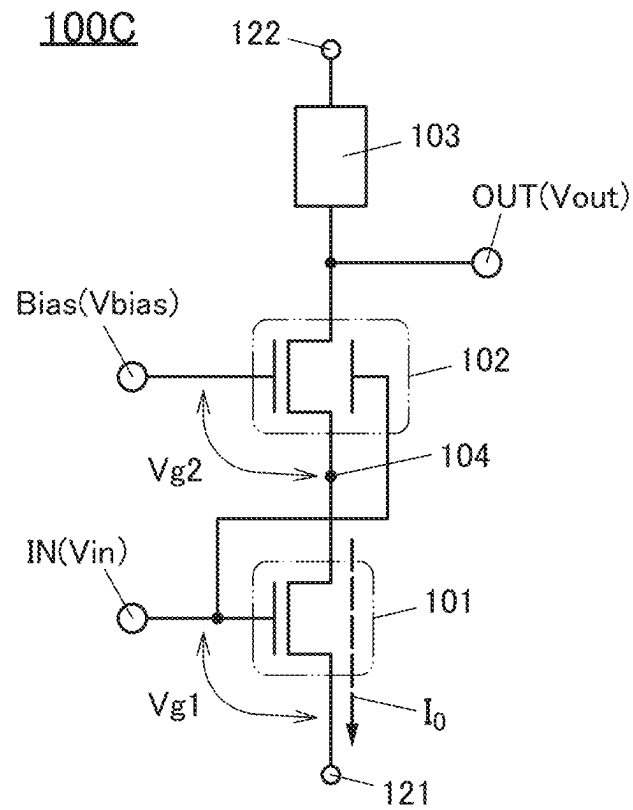


FIG. 6B

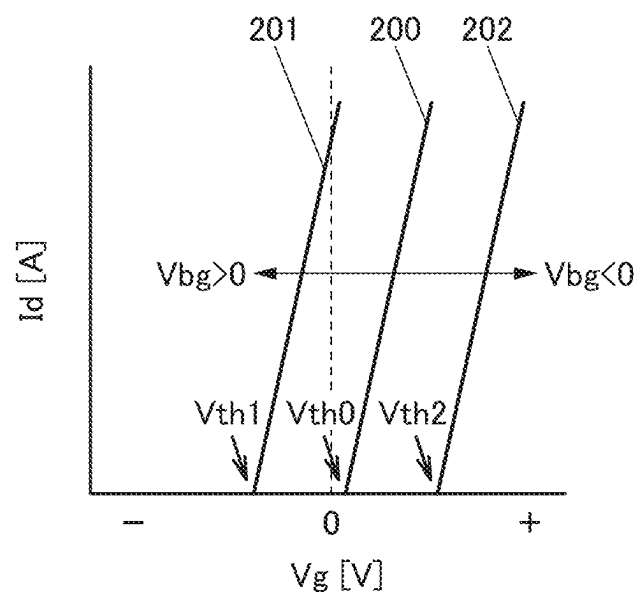


FIG. 7A

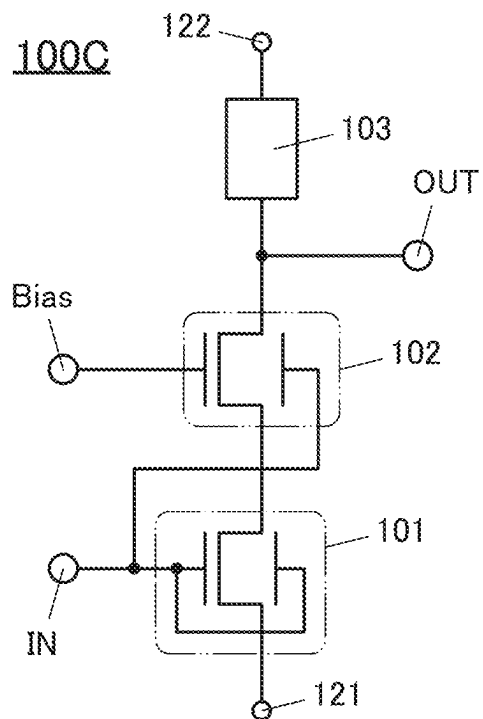


FIG. 7B

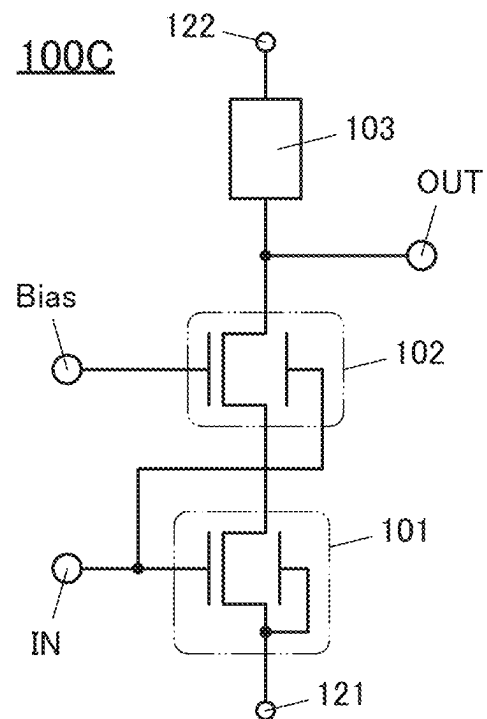


FIG. 7C

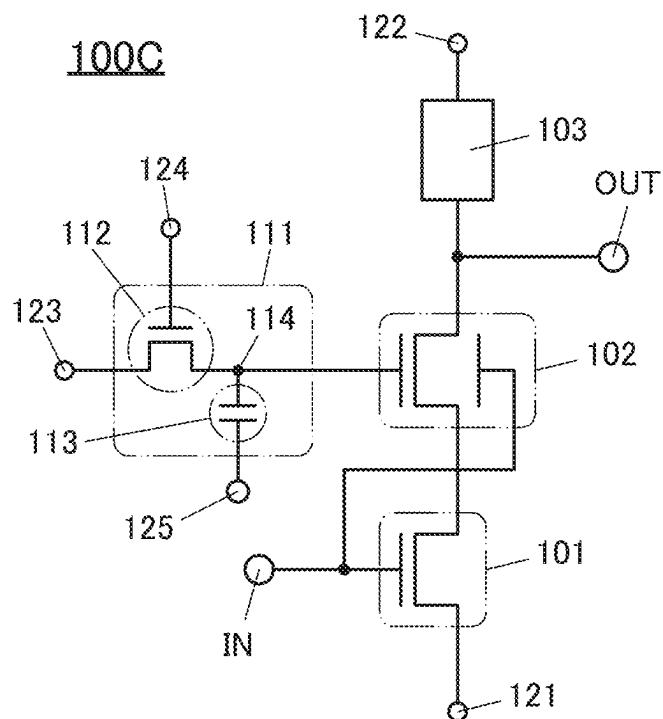


FIG. 8A

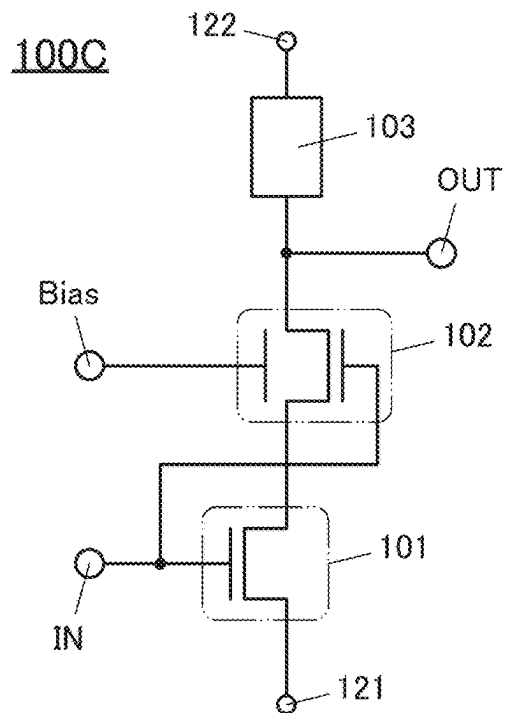


FIG. 8B

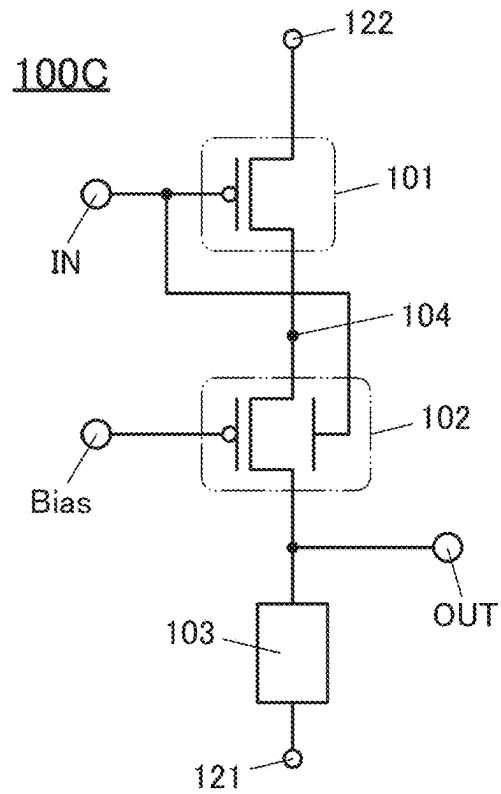


FIG. 8C

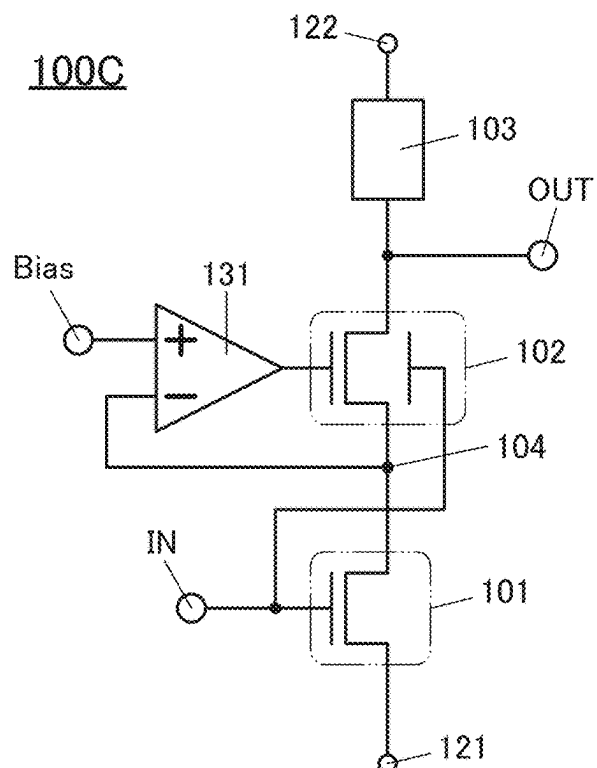


FIG. 9A

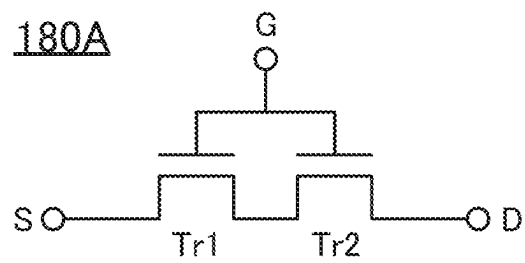


FIG. 9B

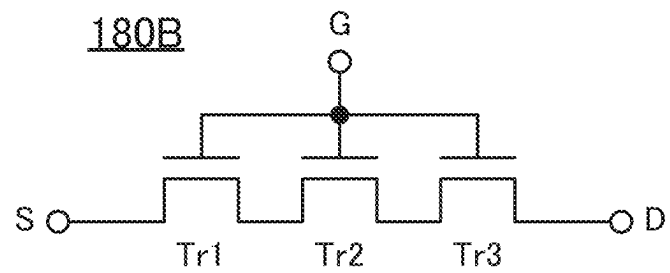


FIG. 10A

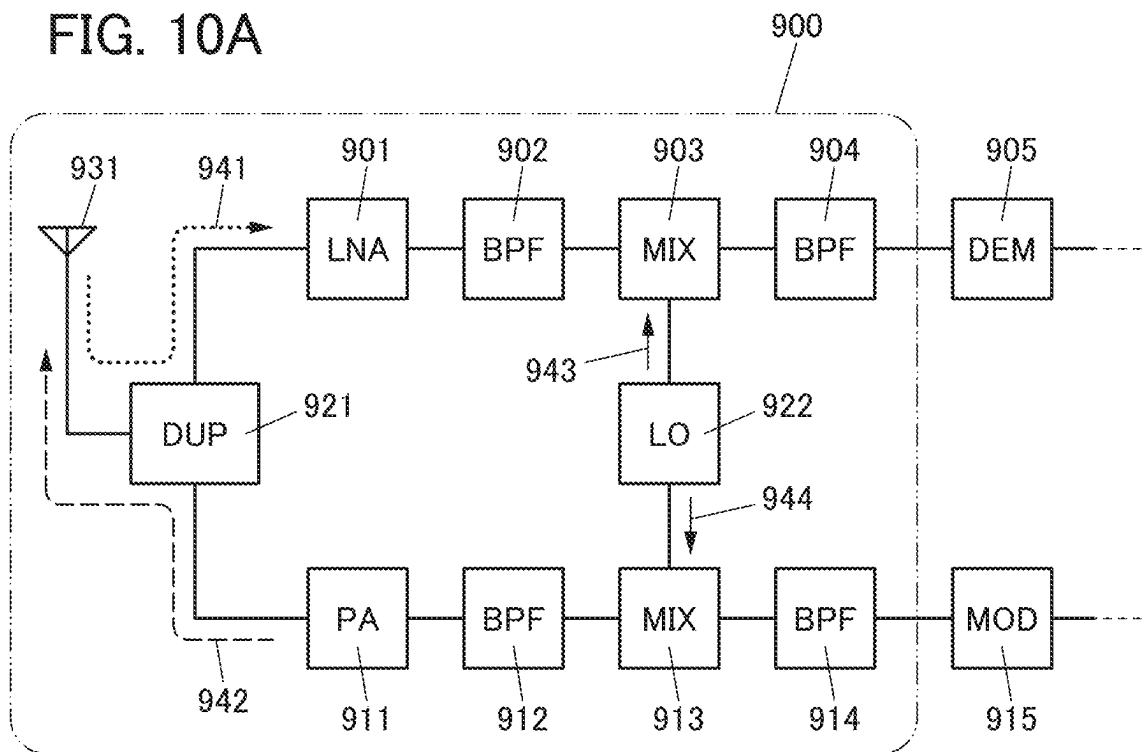


FIG. 10B

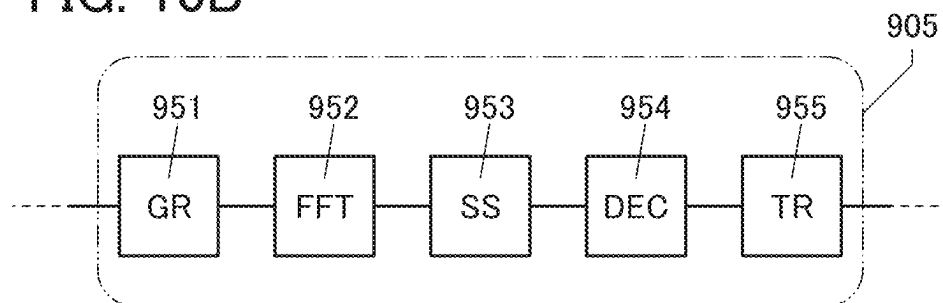


FIG. 10C

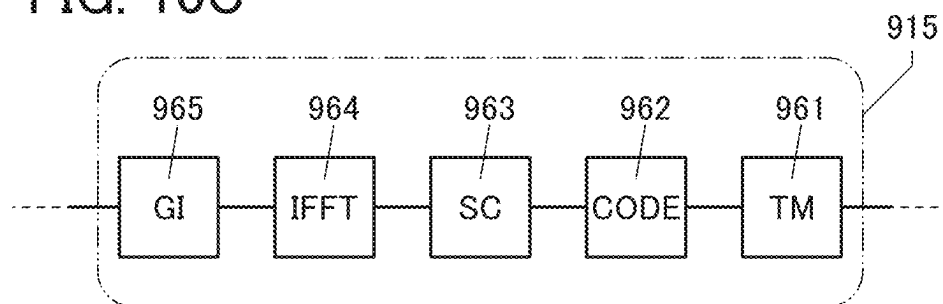


FIG. 11A

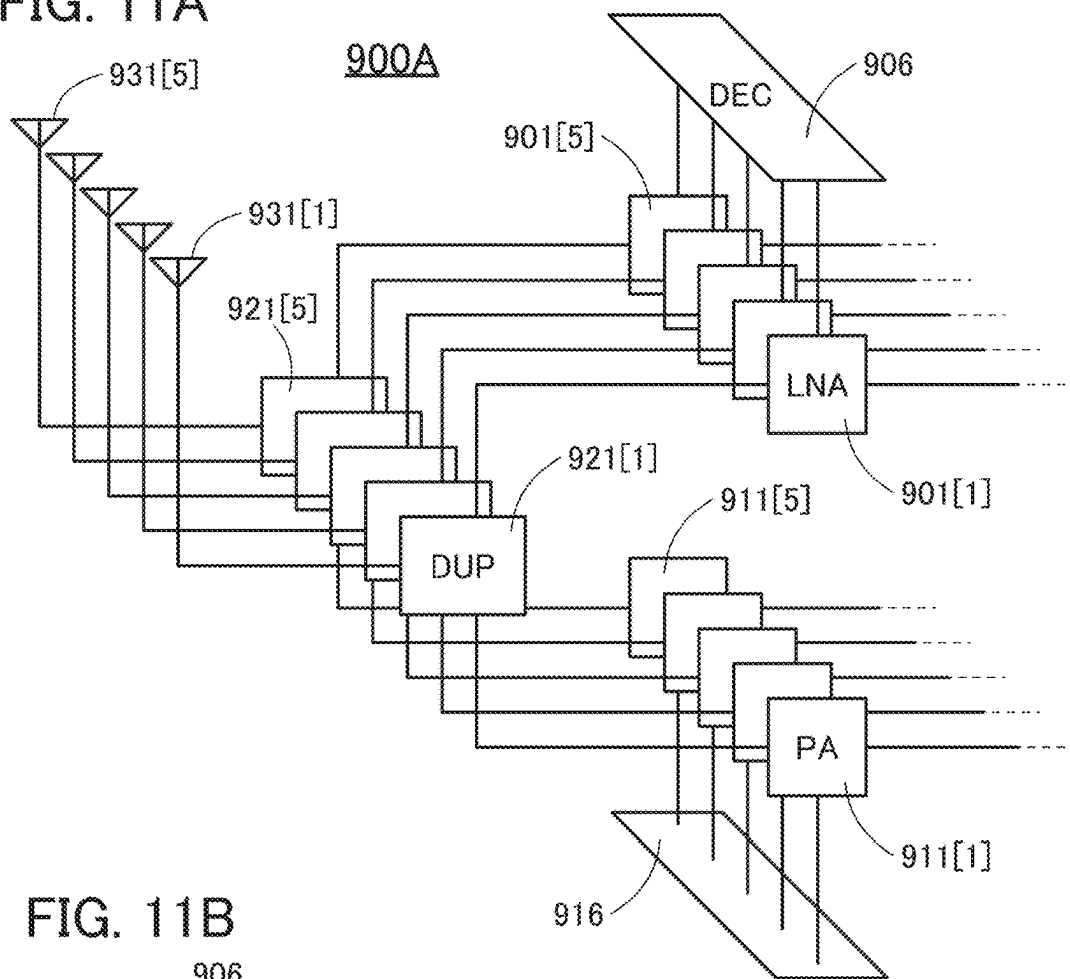


FIG. 11B

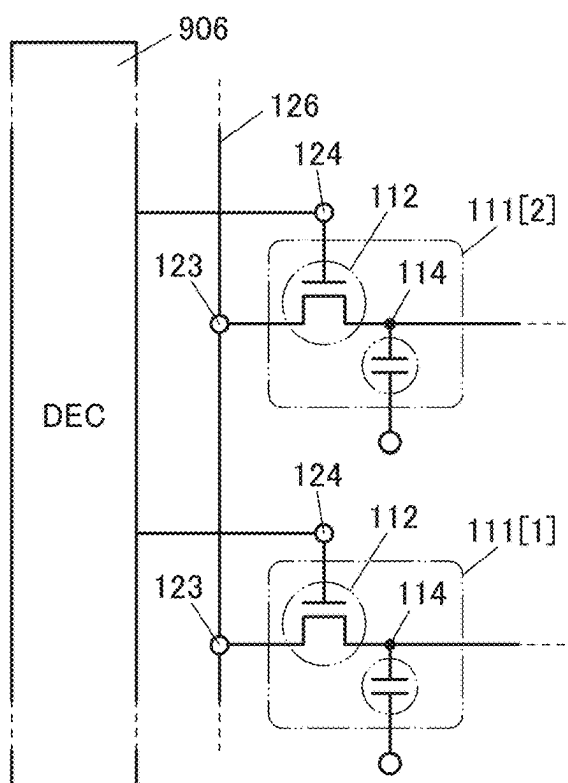


FIG. 12A

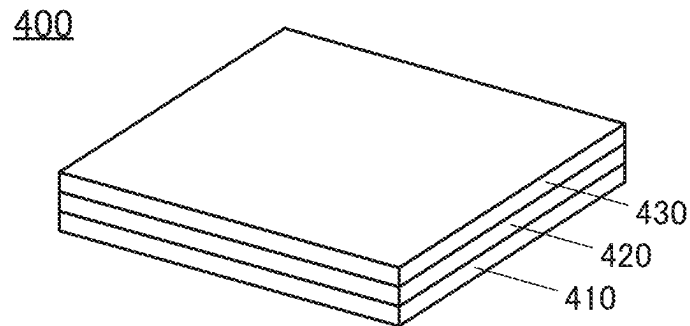


FIG. 12B

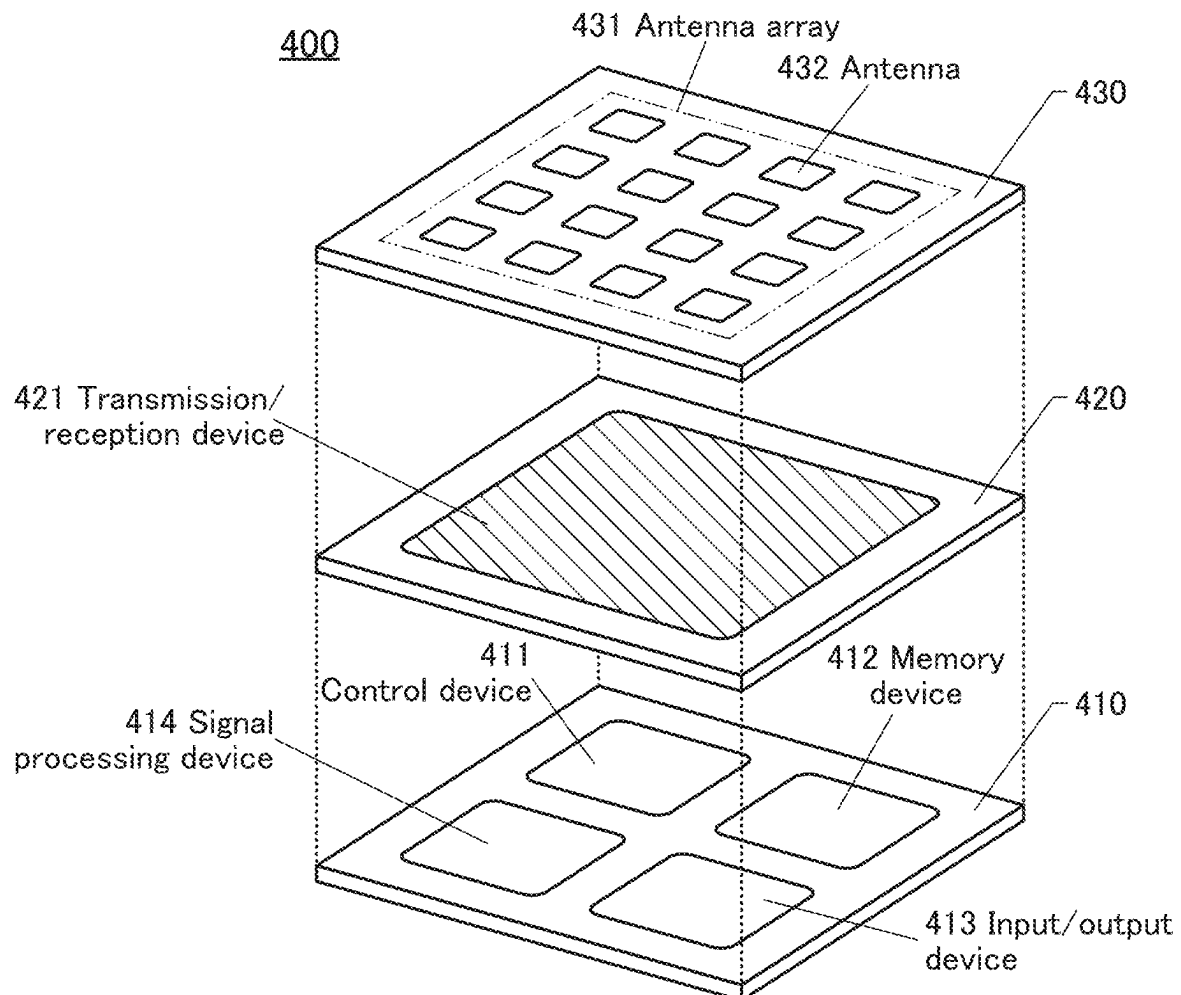


FIG. 13A

400A

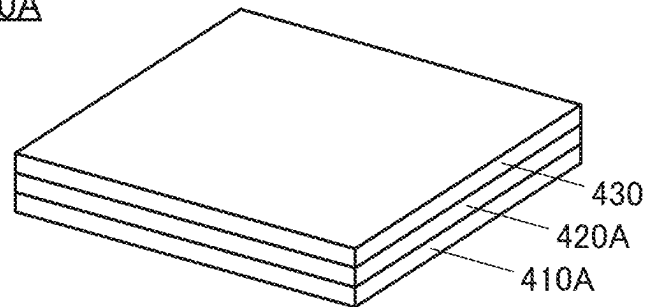


FIG. 13B

400A

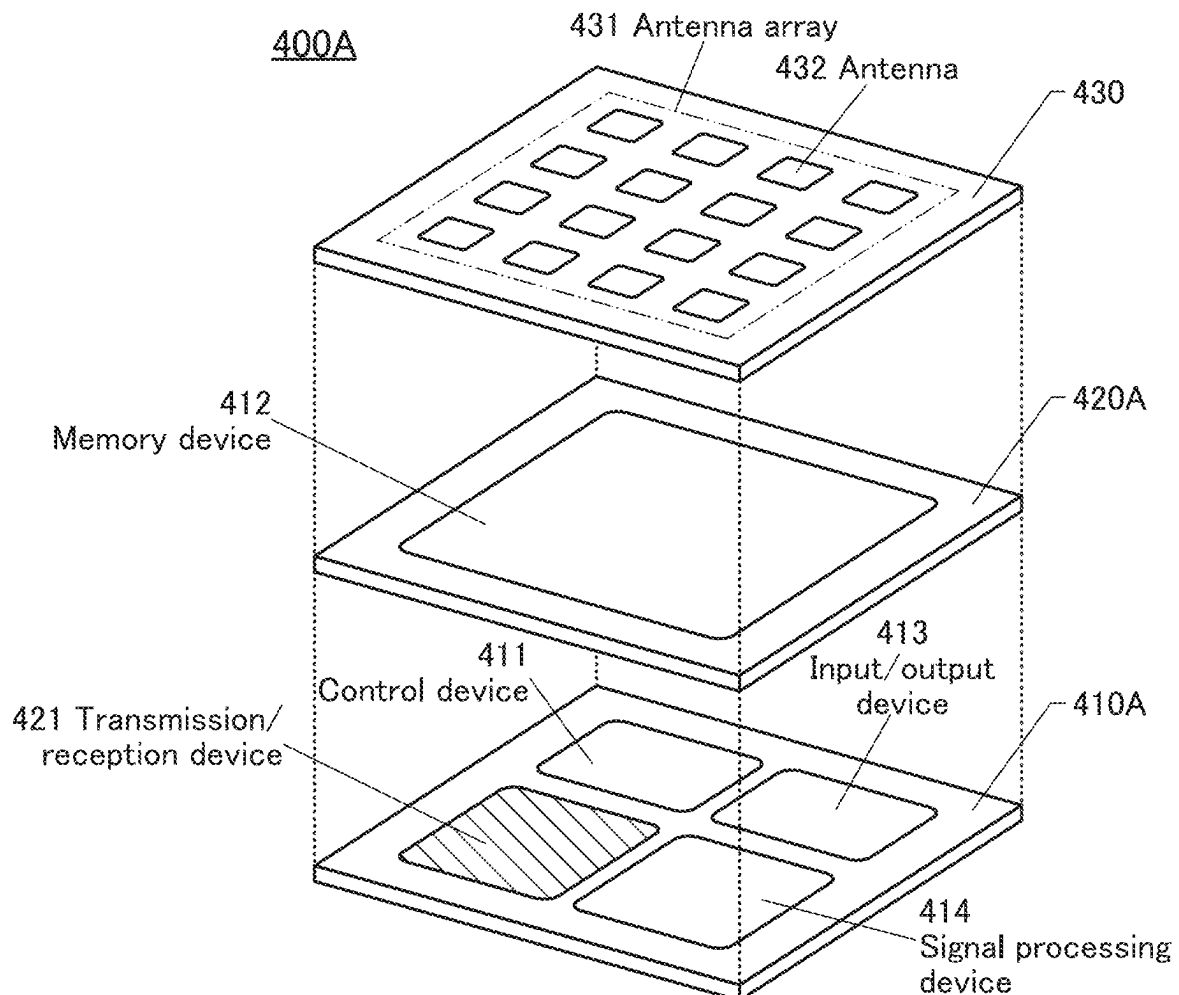


FIG. 14

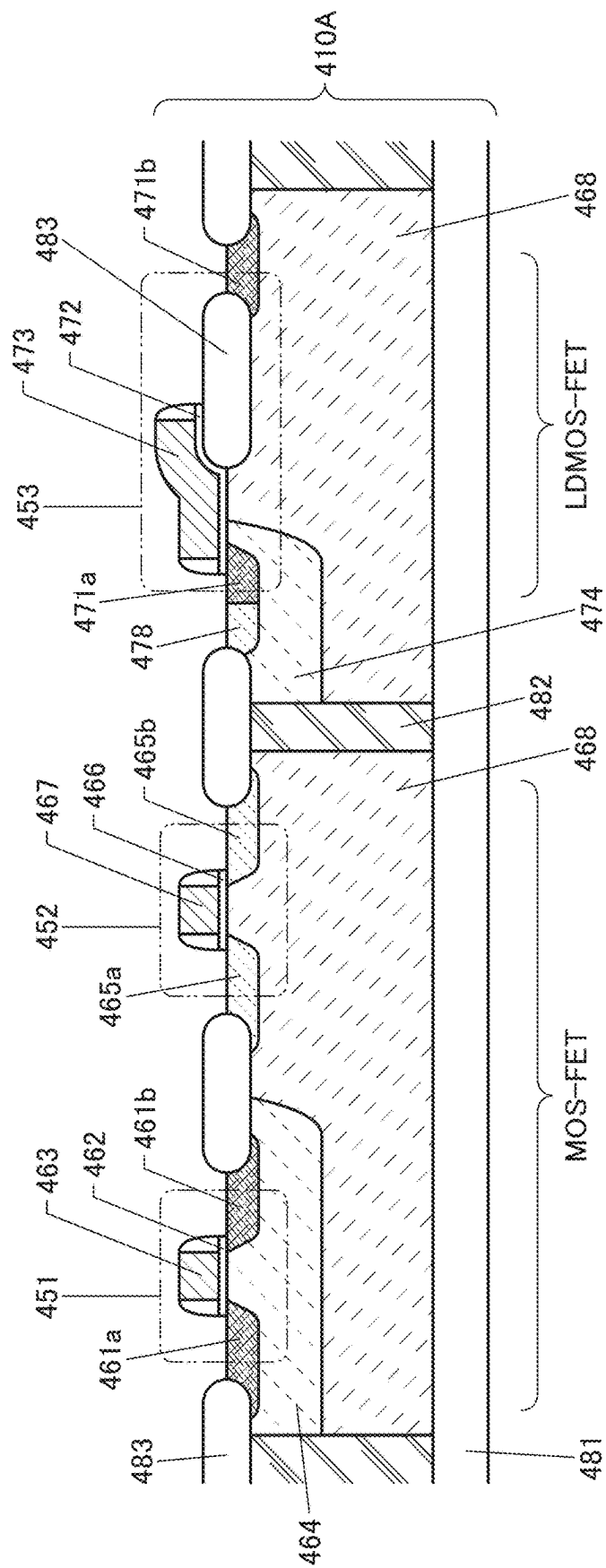


FIG. 15A

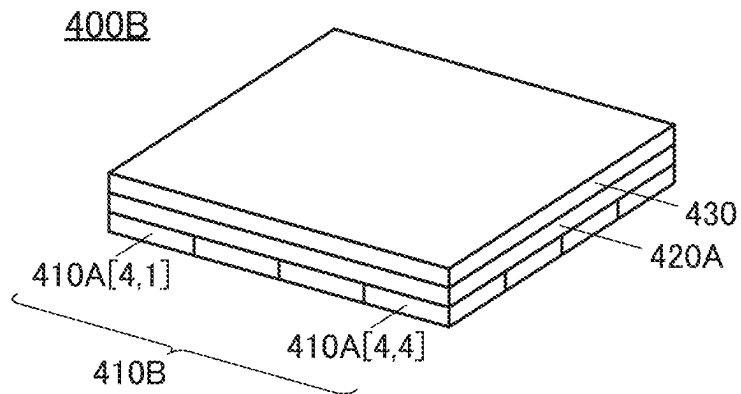


FIG. 15B

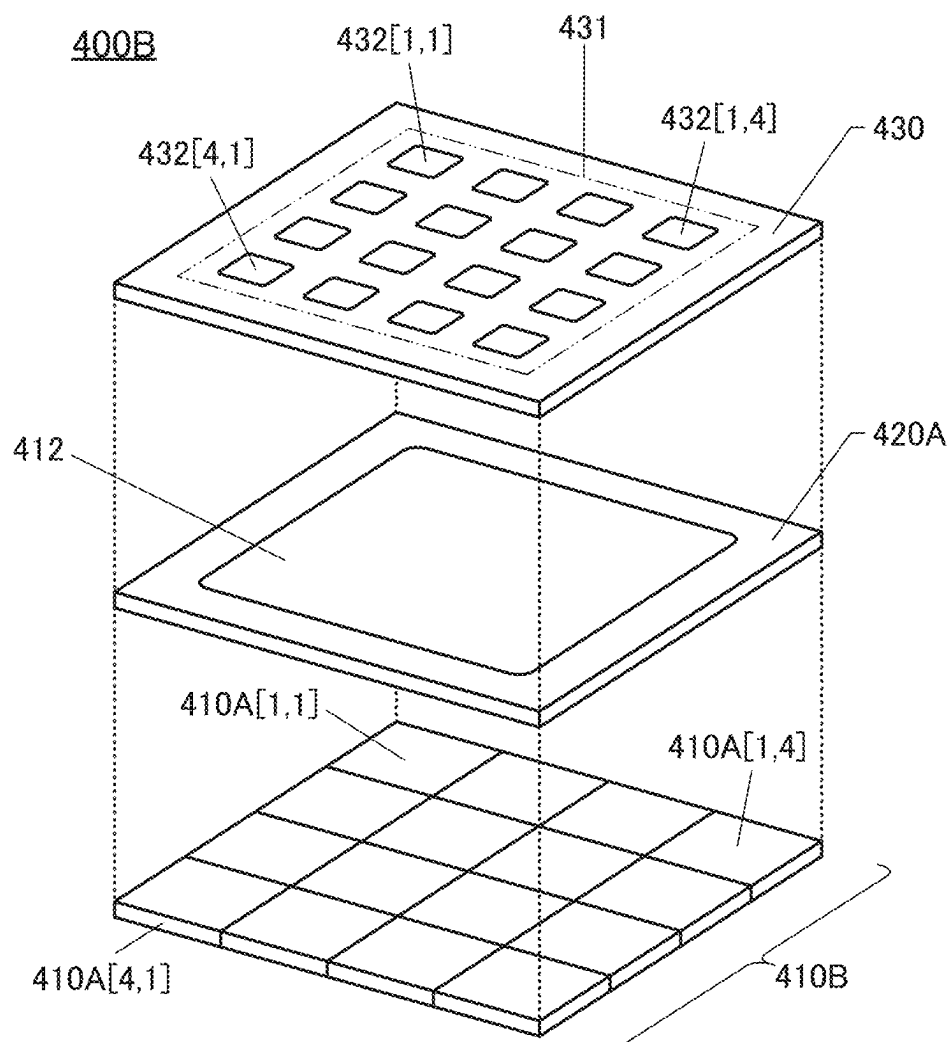


FIG. 16A

400C

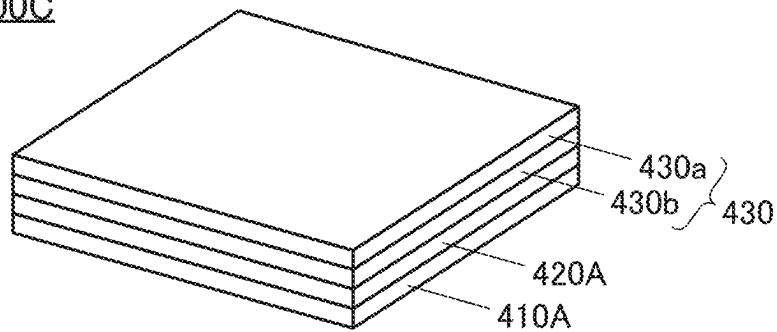


FIG. 16B

400C

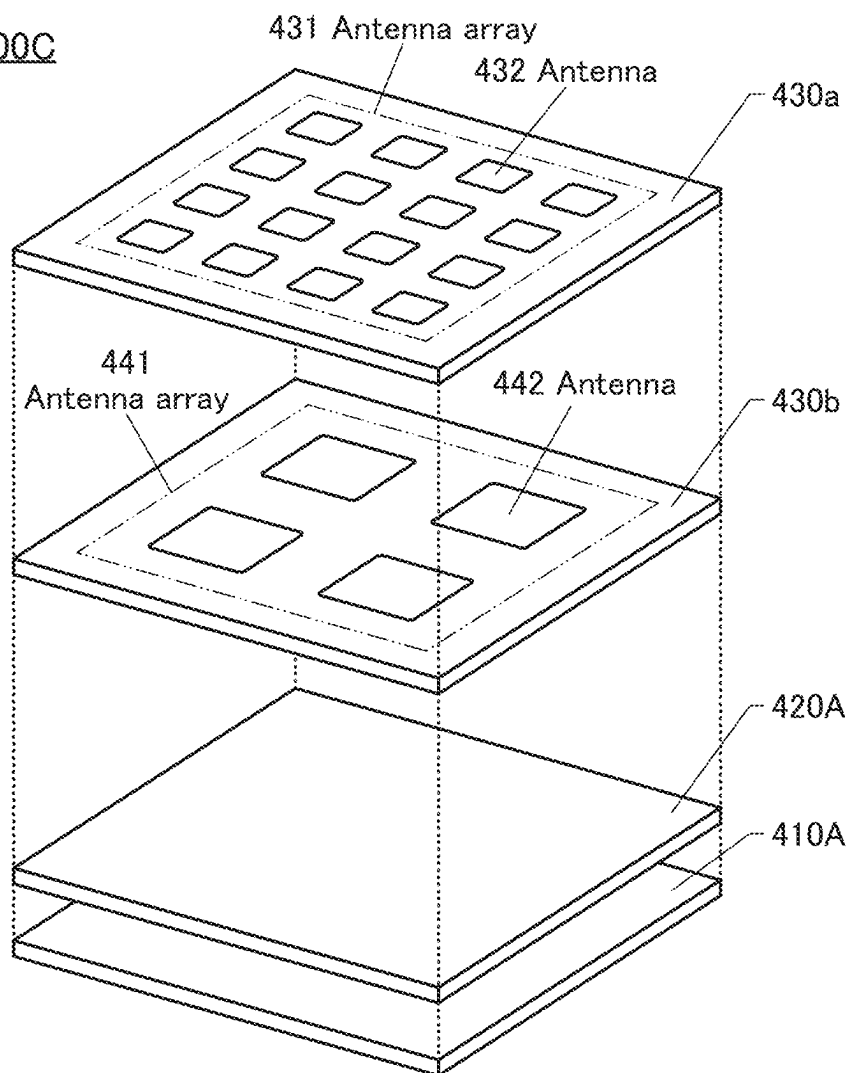


FIG. 17

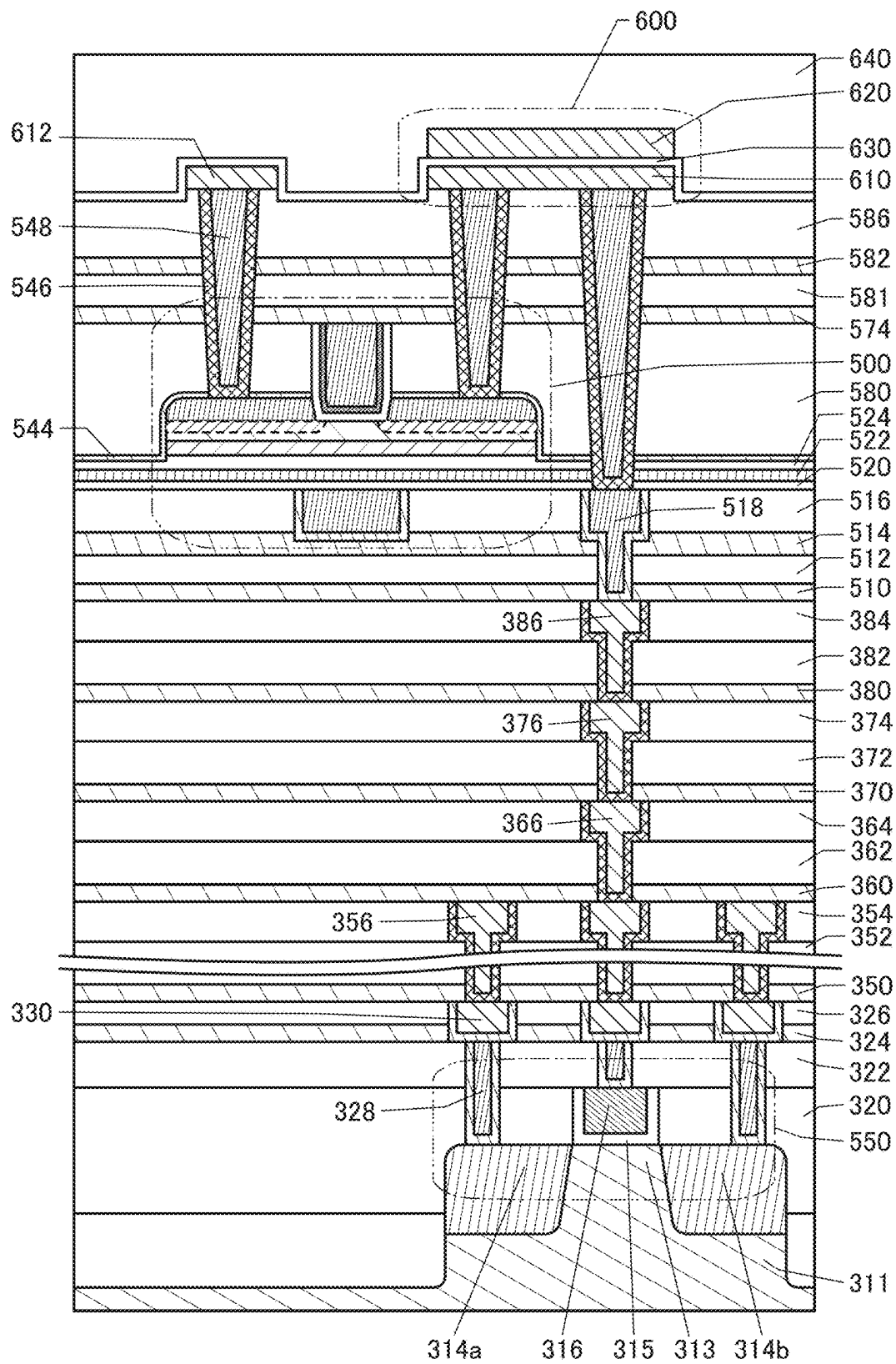


FIG. 18

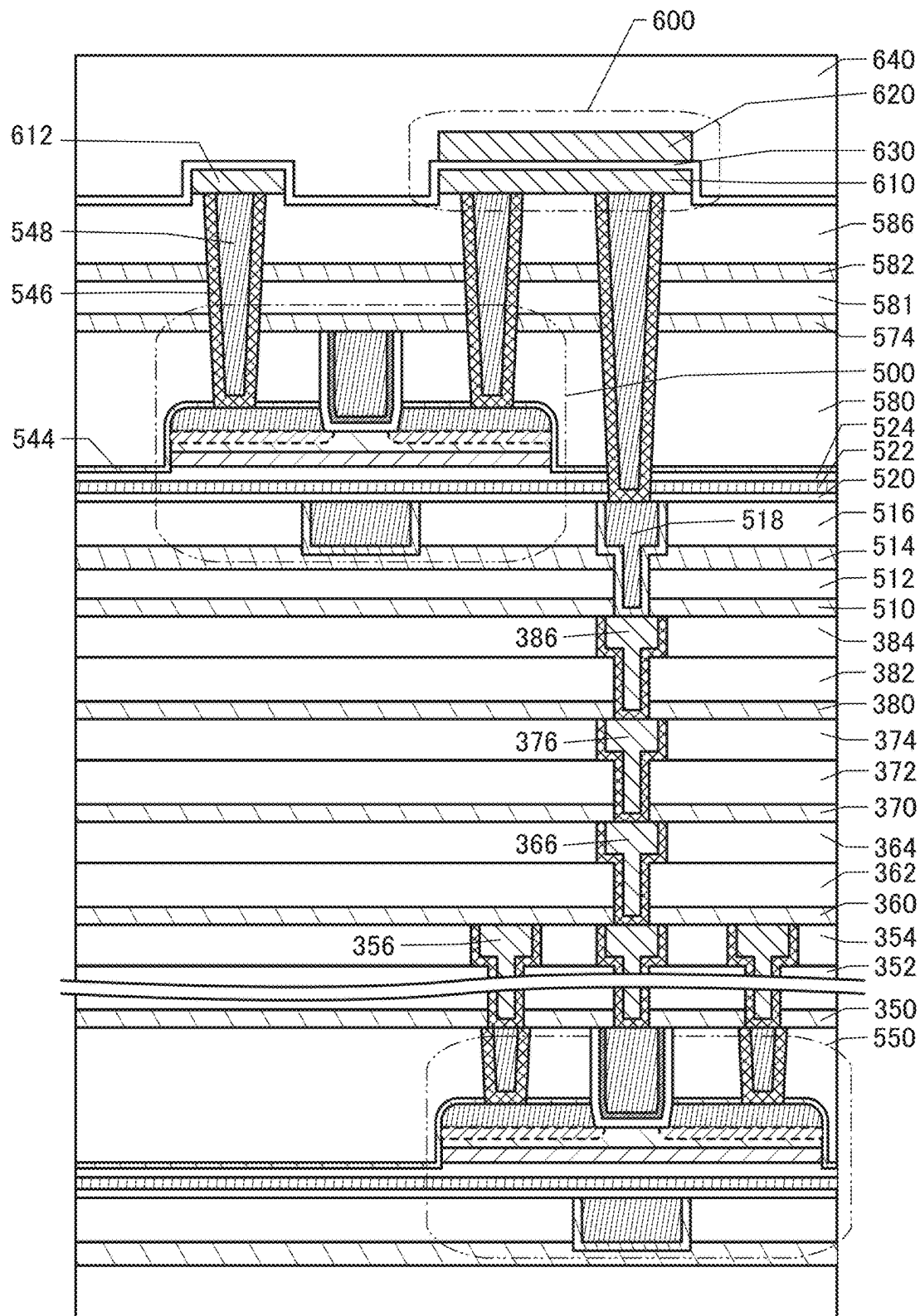


FIG. 19A

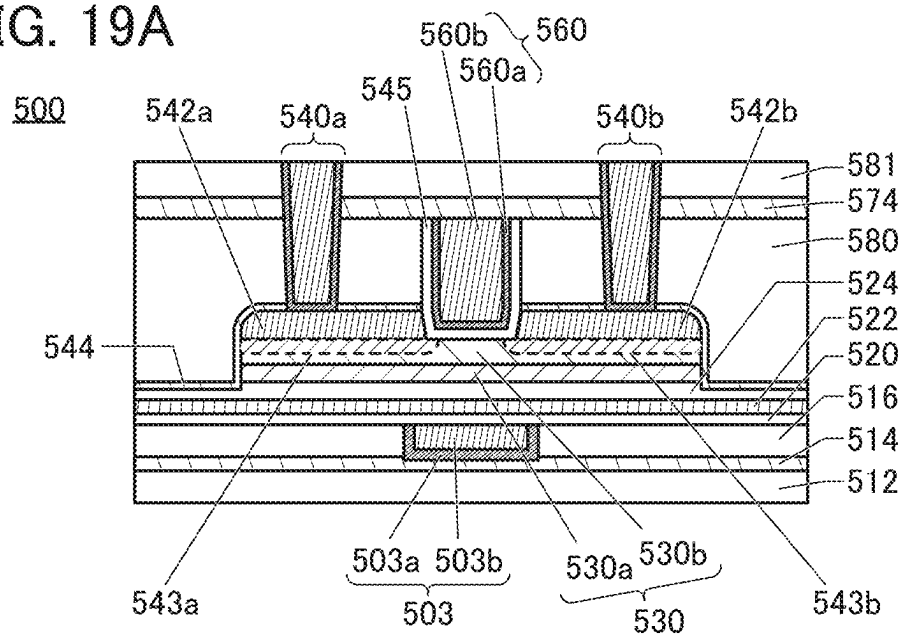


FIG. 19B

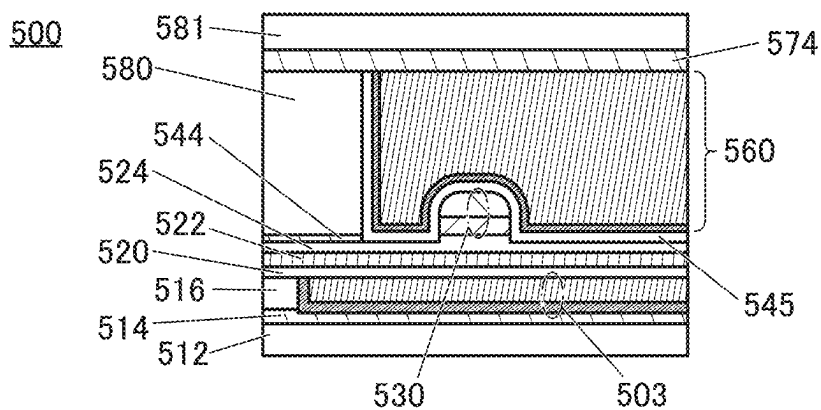


FIG. 19C

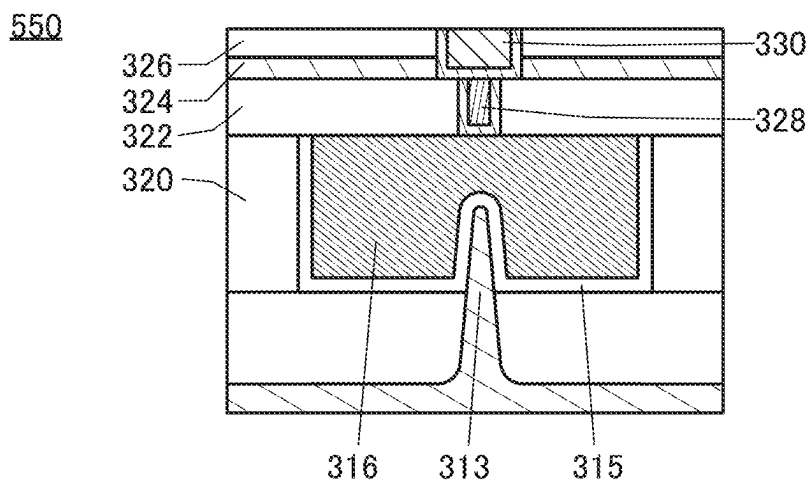


FIG. 20A

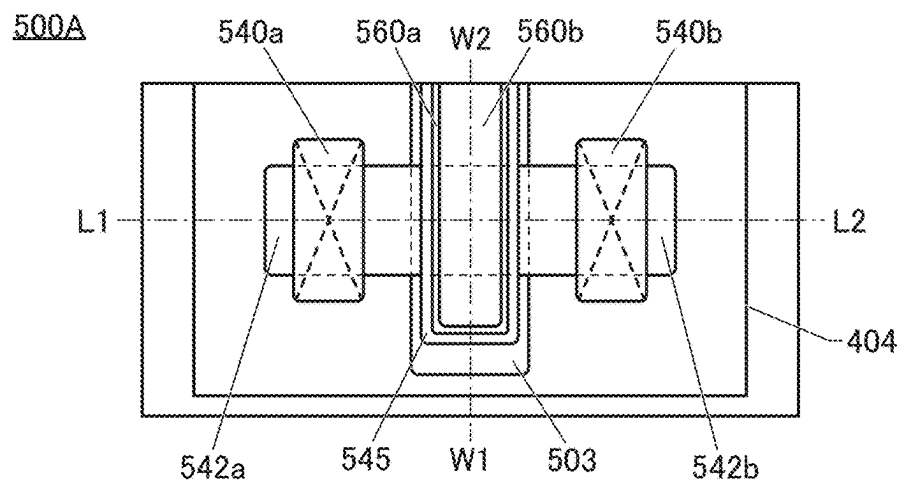


FIG. 20B

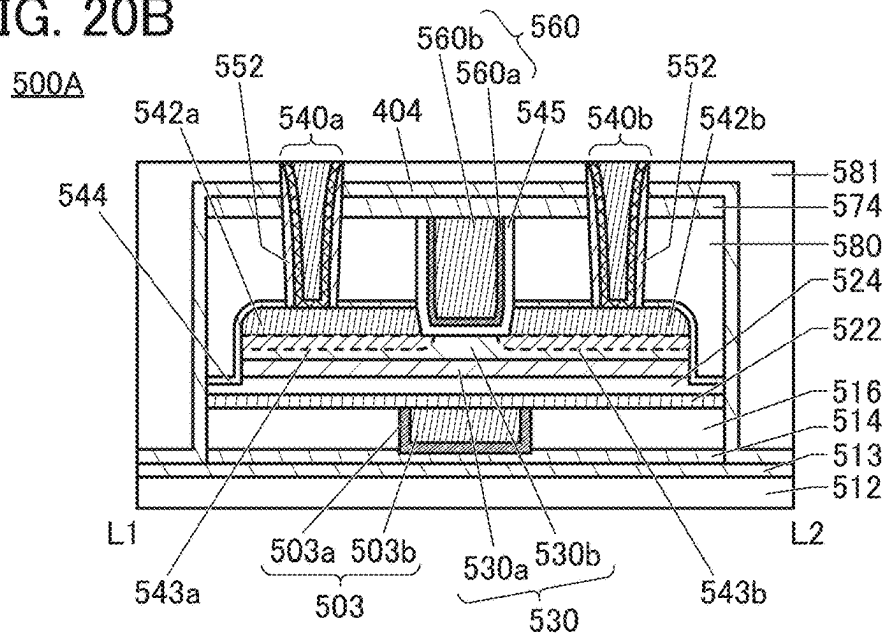


FIG. 20C

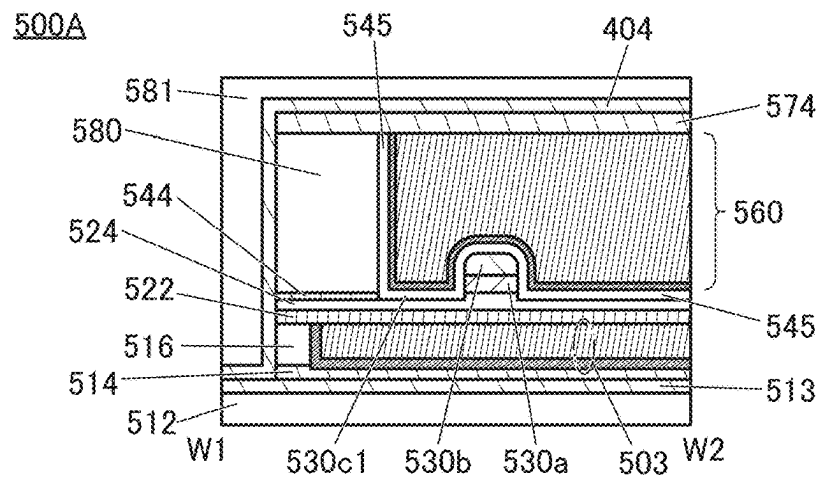


FIG. 21A

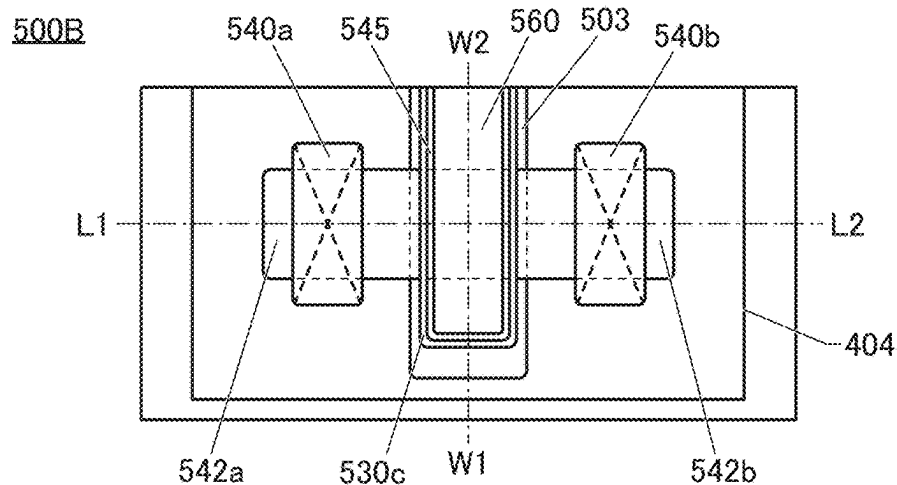


FIG. 21B

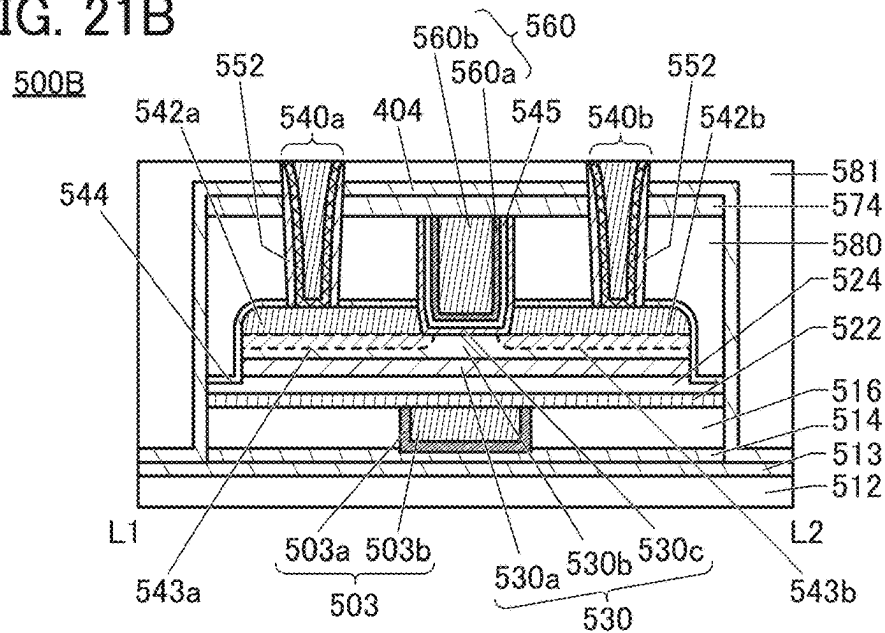


FIG. 21C

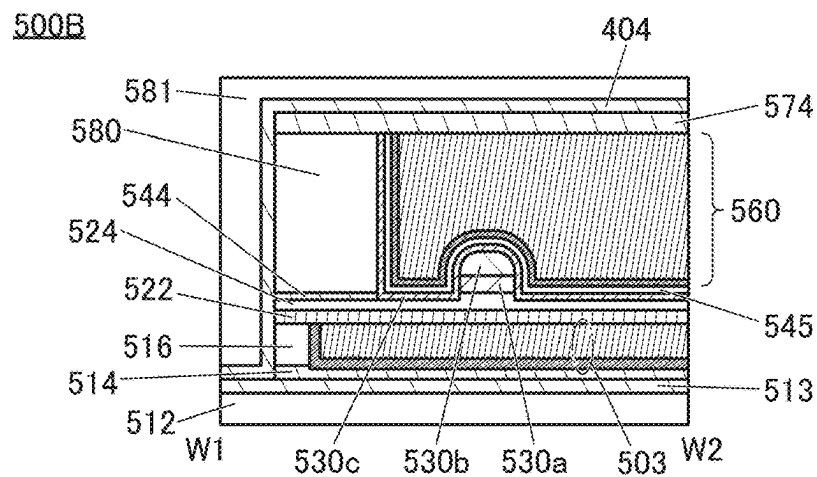


FIG. 22A

500C

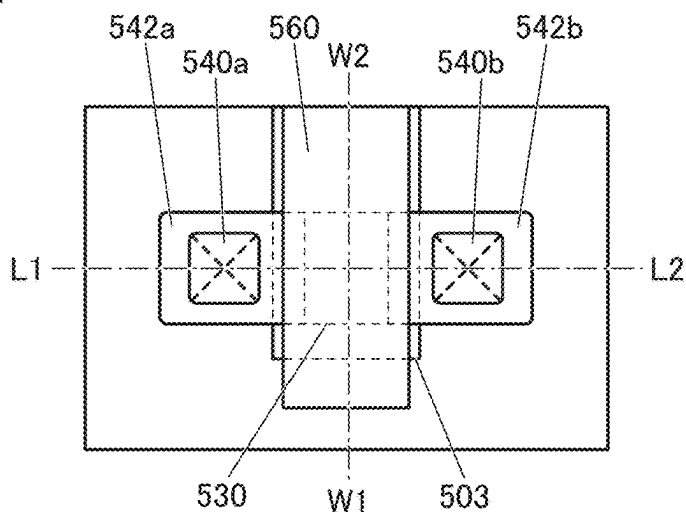


FIG. 22B

500C

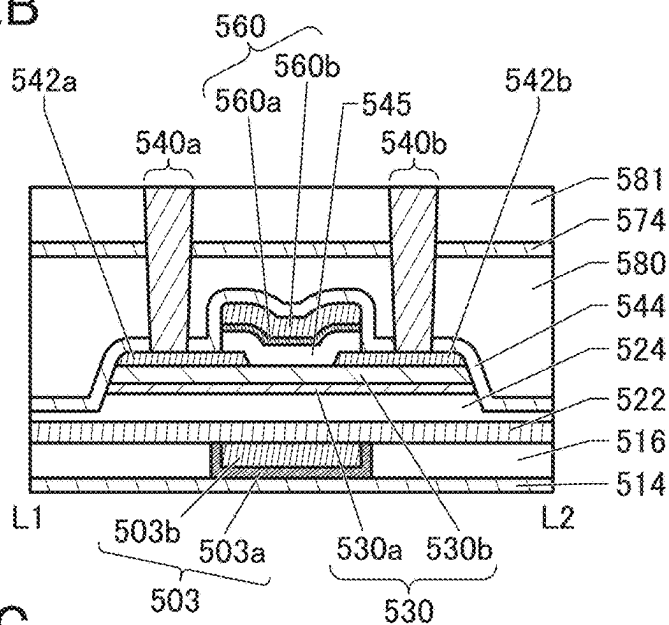


FIG. 22C

500C

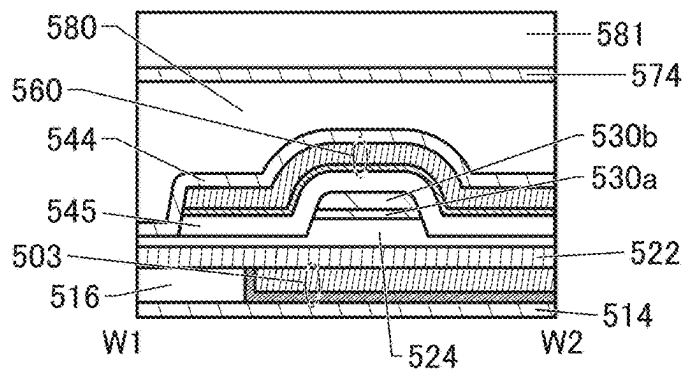


FIG. 23A

Intermediate state New crystalline phase		
Amorphous	Crystalline	Crystal
• completely amorphous	• CAAC • nc • CAC excluding single crystal and poly crystal	• single crystal • poly crystal

FIG. 23B

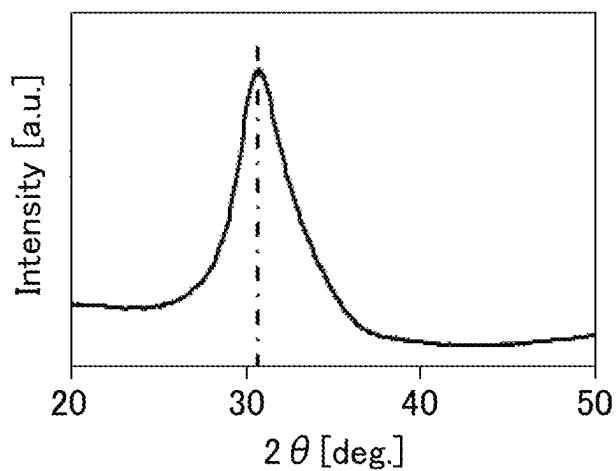


FIG. 23C

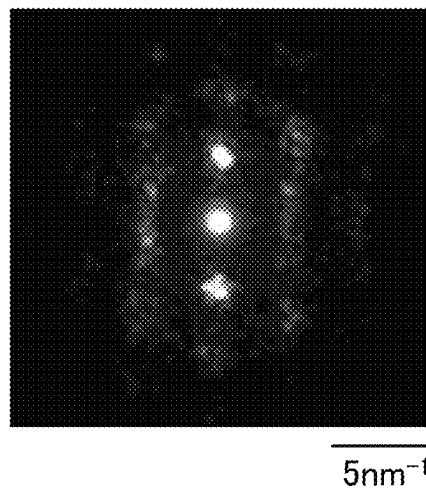


FIG. 24A

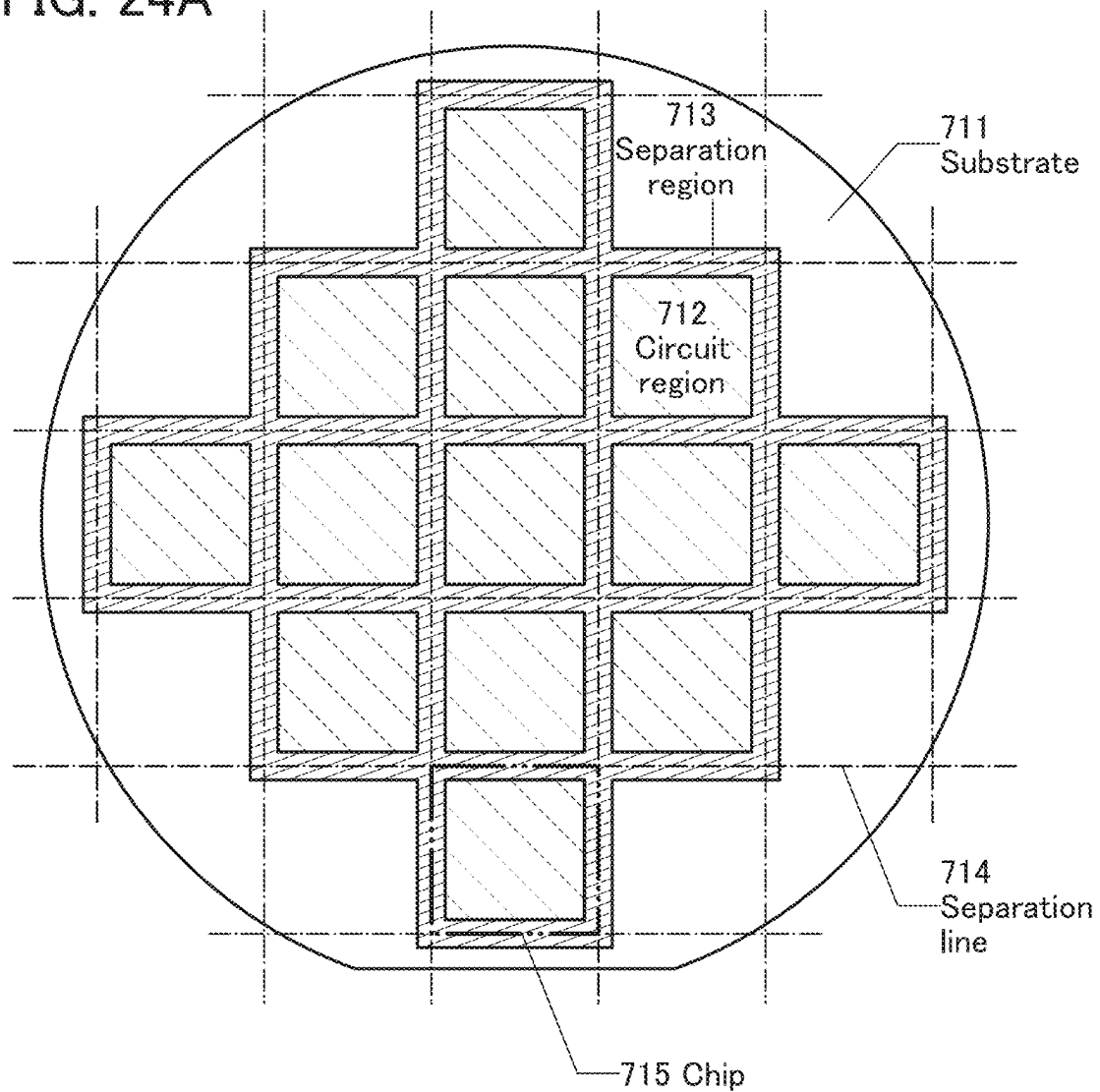


FIG. 24B

715 Chip

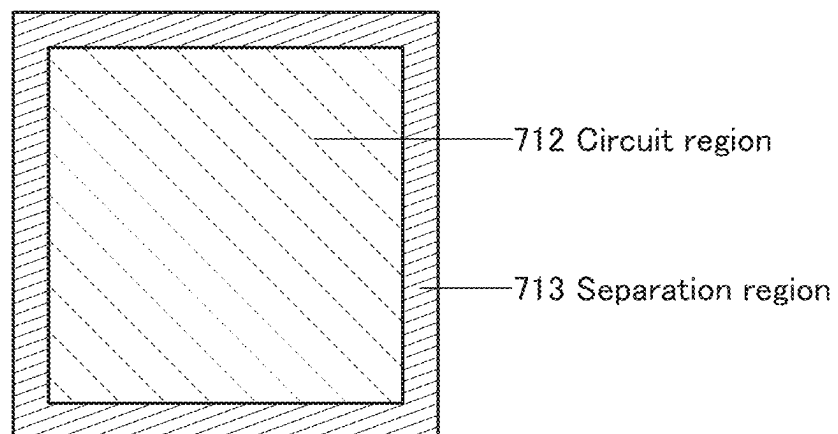


FIG. 25A

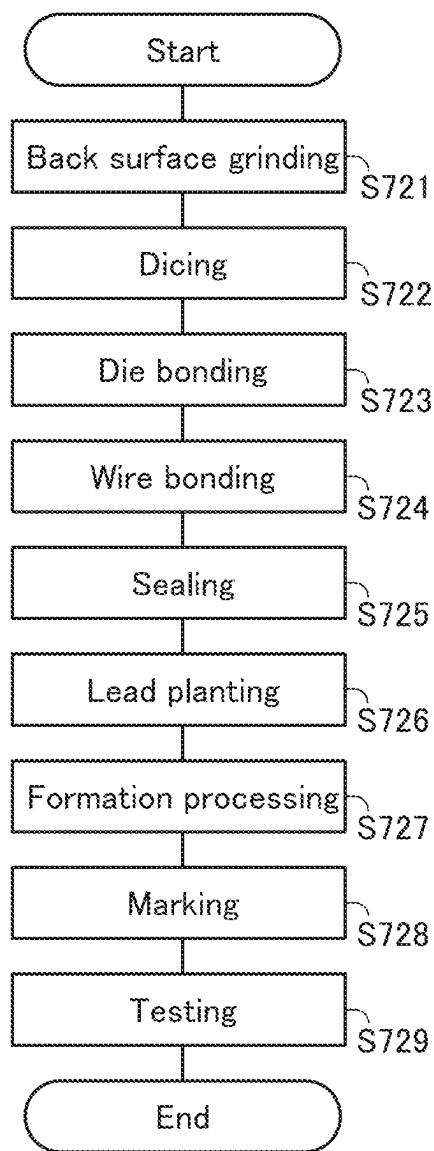


FIG. 25B

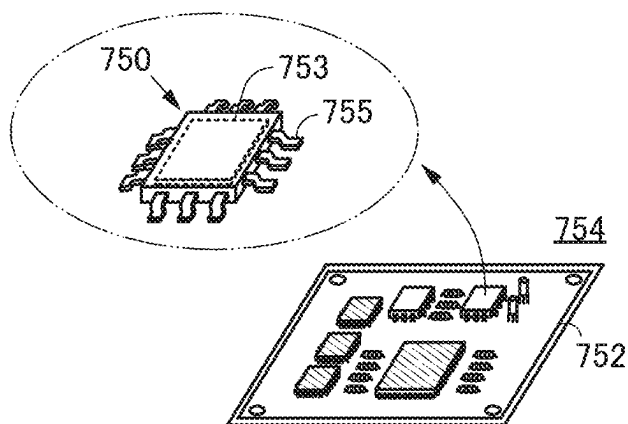


FIG. 26

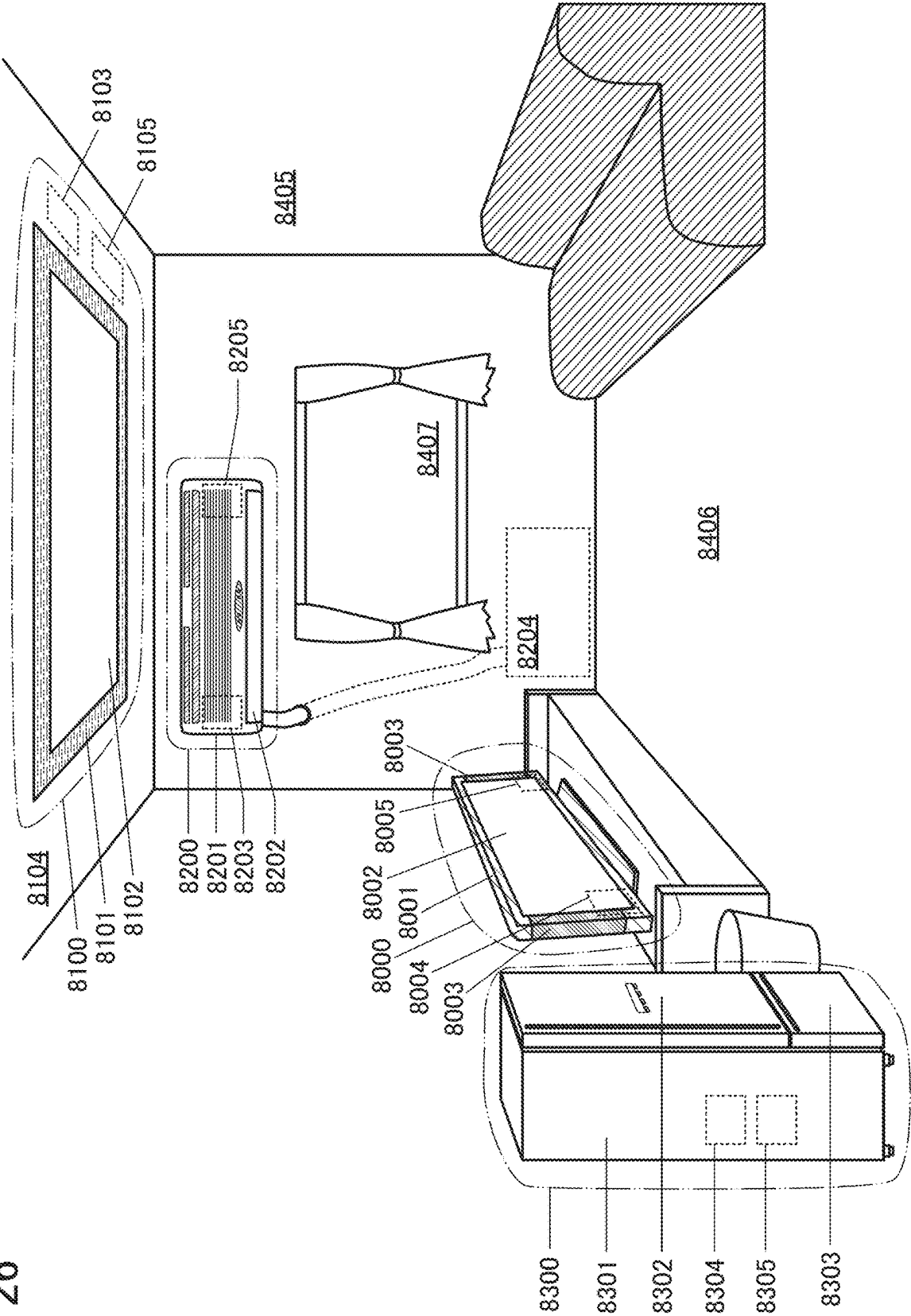


FIG. 27A

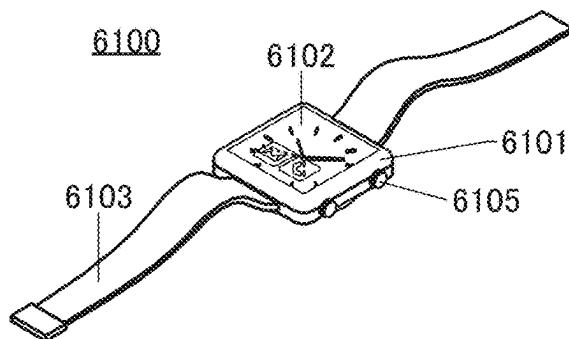


FIG. 27B

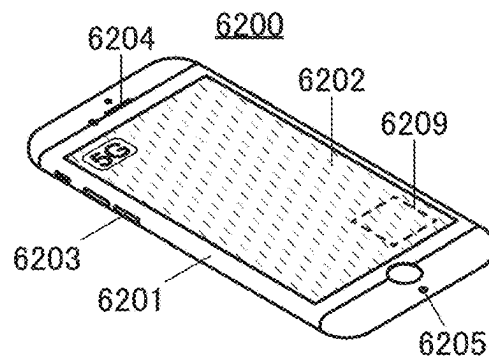


FIG. 27C

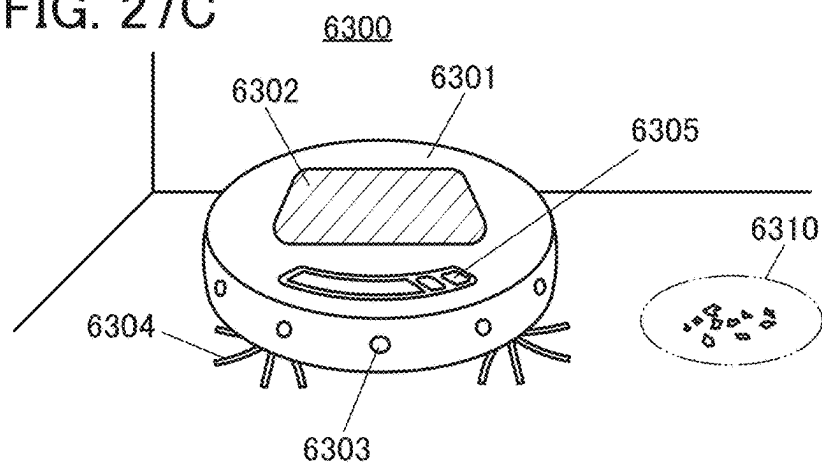


FIG. 27D

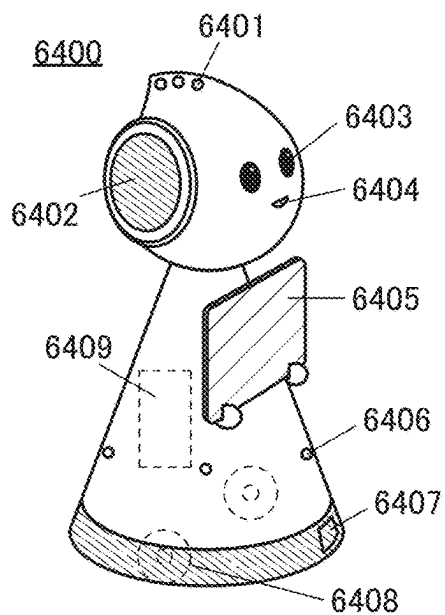


FIG. 27E

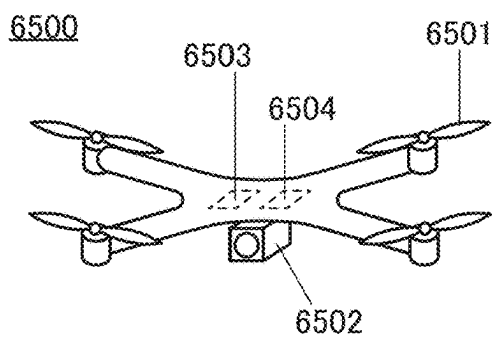


FIG. 27F



FIG. 28

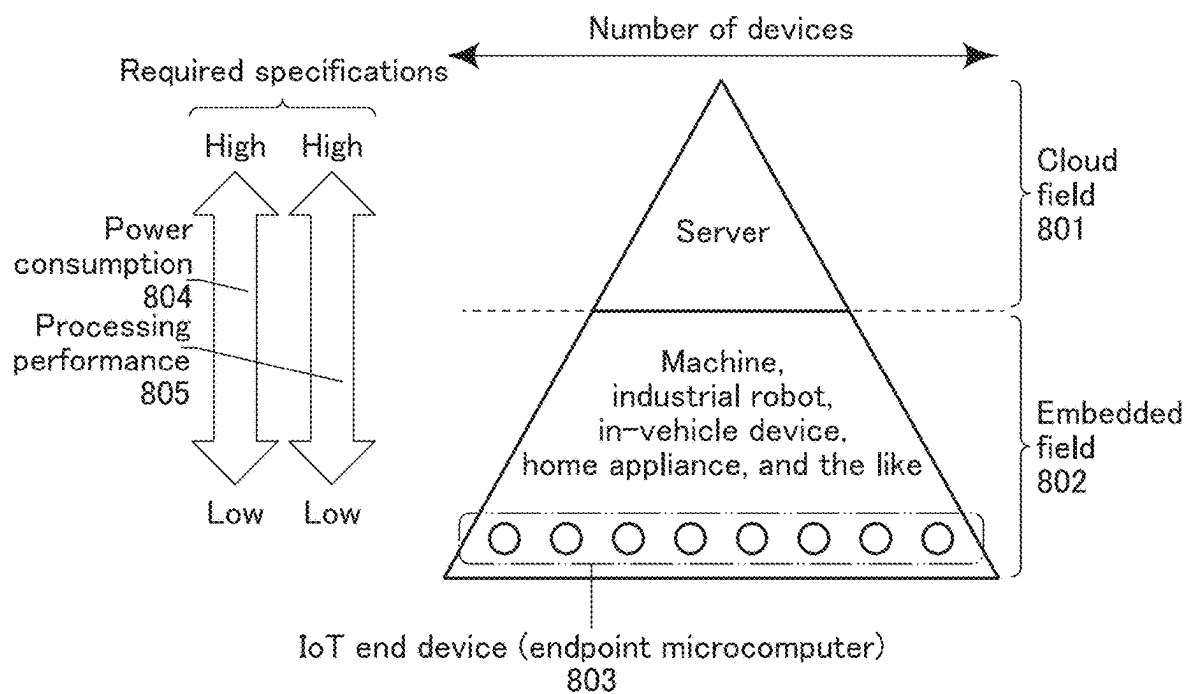


FIG. 29

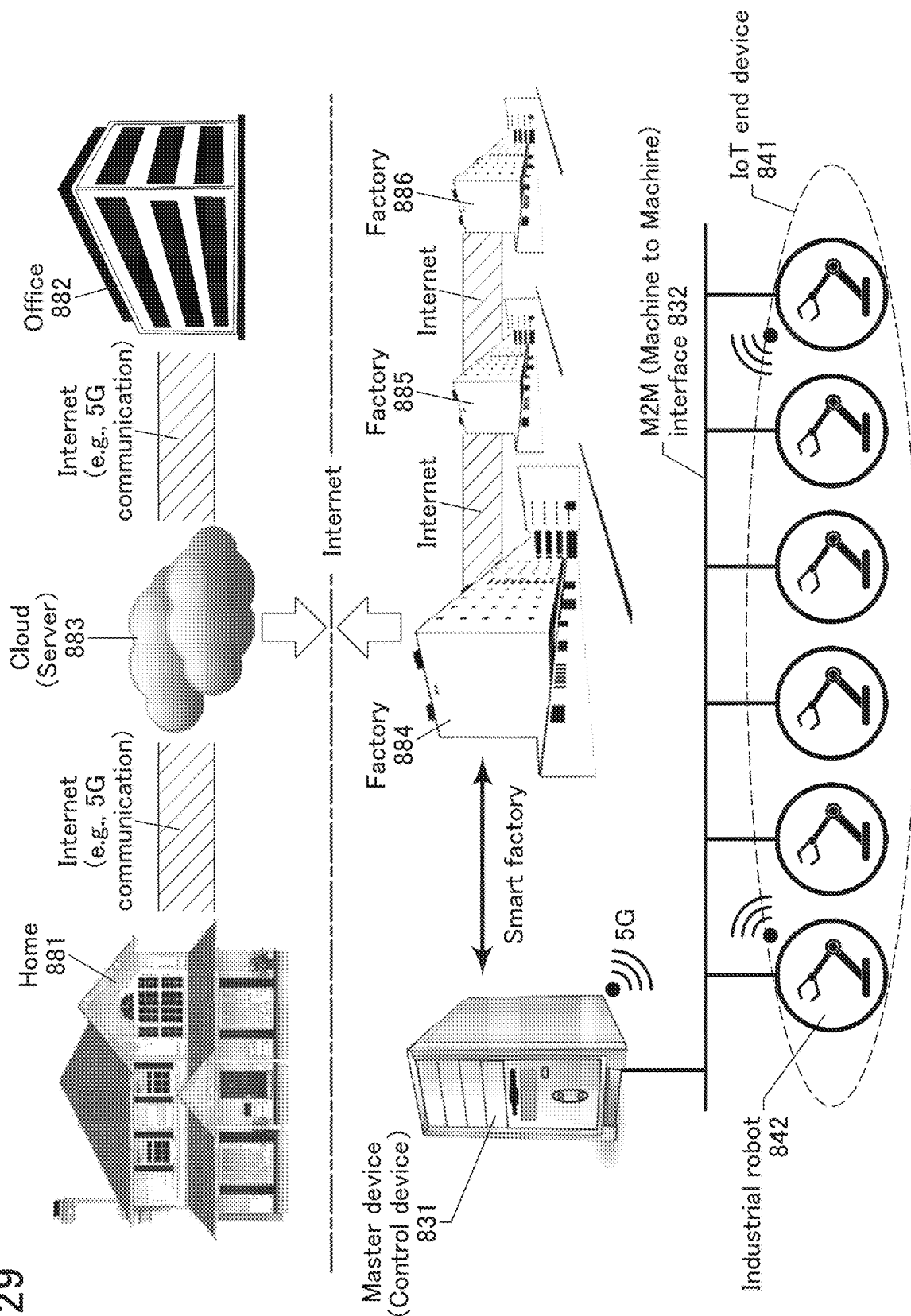


FIG. 30A

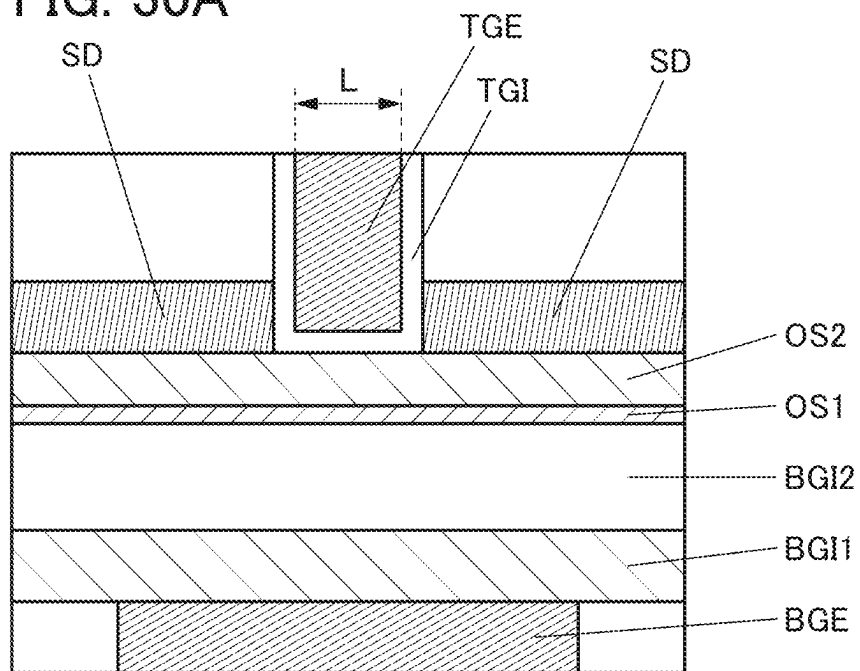


FIG. 30B

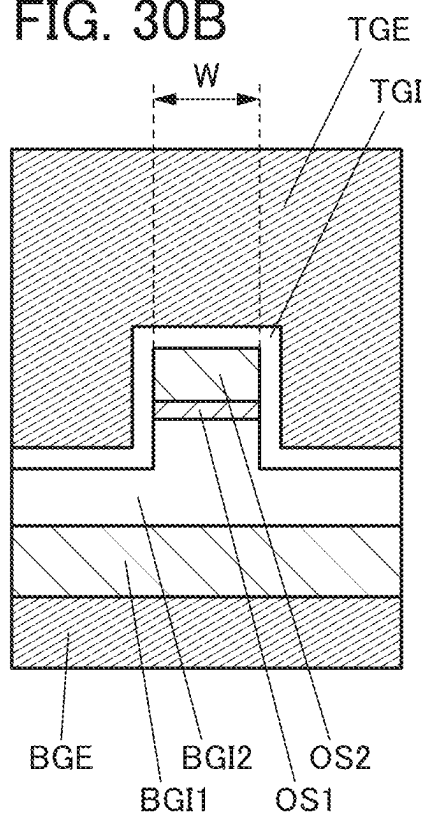


FIG. 30C

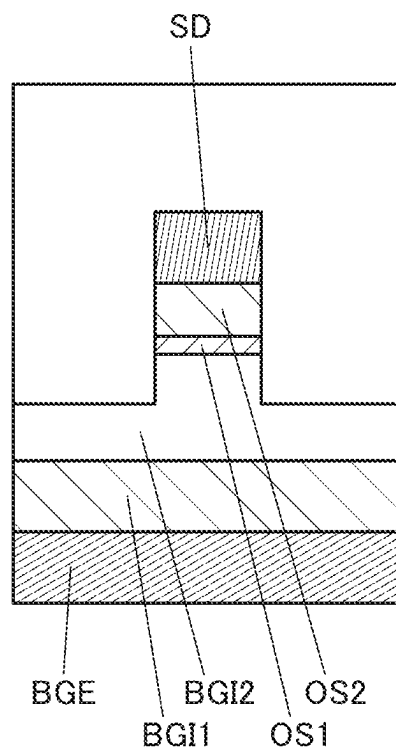
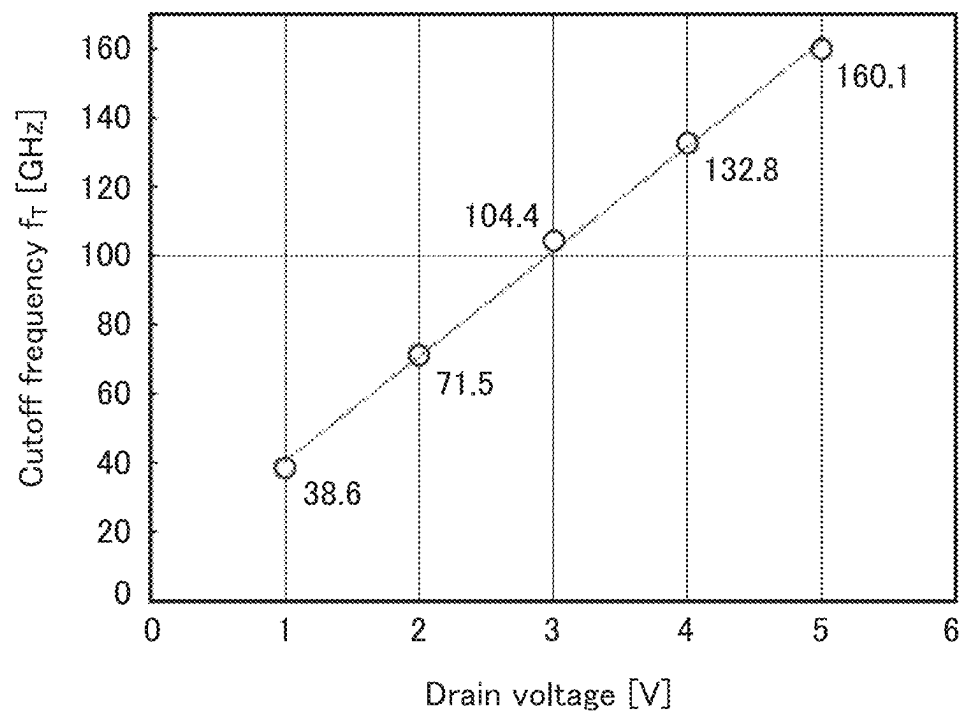


FIG. 31



1

SEMICONDUCTOR DEVICE AND ELECTRONIC DEVICE

TECHNICAL FIELD

One embodiment of the present invention relates to a semiconductor device.

Note that one embodiment of the present invention is not limited to the above technical field. The technical field of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. One embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter.

In this specification and the like, a semiconductor device generally means a device that can function by utilizing semiconductor characteristics. Thus, a semiconductor element such as a transistor or a diode and a circuit including a semiconductor element are semiconductor devices.

A display device, a light-emitting device, a lighting device, an electro-optical device, a communication device, an electronic device, and the like may include a semiconductor element or a semiconductor circuit. Therefore, a display device, a light-emitting device, a lighting device, an electro-optical device, an imaging device, a communication device, an electronic device, and the like are referred to as a semiconductor device in some cases.

BACKGROUND ART

Information terminals that are easy to carry, typified by smartphones, tablet terminals, and the like, have come into widespread use. With the widespread use of information terminals, various communication standards have been established. For example, the use of an LTE-Advanced standard called the fourth-generation mobile communication system (4G) has started.

With the development of information technology such as IoT (Internet of Things), the amount of data handled in information terminals has been recently showing an increasing tendency. In addition, the communication speed of electronic devices such as information terminals needs to be improved.

A base station needs to be connected to approximately one million information terminals per 1 km² simultaneously in order to achieve IoT. However, the number of feasible simultaneous connections with information terminals at 4G is approximately 20,000 per 1 km².

In order to be compatible with various kinds of information technology such as IoT, a new communication standard called the fifth-generation mobile communication system (5G) that achieves higher transmission speed, more simultaneous connections, and shorter delay time than 4G has been examined. Note that 5G uses communication frequencies of a 3.7 GHz band, a 4.5 GHz band, and a 28 GHz band.

A 5G compatible semiconductor device is manufactured using a semiconductor containing one kind of element such as Si as its main component or a compound semiconductor containing a plurality of kinds of elements such as Ga and As as its main components. Furthermore, an oxide semiconductor, which is one kind of metal oxide, has attracted attention.

A CAAC (c-axis aligned crystalline) structure and an nc (nanocrystalline) structure, which are neither single crystal nor amorphous, have been found in an oxide semiconductor (see

2

Non-Patent Document 1 and Non-Patent Document 2 disclose a technique for manufacturing a transistor using an oxide semiconductor having a CAAC structure.

REFERENCE

Non-Patent Document

[Non-Patent Document 1] S. Yamazaki et al., "SID Symposium Digest of Technical Papers", 2012, volume 43, issue 1, pp. 183-186

[Non-Patent Document 2] S. Yamazaki et al., "Japanese Journal of Applied Physics", 2014, volume 53, Number 4S, pp. 04ED18-1-04ED18-10

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

Improvement of communication speed tends to increase power consumption. Thus, electronic devices such as information terminals need to improve communication speed and reduce power consumption.

An object of one embodiment of the present invention is to provide a semiconductor device or the like with reduced power consumption. Another object is to provide a semiconductor device or the like that stably operates. Another object is to provide a semiconductor device or the like with high reliability. Another object is to provide a semiconductor device or the like with high productivity. Another object is to provide a novel semiconductor device or the like.

Note that the description of these objects does not preclude the existence of other objects. In one embodiment of the present invention, there is no need to achieve all the objects. Note that objects other than these will be apparent from the description of the specification, the drawings, the claims, and the like, and objects other than these can be derived from the description of the specification, the drawings, the claims, and the like.

Means for Solving the Problems

In a semiconductor device including a cascode circuit including a first transistor provided on a low power supply potential side and a second transistor provided on a high power supply potential side, a source or a drain of a third transistor and a capacitor are connected to a gate of the second transistor. An OS transistor is used as the third transistor.

One embodiment of the present invention is a semiconductor device including a first transistor, a second transistor, a third transistor, a capacitor, and a functional element. One of a source and a drain of the first transistor is electrically connected to a first terminal. The other of the source and the drain of the first transistor is electrically connected to one of a source and a drain of the second transistor. A gate of the first transistor is electrically connected to a second terminal. The other of the source and the drain of the second transistor is electrically connected to a third terminal. The other of the source and the drain of the second transistor is electrically connected to the functional element. One of a source and a drain of the third transistor is electrically connected to a fourth terminal. The other of the source and the drain of the third transistor is electrically connected to the gate of the second transistor. A gate of the third transistor is electrically connected to a fifth terminal. The other of the source and the

drain of the third transistor is provided with the capacitor. A semiconductor layer of the third transistor includes an oxide semiconductor.

The second transistor may include a back gate. The gate of the first transistor may be electrically connected to the back gate of the second transistor.

The functional element is preferably a resistor, a constant current source, or a parallel resonant circuit. The oxide semiconductor preferably contains at least one of In and Zn.

At least one of the first to the third transistors may be a multi-gate transistor. A semiconductor layer of the second transistor may include an oxide semiconductor. A semiconductor layer of the third transistor may include an oxide semiconductor.

The semiconductor device may be electrically connected to an antenna.

Another embodiment of the present invention is an electronic device including the semiconductor device described above and a speaker, a microphone, or a secondary battery.

Another embodiment of the present invention is a semiconductor device including a first layer, a second layer, and a third layer. The first layer includes a transmission/reception device and a signal processing device. The second layer includes a memory device. The third layer includes an antenna array. The transmission/reception device includes a fourth transistor. The signal processing device includes a fifth transistor. The memory device includes a memory element. The memory element includes a sixth transistor and a capacitor. The sixth transistor includes an oxide semiconductor in a semiconductor layer. The antenna array includes a plurality of antennas. The first layer and the third layer include regions overlapping with each other with the second layer therebetween.

An LDMOS-FET may be used as the fourth transistor. A MOS-FET may be used as the fifth transistor.

The oxide semiconductor preferably contains at least one of In and Zn.

The signal processing device may include a demodulator and a modulator. The semiconductor device may have a function of transmitting a radio wave with beamforming. The semiconductor device may have a function of transmitting a radio wave with spatial multiplex transmission. The semiconductor device may have a function of receiving a radio wave with spatial multiplex transmission.

Effect of the Invention

According to one embodiment of the present invention, a semiconductor device or the like with reduced power consumption can be provided. A semiconductor device or the like which stably operates can be provided. A semiconductor device or the like with high reliability can be provided. A semiconductor device or the like with high productivity can be provided. A novel semiconductor device or the like can be provided.

Note that the description of these effects does not preclude the existence of other effects. One embodiment of the present invention does not have to have all these effects. Note that effects other than these are apparent from the description of the specification, the drawings, the claims, and the like, and effects other than these can be derived from the description of the specification, the drawings, the claims, and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a circuit diagram of a semiconductor device. FIG. 1B is a conceptual diagram of an operation of a semiconductor device.

FIG. 2A and FIG. 2B are circuit diagrams of a semiconductor device.

FIG. 3A is a circuit diagram of a semiconductor device. FIG. 3B is a graph showing a relation between the frequency of a signal passing through a parallel resonant circuit and the impedance of the parallel resonant circuit.

FIG. 4A to FIG. 4D are diagrams illustrating configuration examples of functional elements.

FIG. 5 is a circuit diagram of a semiconductor device.

FIG. 6A is a circuit diagram of a semiconductor device. FIG. 6B is a graph showing a relation between the back gate voltage and Id-Vg characteristics.

FIG. 7A to FIG. 7C are circuit diagrams of semiconductor devices.

FIG. 8A to FIG. 8C are circuit diagrams of semiconductor devices.

FIG. 9A and FIG. 9B are diagrams illustrating circuit symbols of transistors.

FIG. 10A is a diagram illustrating a structure example of a wireless transceiver. FIG. 10B is a diagram illustrating a structure example of a demodulator. FIG. 10C is a diagram illustrating a structure example of a modulator.

FIG. 11A and FIG. 11B are diagrams illustrating a structure example of a wireless transceiver.

FIG. 12A is a perspective view of a semiconductor device 400. FIG. 12B is a perspective view illustrating a structure of the semiconductor device 400.

FIG. 13A is a perspective view of a semiconductor device 400A. FIG. 13B is a perspective view illustrating a structure of the semiconductor device 400A.

FIG. 14 is a diagram illustrating a cross-sectional structure example of a MOS-FET and an LDMOS-FET.

FIG. 15A is a perspective view of a semiconductor device 400B. FIG. 15B is a perspective view illustrating a structure of the semiconductor device 400B.

FIG. 16A is a perspective view of a semiconductor device 400C. FIG. 16B is a perspective view illustrating a structure of the semiconductor device 400C.

FIG. 17 is a diagram illustrating a structure example of a semiconductor device.

FIG. 18 is a diagram illustrating a structure example of a semiconductor device.

FIG. 19A to FIG. 19C are diagrams illustrating a structure example of a transistor.

FIG. 20A to FIG. 20C are diagrams illustrating a structure example of a transistor.

FIG. 21A to FIG. 21C are diagrams illustrating a structure example of a transistor.

FIG. 22A to FIG. 22C are diagrams illustrating a structure example of a transistor.

FIG. 23A is a diagram showing the classification of crystal structures of IGZO. FIG. 23B is a graph showing an XRD spectrum of a CAAC-IGZO film. FIG. 23C is an image showing nanobeam electron diffraction patterns of a CAAC-IGZO film.

FIG. 24A is a top view of a semiconductor wafer. FIG. 24B is a top view of a chip.

FIG. 25A is a flowchart showing an example of a fabricating process of an electronic component.

FIG. 25B is a schematic perspective view of the electronic component.

FIG. 26 is a diagram illustrating examples of electronic devices.

FIG. 27A to FIG. 27F are diagrams illustrating examples of electronic devices.

FIG. 28 is a diagram illustrating a hierarchical structure of an IoT network and tendencies of required specifications.

5

FIG. 29 is a conceptual diagram of factory automation. FIG. 30A to FIG. 30C are diagrams illustrating a structure of an OS-FET used for calculation of the cutoff frequency. FIG. 31 is a graph showing the calculation results of the cutoff frequency of an OS-FET.

MODE FOR CARRYING OUT THE INVENTION

Embodiments are described in detail with reference to the drawings. However, the present invention is not limited to the following description, and it is readily appreciated by those skilled in the art that modes and details can be modified in various ways without departing from the spirit and the scope of the present invention. Thus, the present invention should not be construed as being limited to the description in the following embodiments. Note that in the structures of the invention described below, the same portions or portions having similar functions are denoted by the same reference numerals in different drawings, and description thereof is not repeated.

In addition, the position, size, range, and the like of each component illustrated in the drawings and the like do not represent the actual position, size, range, and the like in some cases for easy understanding of the invention. Therefore, the disclosed invention is not necessarily limited to the position, size, range, or the like disclosed in drawings and the like. For example, in an actual manufacturing process, a resist mask or the like might be unintentionally reduced in size by treatment such as etching, which might not be reflected in the drawings for easy understanding.

Furthermore, in a top view (also referred to as a “plan view”), a perspective view, or the like, the description of some components might be omitted for easy understanding of the drawings.

In addition, in this specification and the like, the terms “electrode” and “wiring” do not functionally limit these components. For example, an “electrode” is used as part of a “wiring” in some cases, and vice versa. Furthermore, the term “electrode” or “wiring” also includes the case where a plurality of “electrodes” or “wirings” are formed in an integrated manner, for example.

In this specification and the like, a “terminal” in an electric circuit refers to a portion that inputs or outputs a current, inputs or outputs a voltage, or receives or transmits a signal. Accordingly, part of a wiring or an electrode functions as a terminal in some cases.

Note that the term “over” or “under” in this specification and the like does not necessarily mean that a component is placed directly over and in contact with or directly under and in contact with another component. For example, the expression “electrode B over insulating layer A” does not necessarily mean that the electrode B is formed on and in direct contact with the insulating layer A, and does not exclude the case where another component is provided between the insulating layer A and the electrode B.

In addition, functions of a source and a drain are interchanged with each other depending on operation conditions and the like, for example, when a transistor of different polarity is employed or when the current direction is changed in a circuit operation; therefore, it is difficult to define which is the source or the drain. Thus, the terms “source” and “drain” can be interchangeably used in this specification.

In this specification and the like, the expression “electrically connected” includes the case where components are directly connected to each other and the case where components are connected through an “object having any elec-

6

tric function”. Here, there is no particular limitation on the “object having any electric function” as long as electric signals can be transmitted and received between components that are connected through the object. Thus, even when the expression “electrically connected” is used, there is a case where no physical connection portion is made and a wiring is just extended in an actual circuit.

Furthermore, in this specification and the like, “parallel” indicates a state where two straight lines are placed at an angle of greater than or equal to -10° and less than or equal to 10° , for example. Accordingly, the case where the angle is greater than or equal to -5° and less than or equal to 5° is also included. Moreover, “perpendicular” and “orthogonal” indicate a state where two straight lines are placed at an angle of greater than or equal to 80° and less than or equal to 100° , for example. Accordingly, the case where the angle is greater than or equal to 85° and less than or equal to 95° is also included.

In this specification and the like, the terms “identical”, “same”, “equal”, “uniform”, and the like used in describing calculation values and measurement values allow for a margin of error of $\pm 20\%$ unless otherwise specified.

In addition, a voltage refers to a potential difference between a certain potential and a reference potential (e.g., a ground potential or a source potential) in many cases. Therefore, the terms “voltage” and “potential” can be replaced with each other in many cases. In this specification and the like, the terms “voltage” and “potential” can be replaced with each other unless otherwise specified.

Note that a “semiconductor” has characteristics of an “insulator” when conductivity is sufficiently low, for example. Thus, a “semiconductor” can be replaced with an “insulator”. In that case, a “semiconductor” and an “insulator” cannot be strictly distinguished from each other because a border therebetween is not clear. Accordingly, a “semiconductor” and an “insulator” described in this specification can be replaced with each other in some cases.

Furthermore, a “semiconductor” has characteristics of a “conductor” when conductivity is sufficiently high, for example. Thus, a “semiconductor” can be replaced with a “conductor”. In that case, a “semiconductor” and a “conductor” cannot be strictly distinguished from each other because a border therebetween is not clear. Accordingly, a “semiconductor” and a “conductor” in this specification can be replaced with each other in some cases.

Note that ordinal numbers such as “first” and “second” in this specification and the like are used in order to avoid confusion among components and do not denote the priority or the order such as the order of steps or the stacking order. A term without an ordinal number in this specification and the like might be provided with an ordinal number in the scope of claims in order to avoid confusion among components. Furthermore, a term with an ordinal number in this specification and the like might be provided with a different ordinal number in the scope of claims. Furthermore, even when a term is provided with an ordinal number in this specification and the like, the ordinal number might be omitted in the scope of claims and the like.

Note that in this specification and the like, an “on state” of a transistor refers to a state in which a source and a drain of the transistor are electrically short-circuited (also referred to as a “conduction state”). Furthermore, an “off state” of the transistor refers to a state in which the source and the drain of the transistor are electrically disconnected (also referred to as a “non-conduction state”).

In addition, in this specification and the like, an “on-state current” sometimes refers to a current that flows between a

source and a drain when a transistor is in an on state. Furthermore, an “off-state current” sometimes refers to a current that flows between a source and a drain when a transistor is in an off state.

In this specification and the like, a high power supply potential VDD (hereinafter, also simply referred to as “VDD”, “H potential”, or “H”) is a power supply potential higher than a low power supply potential VSS (hereinafter, also simply referred to as “VSS”, “L potential”, or “L”). The potential VSS refers to a power supply potential at a potential lower than the potential VDD. A ground potential (hereinafter, also simply referred to as “GND” or “GND potential”) can be used as VDD or VSS. For example, in the case where VDD is a ground potential, VSS is a potential lower than the ground potential, and in the case where VSS is a ground potential, VDD is a potential higher than the ground potential.

In addition, in this specification and the like, a gate refers to part or the whole of a gate electrode and a gate wiring. A gate wiring refers to a wiring for electrically connecting at least one gate electrode of a transistor to another electrode or another wiring.

Furthermore, in this specification and the like, a source refers to part or the whole of a source region, a source electrode, or a source wiring. A source region refers to a region in a semiconductor layer, where the resistivity is lower than or equal to a given value. A source electrode refers to part of a conductive layer that is connected to a source region. A source wiring refers to a wiring for electrically connecting at least one source electrode of a transistor to another electrode or another wiring.

Moreover, in this specification and the like, a drain refers to part or the whole of a drain region, a drain electrode, or a drain wiring. A drain region refers to a region in a semiconductor layer, where the resistivity is lower than or equal to a given value. A drain electrode refers to part of a conductive layer that is connected to a drain region. A drain wiring refers to a wiring for electrically connecting at least one drain electrode of a transistor to another electrode or another wiring.

Embodiment 1

A semiconductor device of one embodiment of the present invention is described with reference to drawings. FIG. 10A is a block diagram illustrating a structure of a wireless transceiver 900 that is a type of semiconductor device.

Note that the structure of the semiconductor device described in this specification and the like is just an example, and all of the components are not necessarily included. It is acceptable as long as the semiconductor device includes necessary components among the components described in this specification and the like. A component other than the components described in this specification and the like may be included.

The wireless transceiver 900 includes a low noise amplifier 901 (LNA), a band pass filter 902 (BPF), a mixer 903 (MIX), a band pass filter 904, a power amplifier 911 (PA), a band pass filter 912, a mixer 913, a band pass filter 914, a duplexer 921 (DUP), a local oscillator 922 (LO), and an antenna 931.

<Reception>

A signal 941 transmitted from another semiconductor device, a base station, or the like is input to the low noise amplifier 901 through the antenna 931 and the duplexer 921 as a received signal. The duplexer 921 has a function of transmitting and receiving a radio signal with one antenna.

The low noise amplifier 901 has a function of amplifying a weak received signal to a signal with strength that can be processed in the wireless transceiver 900. The signal 941 amplified by the low noise amplifier 901 is supplied to the mixer 903 through the band pass filter 902.

The band pass filter 902 has a function of attenuating frequency components other than frequency components in a necessary frequency band among frequency components contained in the signal 941 to allow the necessary frequency components in the necessary frequency band to pass through.

The mixer 903 has a function of mixing the signal 941 passed through the band pass filter 902 and a signal 943 generated in the local oscillator 922 by a superheterodyne system. The mixer 903 mixes the signal 941 and the signal 943, and supplies a signal that has a frequency component of the difference therebetween and a frequency component of the sum thereof, to the band pass filter 904. Note that instead of a superheterodyne system, a homodyne system may be used.

The band pass filter 904 has a function of passing one frequency of two frequency components. For example, a frequency component of the difference is passed. The band pass filter 904 also has a function of removing a noise component generated in the mixer 903. The signal passed through the band pass filter 904 is supplied to a demodulator 905 (DEM). The demodulator 905 has a function of converting the supplied signal into a control signal, a data signal, or the like and outputting the signal. The signal output from the demodulator 905 is supplied to a variety of processing devices (e.g., an arithmetic device or a memory device).

<Transmission>

A basic signal is supplied from a modulator 915 (MOD) to the band pass filter 914. The modulator 915 has a function of generating a basic signal for transmitting a control signal, a data signal, or the like from the wireless transceiver 900 to another semiconductor device, a base station, or the like. The basic signal is supplied to the mixer 913 through the band pass filter 914.

The band pass filter 914 has a function of removing a noise component contained in the basic signal.

The mixer 913 has a function of mixing the basic signal passed through the band pass filter 914 and a signal 944 generated in the local oscillator 922 by a superheterodyne system. The mixer 913 mixes the basic signal and the signal 944, and a signal having a frequency component of the difference therebetween and a frequency component of the sum thereof is supplied to the band pass filter 912.

The band pass filter 912 has a function of passing one frequency of two frequency components. For example, a frequency component of the sum is passed. The band pass filter 912 also has a function of removing a noise component generated in the mixer 913. The signal passed through the band pass filter 912 is supplied to the power amplifier 911.

The power amplifier 911 has a function of amplifying the supplied signal to generate a signal 942. The signal 942 is emitted from the antenna 931 to the outside through the duplexer 921.

[Demodulator]

FIG. 10B illustrates a structure example of the demodulator 905. The demodulator 905 illustrated in FIG. 10B includes a guard interval removing circuit 951 (GR), a fast Fourier transform circuit 952 (FFT), a segment separation circuit 953 (SS), an error correction circuit 954 (DEC), and a transport stream structure circuit 955 (TR).

[Modulator]

FIG. 10C illustrates a structure example of the modulator 915. The modulator 915 illustrated in FIG. 10C includes a transmodulation circuit 961 (TM), a data rearrangement circuit 962 (CODE), a signal synthesizing circuit 963 (SC), a fast inverse Fourier transform circuit 964 (IFFT), and a guard interval insertion circuit 965 (GI).

The demodulator 905 and the modulator 915 are included in a signal processing device 414 described later. The signal processing device 414 may include an analog to digital converter circuit (ADC) or a digital to analog converter circuit (DAC).

<Amplifier Circuit>

FIG. 1A illustrates a circuit diagram of a semiconductor device 100 that can be used in the low noise amplifier 901 and the power amplifier 911. The semiconductor device 100 includes a transistor 101, a transistor 102, a functional element 103, a transistor 112, and a capacitor 113.

One of a source and a drain of the transistor 101 is electrically connected to a terminal 121, and the other is electrically connected to one of a source and a drain of the transistor 102 through a node 104. A gate of the transistor 101 is electrically connected to a terminal IN. The other of the source and the drain of the transistor 102 is electrically connected to a terminal OUT. The other of the source and the drain of the transistor 102 is electrically connected to a terminal 122 through the functional element 103. One of a source and a drain of the transistor 112 is electrically connected to a terminal 123, and the other is electrically connected to a gate of the transistor 102 through a node 114. The other of the source and the drain of the transistor 112 is electrically connected to one electrode of the capacitor 113. A gate of the transistor 112 is electrically connected to a terminal 124. The other electrode of the capacitor 113 is electrically connected to a terminal 125. The functional element 103 may be a resistor or a constant current source.

The semiconductor device 100 functions as a cascode circuit. Thus, a mirror effect can be inhibited and excellent high-frequency characteristics and high gain can be achieved. Furthermore, the semiconductor device has characteristics of being less likely to be affected by fluctuation of the impedance of a circuit connected to a subsequent stage by a high isolation. In the semiconductor device 100, the voltage of the node 104 is less likely to change even when the gate voltage of the transistor 101 changes. Thus, the transistor 101 can be stably operated in a saturation region.

FIG. 1B is a conceptual diagram of an operation of the semiconductor device 100. In FIG. 1B, the vertical axis represents voltage and the horizontal axis represents time. The semiconductor device 100 has a function of amplifying a voltage V_{in} supplied to the terminal IN and outputting the voltage to the terminal OUT as a voltage V_{out} . Note that the voltage V_{in} is a signal in which a DC bias is added to an AC signal. The semiconductor device 100 has a function of amplifying an AC signal included in the voltage V_{in} and outputting the voltage as the voltage V_{out} .

VSS is supplied to the terminal 121 and VDD is supplied to the terminal 122. In the case where the semiconductor device 100 is used in the low noise amplifier 901, the signal 941 is supplied to the terminal IN of the semiconductor device 100. In the case where the semiconductor device 100 is used in the power amplifier 911, the signal passed through the band pass filter 912 is supplied to the terminal IN of the semiconductor device 100.

The signal 941 and the signal passed through the band pass filter 912 are both AC signals. For example, in the case where the signal 941 is amplified by the semiconductor

device 100, the signal 941 to which a DC bias is added is supplied to the terminal IN. The DC bias is a signal to keep the on state of the transistor 101.

The transistor 102 has a function of determining an amplification factor (also referred to as "gain") of the semiconductor device 100. As the resistance value of the functional element 103 is larger, the gain of the semiconductor device 100 is larger. The gain of the semiconductor device 100 is also changed by the transconductance (also referred to as " g_m ") of the transistor 102. The g_m of the transistor 102 can be adjusted with the gate voltage of the transistor 102. Accordingly, the gain of the semiconductor device 100 can be adjusted with the voltage of the node 114.

The transistor 112 has a function of writing a voltage to the node 114. Specifically, a voltage for bringing the transistor 112 into an on state is supplied to the terminal 124 to make conductivity between the terminal 123 and the node 114. Then, charge for setting the node 114 to a predetermined voltage is supplied from the terminal 123 to the node 114. After the writing ends, a voltage for bringing the transistor 112 into an off state is supplied to the terminal 124. The charge written to the node 114 can be retained when the transistor 112 is turned off.

For semiconductor layers of the transistor 101, the transistor 102, and the transistor 112, a single crystal semiconductor, a polycrystalline semiconductor, a microcrystalline semiconductor, an amorphous semiconductor, or the like can be used alone or in combination. As a semiconductor material, silicon, germanium, or the like can be used, for example. Alternatively, a compound semiconductor such as silicon germanium, silicon carbide, gallium arsenide, an oxide semiconductor, or a nitride semiconductor may be used.

Alternatively, gallium arsenide, aluminum gallium arsenide, indium gallium arsenide, gallium nitride, indium phosphide, silicon germanium, or the like that can be used for a high electron mobility transistor (HEMT) may be used.

Note that semiconductor layers may be stacked. In the case of stacking semiconductor layers, semiconductors having different crystal states may be used or different semiconductor materials may be used.

A transistor including an oxide semiconductor (OS), which is one kind of metal oxide, in a semiconductor layer where a channel is formed (such a transistor is also referred to as an "OS transistor" or "OS-FET") is preferably used as the transistor 112. An oxide semiconductor has a band gap of 2 eV or more and thus has an extremely low off-state current. When an OS transistor is used as the transistor 112, charge written to the node 114 can be retained for a long time.

When the transistor 112 is turned off, the node 114 is brought into an electrically floating state (also referred to as a "floating state"). When the transistor 112 is in the floating state, the voltage of the node 114 is likely to change following an ambient potential change. The capacitor 113 has a function of making the node 114 to be less likely to be affected by the ambient potential change.

Thus, it can be said that the semiconductor device 100 has a configuration in which a memory element 111 including the transistor 112 and the capacitor 113 is connected to the gate of the transistor 102. In particular, in the case where an OS transistor is used as the transistor 112, the memory element 111 can be referred to as an "OS memory".

An OS memory can retain written data in a period of one year or longer, or 10 years or longer even after power supply is stopped. Thus, the OS memory can be regarded as a nonvolatile memory.

11

An OS memory employs a method in which charge is written to the node through the OS transistor; hence, a high voltage, which a conventional flash memory requires, is unnecessary and a high-speed writing operation can be achieved. Furthermore, the number of data writing and reading in an OS memory is substantially unlimited because charge injection and extraction into/from a floating gate or a charge trap layer are not performed. An OS memory is less likely to degrade than a conventional flash memory and can have high reliability.

Unlike a magnetic memory, a resistive random access memory, or the like, an OS memory has no change in the structure at the atomic level. Thus, an OS memory has higher rewrite endurance than a magnetic memory and a resistive random access memory.

When the gate of the transistor **102** is provided with the memory element **111**, it is unnecessary to supply power continuously to the gate of the transistor **102**. Thus, the power consumption of the semiconductor device **100** can be reduced.

OS transistors may be used as the transistor **101** and the transistor **102**. The off-state current of an OS transistor hardly increases even in a high-temperature environment. Specifically, the off-state current hardly increases even at an environment temperature higher than or equal to room temperature and lower than or equal to 200° C. In addition, the on-state current of an OS transistor is less likely to decrease even in the high-temperature environment. Furthermore, the withstand voltage between a source and a drain of an OS transistor is high. A semiconductor device in which OS transistors are used as transistors achieves stable operation and high reliability even in a high-temperature environment.

Each of the transistor **101**, the transistor **102**, and the transistor **112** may be a double-gate transistor. FIG. **9A** illustrates a circuit symbol example of a double-gate transistor **180A**.

The transistor **180A** has a structure in which a transistor **Tr1** and a transistor **Tr2** are connected in series. FIG. **9A** illustrates a state in which one of a source and a drain of the transistor **Tr1** is electrically connected to a terminal S, the other of the source and the drain of the transistor **Tr1** is electrically connected to one of a source and a drain of the transistor **Tr2**, and the other of the source and the drain of the transistor **Tr2** is electrically connected to a terminal D. FIG. **9A** illustrates a state in which gates of the transistor **Tr1** and the transistor **Tr2** are electrically connected to each other and electrically connected to a terminal G.

The transistor **180A** illustrated in FIG. **9A** has a function of switching a conduction state and a non-conduction state between the terminal S and the terminal D by changing the potential of the terminal G. Thus, the transistor **180A** which is a double-gate transistor functions as one transistor including the transistor **Tr1** and the transistor **Tr2**. In other words, it can be said that in FIG. **9A**, one of a source and a drain of the transistor **180A** is electrically connected to the terminal S, the other of the source and the drain thereof is electrically connected to the terminal D, and a gate thereof is electrically connected to the terminal G.

Each of the transistor **101**, the transistor **102**, and the transistor **112** may be a triple-gate transistor. FIG. **9B** illustrates a circuit symbol example of a triple-gate transistor **180B**.

The transistor **180B** has a structure in which the transistor **Tr1**, the transistor **Tr2**, and a transistor **Tr3** are connected in series. FIG. **9B** illustrates a state where the one of the source and the drain of the transistor **Tr1** is electrically connected

12

to the terminal S, the other of the source and the drain of the transistor **Tr1** is electrically connected to the one of the source and the drain of the transistor **Tr2**, the other of the source and the drain of the transistor **Tr2** is electrically connected to one of a source and a drain of the transistor **Tr3**, and the other of the source and the drain of the transistor **Tr3** is electrically connected to the terminal D. FIG. **9B** illustrates a state in which gates of the transistor **Tr1**, the transistor **Tr2**, and the transistor **Tr3** are electrically connected to each other and electrically connected to the terminal G.

The transistor **180B** illustrated in FIG. **9B** has a function of switching a conduction state and a non-conduction state between the terminal S and the terminal D by changing the potential of the terminal G. Thus, the transistor **180B** which is a triple-gate transistor functions as one transistor including the transistor **Tr1**, the transistor **Tr2**, and the transistor **Tr3**. In other words, it can be said that in FIG. **9B**, one of a source and a drain of the transistor **180B** is electrically connected to the terminal S, the other of the source and the drain thereof is electrically connected to the terminal D, and a gate thereof is electrically connected to the terminal G.

Like the transistor **180A** and the transistor **180B**, a transistor including a plurality of gates electrically connected to each other is referred to as a “multi-gate type transistor” or a “multi-gate transistor” in some cases.

At least one of the transistor **101**, the transistor **102**, and the transistor **112** may include a back gate.

The back gate is placed so that a channel formation region of a semiconductor layer is sandwiched between the gate and the back gate. By changing the potential of the back gate, the threshold voltage of the transistor can be changed. The potential of the back gate may be the same as the potential of the gate or may be a GND potential or a given potential. The back gate can function in a manner similar to that of the gate. Thus, the gate and the back gate can be used interchangeably. For example, one of a gate and a back gate is referred to as a “first gate” and the other is referred to as a “second gate”, in some cases.

In general, the gate and the back gate are formed using conductive layers and thus each have a function of preventing an electric field generated outside the transistor from influencing the semiconductor layer in which a channel is formed (particularly, a function of preventing static electricity). That is, a change in the electrical characteristics of the transistor due to the influence of an external electric field such as static electricity can be prevented.

The threshold voltage of the transistor **102** may be smaller than the threshold voltage of the transistor **101**. The transistor **102** may be a normally-on transistor.

The ratio of the channel width *W* to the channel length *L* (also referred to as “*W/L* ratio”) of the transistor **102** is preferably larger than the *W/L* ratio of the transistor **101**. In the case where the channel length *L* of the transistor **102** and the channel length *L* of the transistor **101** are the same, the channel width *W* of the transistor **102** is preferably larger than the channel width *W* of the transistor **101**.

FIG. **2A** and FIG. **2B** illustrate circuit diagrams of the semiconductor device **100** in the case where the transistor **101**, the transistor **102**, and the transistor **112** each include a back gate. Although FIG. **2A** illustrates an example in which a gate and a back gate of each transistor are electrically connected, one embodiment of the present invention is not limited thereto. FIG. **2B** illustrates an example in which one of the source and the drain of the transistor **101** and its back gate are electrically connected. In addition, FIG. **2B**

13

illustrates an example in which one of the source and the drain of the transistor **102** and its back gate are electrically connected.

The semiconductor device **100** can be used for a variety of circuits. For example, the semiconductor device **100** may be used for a source-grounded amplifier circuit included in an operational amplifier.

Modification Example 1

FIG. **3A** illustrates a circuit diagram of a semiconductor device **100A**. The semiconductor device **100A** is a modification example of the semiconductor device **100** illustrated in FIG. **1A**. In order to reduce repeated description, differences of the semiconductor device **100A** from the semiconductor device **100** are mainly described.

As the functional element **103**, one kind or a combination of various kinds of a resistor, a coil, a capacitor, a diode, a transistor, and the like can be used. FIG. **4A** illustrates a circuit diagram in the case where a resistor is used as the functional element **103**. FIG. **4B** illustrates a circuit diagram in the case where a transistor is used as the functional element **103**. The transistor used as the functional element **103** may be a p-channel transistor or an n-channel transistor. A gate of the transistor used as the functional element **103** is electrically connected to a terminal **127**. A constant voltage is supplied to the gate of the transistor through the terminal **127**.

FIG. **4C** and FIG. **4D** illustrate circuit diagrams in the case where a diode-connected transistor is used as the functional element **103**. A gate and a drain of the transistor are electrically connected to each other, whereby the transistor can function as a diode. FIG. **4C** illustrates a configuration example in which a p-channel transistor is diode-connected. FIG. **4D** illustrates a configuration example in which an n-channel transistor is diode-connected.

An example where a parallel resonant circuit (an LC tank circuit) in which a coil **105** and a capacitor **106** are connected in parallel is used as the functional element **103** in the semiconductor device **100A** is illustrated. When the inductance of the coil **105** is L_t and the capacitance of the capacitor **106** is C_t , the resonance frequency f_0 is expressed by Formula 1.

[Formula 1]

$$f_0 = \frac{1}{2\pi\sqrt{L_t C_t}} \quad (1)$$

FIG. **3B** is a graph showing a relation between the frequency of a signal passing through the parallel resonant circuit and the impedance of the parallel resonant circuit. In FIG. **3B**, the horizontal axis represents the frequency and the vertical axis represents the impedance. The parallel resonant circuit has a maximum impedance when the resonance frequency is f_0 . That is, when the frequency of a signal input to the terminal IN is equal to the resonance frequency f_0 , the gain of the semiconductor device **100A** can be maximized. Accordingly, it is preferable that the resonance frequency f_0 of the parallel resonant circuit be in accordance with the frequency of the signal input to the terminal IN.

The semiconductor device **100A** includes a coil **107** between the gate of the transistor **101** and the terminal IN, and a coil **108** between one of the source and the drain of the transistor **101** and the terminal **121**.

14

In a semiconductor device processing an AC signal, the impedance on a signal transmitting side and the impedance on a signal receiving side need to be equal (matched). Converting the impedance into a constant value is referred to as "impedance conversion" or "impedance matching". Impedance matching is performed so that the impedance becomes 50Ω in many cases.

The input impedance of the semiconductor device **100A** can be adjusted by changing the inductance of the coil **107** and the coil **108**. Here, the inductance of the coil **107** is L_g and the inductance of the coil **108** is L_s . The capacitance value of a parasitic capacitance **109** generated between the source and the gate of the transistor **101** is C_g and the transconductance of the transistor **101** is g_{m1} . Then, an input impedance Z_{in} of the semiconductor device **100A** can be expressed by Formula 2.

[Formula 2]

$$Z_{in} = \frac{g_{m1} L_s}{C_g} + j \left(\omega L_g + \omega L_s - \frac{1}{\omega C_g} \right) \quad (2)$$

Changed are g_{m1} and C_g depending on the channel length and the channel width of the transistor **101**. For example, when the input impedance Z_{in} of the semiconductor device **100A** matches with 50Ω , L_g and L_s can be determined such that a real part is 50 and an imaginary part is 0.

Like the semiconductor device **100** illustrated in FIG. **2A** and FIG. **2B**, a transistor including a back gate may be used as the transistor **101**, the transistor **102**, and the transistor **112**.

Modification Example 2

FIG. **5** illustrates a circuit diagram of a semiconductor device **100B**. The semiconductor device **100B** is a modification example of the semiconductor device **100** illustrated in FIG. **1A**. In order to reduce repeated description, differences of the semiconductor device **100B** from the semiconductor device **100** are mainly described.

The semiconductor device **100B** includes an operational amplifier **131** between the gate of the transistor **102** and the memory element **111**. A non-inverting input of the operational amplifier **131** is electrically connected to the node **114** and an inverting input thereof is electrically connected to the node **104**. An output of the operational amplifier **131** is electrically connected to the gate of the transistor **102**.

The operational amplifier **131** changes an output voltage such that a voltage applied to the non-inverting input and a voltage applied to the inverting input are equal to each other. In the semiconductor device **100B**, the operational amplifier **131** operates such that the voltage of the node **104** is equal to the voltage of the node **114**. Thus, the voltage of the node **104** is less likely to change in the semiconductor device **100B** than in the semiconductor device **100**. The semiconductor device **100B** operates more stably than the semiconductor device **100**.

When the non-inverting input of the operational amplifier **131** is provided with the memory element **111**, it is unnecessary to supply power continuously to the non-inverting input of the operational amplifier **131**. Thus, the power consumption of the semiconductor device **100B** can be reduced.

Modification Example 3

FIG. **6A** illustrates a circuit diagram of a semiconductor device **100C**. The semiconductor device **100C** is a modification example of the semiconductor device **100**.

15

cation example of the semiconductor device **100B** illustrated in FIG. **5**. In order to reduce repeated description, differences of the semiconductor device **100C** from the semiconductor device **100B** are mainly described.

In the semiconductor device **100C**, the gate of the transistor **101** is electrically connected to the terminal **IN** and the back gate of the transistor **102**. The semiconductor device **100C** has a function of supplying an input signal supplied to the terminal **IN** to the gate of the transistor **101** and the back gate of the transistor **102**. The gate of the transistor **102** is electrically connected to a terminal **Bias**.

Here, a relation between a voltage (V_{bg}) supplied to a back gate of a transistor including a back gate and transistor characteristics is described with reference to FIG. **6B**. FIG. **6B** is a graph showing Id-Vg characteristics, one of electrical characteristics of a transistor. In FIG. **6B**, the horizontal axis represents a gate voltage (V_g) on a linear scale and the vertical axis represents a current (I_d) flowing between a source and a drain on a linear scale.

Id-Vg characteristics **200** shown in FIG. **6B** are Id-Vg characteristics when V_{bg} is 0 V. Id-Vg characteristics **201** are Id-Vg characteristics when V_{bg} is a positive voltage. Id-Vg characteristics **202** are Id-Vg characteristics when V_{bg} is a negative voltage.

In FIG. **6B**, the threshold voltage of the Id-Vg characteristics **200** is denoted by V_{th0} , the threshold voltage of the Id-Vg characteristics **201** is denoted by V_{th1} , and the threshold voltage of the Id-Vg characteristics **202** is denoted by V_{th2} .

Based on the Id-Vg characteristics **200** with V_{bg} of 0 V, the Id-Vg characteristics shift in the negative direction when V_{bg} is a positive voltage (the Id-Vg characteristics **201**). Thus, V_{th1} also shifts in the negative direction. The Id-Vg characteristics shift in the positive direction (the Id-Vg characteristics **202**) when V_{bg} is a negative voltage. Thus, V_{th1} also shifts in the positive direction. The shift amount of the threshold voltage is changed depending on V_{bg} . FIG. **6B** shows that the threshold voltage of the transistor is changed depending on V_{bg} .

Next, the operation of the semiconductor device **100C** is described. As a prerequisite, V_{SS} is supplied to the terminal **121** and V_{DD} is supplied to the terminal **122**. Thus, in the transistor **101**, the source or the drain on the terminal **121** side functions as a source, and the source or the drain on the node **104** side functions as a drain. In the transistor **102**, the source or the drain on the node **104** side functions as a source, and the source or the drain on the functional element **103** side functions as a drain. Furthermore, a voltage V_{bias} that is a fixed voltage is supplied to the terminal **Bias**. Note that the voltage V_{bias} is a voltage for bringing the transistor **102** into an on state.

In FIG. **6A**, the gate voltage (a voltage between a gate and a source) of the transistor **101** is V_{g1} and the gate voltage of the transistor **102** is V_{g2} . V_{g1} is a potential difference between the terminal **IN** and the terminal **121** when the terminal **121** is regarded as a reference. V_{g2} is a potential difference between the terminal **Bias** and the node **104** when the node **104** is regarded as a reference.

Note that since the semiconductor device **100C** is an amplifier circuit, it is preferable that the transistor **101** and the transistor **102** operate in a saturation region.

When the voltage V_{in} increases, the I_d of the transistor **101** increases. The I_d of the transistor **101** and a current I_o flowing between the terminal **122** and the terminal **121** are equal to each other. That is, the current I_o increases in accordance with a rise of the voltage V_{in} .

16

In accordance with the increase in the current I_o , the transistor **102** operates to increase the I_d thereof. That is, the transistor **102** operates to increase V_{g2} . At this time, the voltage V_{bias} is fixed and thus the potential of the node **104** that is on the source side decreases. This means that as the voltage V_{in} increases, the potential of the node **104** decreases and thus the operation region of the transistor **101** gets closer to a linear region.

In the semiconductor device **100C** of one embodiment of the present invention, by supplying the voltage V_{in} to the back gate of the transistor **102** and the gate of the transistor **101** simultaneously, whereby the threshold voltage of the transistor **102** is changed actively and a decrease in the potential of the node **104** can be reduced.

Specifically, in the semiconductor device **100C**, the back gate voltage of the transistor **102** rises (shifts in the positive direction) in accordance with a rise of the voltage V_{in} . As described above, when the back gate voltage shifts in the positive direction, the threshold voltage of the transistor **102** shifts in the negative direction (see FIG. **6B**). The shift amount of the threshold voltage is changed depending on the back gate voltage, whereby as the voltage V_{in} rises, the shift amount of the threshold voltage increases.

In the case where the transistor **102** operates in a saturation region, the I_d of the transistor **102** is changed in proportion to the square of the voltage obtained by subtracting the threshold voltage from V_{g2} . When the threshold voltage shifts in the negative direction (the threshold voltage decreases), the amount of change in V_{g2} can be small. Thus, a decrease in the potential of the node **104** can be suppressed.

Note that the semiconductor device **100B** described above has a function of inhibiting a change in the potential of the node **104** by using the operational amplifier **131**. Note that the semiconductor device **100B** using the operational amplifier **131** tends to have a large occupation area. The semiconductor device **100C** described in this embodiment can inhibit the change in the potential of the node **104** without using the operational amplifier **131**. In addition, the semiconductor device **100C** described in this embodiment can operate the transistor **101** in a saturation region regardless of the input signal. According to one embodiment of the present invention, a semiconductor device can operate stably and have higher reliability.

FIG. **7A** and FIG. **7B** illustrate circuit diagrams of the semiconductor device **100C** in the case where the transistor **101** includes a back gate. FIG. **7A** illustrates an example in which the gate and the back gate of the transistor **101** are electrically connected. FIG. **7B** illustrates an example in which one of the source and the drain of the transistor **101** and its back gate are electrically connected. As illustrated in FIG. **7C**, the gate of the transistor **102** may be provided with the memory element **111** like the semiconductor device **100**.

As illustrated in FIG. **8A**, the gate and the back gate of the transistor **102** may be interchanged with each other. In the semiconductor device **100C** illustrated in FIG. **8A**, the terminal **IN** and the gate of the transistor **102** are electrically connected and the terminal **Bias** and the back gate of the transistor **102** are electrically connected. As in the semiconductor device **100C** illustrated in FIG. **8B**, p-channel transistors may be used as the transistor **101** and the transistor **102**. In this case, the transistor **101** is provided on the terminal **122** side. The functional element **103** is electrically connected to the terminal **121**.

As in the semiconductor device **100C** illustrated in FIG. **8C**, the operational amplifier **131** may be provided between the gate of the transistor **102** and the terminal **Bias**. The

17

non-inverting input of the operational amplifier **131** is electrically connected to the terminal Bias and the inverting input thereof is electrically connected to the node **104**. The output of the operational amplifier **131** is electrically connected to the gate of the transistor **102**.

The composition, structure, method, and the like described in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

Embodiment 2

In this embodiment, a wireless transceiver **900A**, which is a modification example of the wireless transceiver **900** described in the above embodiment, is described with reference to FIG. **11A** and FIG. **11B**. In order to reduce repeated description, differences of the wireless transceiver **900A** from the wireless transceiver **900** are mainly described.

The wireless transceiver **900A** includes a plurality of antennas **931** to correspond to a communication standard of 5G. In addition, the wireless transceiver **900A** includes a plurality of duplexers **921**, a plurality of low noise amplifiers **901**, and a plurality of power amplifiers **911**. Furthermore, the wireless transceiver **900A** includes a decoder circuit **906** (DEC) and a decoder circuit **916**.

FIG. **11A** illustrates a case where the wireless transceiver **900A** includes five antennas **931**, five duplexers **921**, five low noise amplifiers **901**, and five power amplifiers **911**. In FIG. **11A**, the first antenna **931** is denoted by an antenna **931[1]** and the fifth antenna **931** is denoted by an antenna **931[5]**. The duplexers **921**, the low noise amplifiers **901**, and the power amplifiers **911** are also denoted in a manner similar to the antennas **931**. Note that the number of each of the antennas **931**, the duplexers **921**, the low noise amplifiers **901**, and the power amplifiers **911** is not limited to five.

The antenna **931[1]** is electrically connected to a duplexer **921[1]**. The duplexer **921[1]** is electrically connected to a low noise amplifier **901[1]** and a power amplifier **911[1]**. The antenna **931[5]** is electrically connected to a duplexer **921[5]**. The duplexer **921[5]** is electrically connected to a low noise amplifier **901[5]** and a power amplifier **911[5]**. Like the antenna **931[1]**, the second to the fourth antennas **931** are also electrically connected to the second to the fourth duplexers **921**, respectively. In addition, like duplexer **921[1]**, the second to the fourth duplexers **921** are also electrically connected to the second to the fourth low noise amplifiers **901** and the second to the fourth power amplifiers **911**, respectively.

The decoder circuit **906** is electrically connected to the plurality of low noise amplifiers **901**. In FIG. **11A**, five low noise amplifiers **901** are connected to the decoder circuit **906**. In addition, the decoder circuit **916** is electrically connected to the plurality of power amplifiers **911**. In FIG. **11A**, five power amplifiers **911** are connected to the decoder circuit **916**.

The decoder circuit **906** has a function of selecting any one or a plurality of the low noise amplifier **901[1]** to the low noise amplifier **901[5]**. Furthermore, the decoder circuit **906** has a function of sequentially selecting the low noise amplifier **901[1]** to the low noise amplifier **901[5]**. Similarly, the decoder circuit **916** has a function of selecting any one or a plurality of the power amplifier **911[1]** to the power amplifier **911[5]**. Furthermore, the decoder circuit **916** has a function of sequentially selecting the power amplifier **911[1]** to the power amplifier **911[5]**.

18

As an example, a connection example of the decoder circuit **906** and the low noise amplifier **901[1]** and a low noise amplifier **901[2]** is illustrated in FIG. **11B**. The decoder circuit **906** is electrically connected to the memory element **111** (denoted by a memory element **111[1]**) included in the low noise amplifier **901[1]** through the terminal **124** electrically connected to the memory element **111[1]**. The decoder circuit **906** is electrically connected to the memory element **111** (denoted by a memory element **111[2]**) included in the low noise amplifier **901[2]** through the terminal **124** electrically connected to the memory element **111[2]**.

The terminal **123** electrically connected to the memory element **111[1]** and the terminal **123** electrically connected to the memory element **111[2]** are electrically connected to a wiring **126**. A voltage (charge) written to the node **114** is supplied through the wiring **126**.

The decoder circuit **906** has a function of supplying a signal to turn on or off the transistor **112** to the terminal **124** electrically connected to a given memory element **111**. The decoder circuit **906** sequentially selects the memory element **111** in each of the low noise amplifiers **901**, whereby a voltage written to the node **114** can be vary among the memory elements **111**. In other words, a voltage suitable for each of the plurality of low noise amplifiers **901** can be written to the node **114**.

The decoder circuit **916** also functions in a manner similar to the decoder circuit **906** with respect to the plurality of power amplifiers **911**.

The composition, structure, method, and the like described in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

Embodiment 3

The wireless transceiver **900** of one embodiment of the present invention may be stacked over a digital circuit. FIG. **12A** illustrates a perspective view of a semiconductor device **400**. The semiconductor device **400** includes a layer **410**, a layer **420**, and a layer **430**. FIG. **12B** is a perspective view for illustrating a structure of the semiconductor device **400**, and the layer **410**, the layer **420**, and the layer **430** are separately illustrated.

The layer **410** includes a digital circuit. For example, the layer **410** includes a control device **411**, a memory device **412**, an input/output device **413**, the signal processing device **414**, and the like. The control device **411** has a function of controlling the operation of the whole semiconductor device **400**.

[Control Device **411**]

As the control device **411**, microprocessors such as a central processing unit (CPU), a DSP (Digital Signal Processor), and a GPU (Graphics Processing Unit) can be used alone or in combination. A structure may be employed in which such a microprocessor is obtained with a PLD (Programmable Logic Device) such as an FPGA (Field Programmable Gate Array) or an FPAA (Field Programmable Analog Array).

[Memory Device **412**]

As the memory device **412**, a memory device using a nonvolatile memory element, such as a flash memory, an MRAM (Magnetoresistive Random Access Memory), a PRAM (Phase change RAM), an ReRAM (Resistive RAM), or an FeRAM (Ferroelectric RAM); a memory device using

a volatile memory element, such as a DRAM (Dynamic RAM) or an SRAM (Static RAM); or the like may be used, for example.

Note that the memory device **412** is not incorporated in the semiconductor device **400**, and a memory device located outside the semiconductor device **400** may be used as the memory device **412**. In that case, the memory device **412** is connected through the input/output device **413**.

[Input/Output Device **413**]

The input/output device **413** is electrically connected to an external port or the like, for example, and has a function of transmitting and receiving a signal to/from the outside. The semiconductor device **400** can transmit and receive a signal to/from another semiconductor device through the input/output device **413**. The input/output device **413** may be electrically connected to an input component such as a button or a switch. Examples of the external port to which the input/output device **413** is electrically connected include a USB terminal or a LAN (Local Area Network) connection terminal.

[Signal Processing Device **414**]

The signal processing device **414** has a function of processing a signal received by a transmission/reception device **421**, and supplying the processed signal to the control device **411**, the memory device **412**, and the input/output device **413**. For example, the signal processing device **414** has a function of returning a signal transmitted in division with spatial multiplex transmission such as MIMO (multiple-input and multiple-output) to the original signal (a demodulation function). In addition, when data is transmitted from the semiconductor device **400** to the outside with spatial multiplex transmission, the signal processing device **414** has a function of converting the data into a transmission signal for spatial multiplex transmission.

Note that MIMO is a technique for dividing a signal to be transmitted into a plurality of communication paths (also referred to as “streams” or “space streams”) and transmitting them at the same time. One stream includes one transmission antenna and one receiving antenna. Thus, the maximum number of streams is the same as the smaller number of the transmission antennas and the receiving antennas. When the number of streams is 10, an apparent transfer speed can be decupled.

The layer **410** can be formed using a single crystal semiconductor, a polycrystalline semiconductor, a microcrystalline semiconductor, an amorphous semiconductor, or the like alone or in combination. As a semiconductor material, silicon, germanium, or the like can be used, for example. Alternatively, a compound semiconductor such as silicon germanium, silicon carbide, gallium arsenide, an oxide semiconductor, a nitride semiconductor, or the like may be used.

Alternatively, gallium arsenide, aluminum gallium arsenide, indium gallium arsenide, gallium nitride, indium phosphide, silicon germanium, or the like that can be used for a HEMT may be used.

The layer **420** includes a high frequency circuit and the like. For example, the layer **420** includes the transmission/reception device **421**. As the transmission/reception device **421**, the semiconductor device **100** described in the other embodiment can be used, for example. The transmission/reception device **421** may include a plurality of semiconductor devices **100** or the like.

The layer **420** can be provided using a semiconductor material that can be formed thin such as an oxide semiconductor or a silicon. Using a technique of forming a thin film can provide the digital circuit in the layer **410** and the high

frequency circuit in the layer **420** three-dimensionally. Thus, the area occupied by the semiconductor device **400** can be reduced.

The mobility of an oxide semiconductor is less likely to decrease compared with that of a silicon or the like in a high-temperature environment. When an OS transistor is used as a transistor included in the layer **420**, the circuit included in the layer **420** can be stably operated even when the temperature in the layer **410** increases. Consequently, the reliability of the semiconductor device can be improved.

The layer **420** may be formed over another substrate and attached to the layer **410**.

The layer **430** includes an antenna array **431**. The antenna array **431** includes a plurality of antennas **432**. The antennas **432** correspond to the antennas **931** described in the other embodiments, for example. In the semiconductor device **400** illustrated in FIG. 12B, the antennas **432** are arranged in a matrix of four rows and four columns.

When the antenna array **431** is provided in the semiconductor device **400**, a communication technology such as beamforming or spatial multiplex transmission in the semiconductor device **400** can be achieved. Note that beamforming is a communication technology for transmitting a radio wave using a plurality of antennas. Strength of directivity or a transmission direction can be adjusted by adjusting phase inversion of a radio wave to be transmitted from each antenna. Directivity of a transmission radio wave is enhanced, whereby the radio wave can be transmitted farther. A radio wave can be transmitted to a specific region by adjusting the transmission direction.

Modification Example 1

FIG. 13A illustrates a perspective view of a semiconductor device **400A**. The semiconductor device **400A** is a modification example of the semiconductor device **400**. Accordingly, in order to avoid repeated description, differences of the semiconductor device **400A** from the semiconductor device **400** are mainly described. The semiconductor device **400A** includes a layer **410A** and a layer **420A** instead of the layer **410** and the layer **420** in the semiconductor device **400**.

FIG. 13B is a perspective view for illustrating a structure of the semiconductor device **400A**. The wireless transceiver **900** of one embodiment of the present invention may be provided in the layer **410A** together with a digital circuit or the like. The memory device **412** may be provided in the layer **420A**.

The layer **410A** includes the control device **411**, the input/output device **413**, the signal processing device **414**, the transmission/reception device **421**, and the like. As the transmission/reception device **421**, the semiconductor device **100** described in the other embodiment can be used, for example. The transmission/reception device **421** may include a plurality of semiconductor devices **100** or the like.

The layer **420A** includes the memory device **412**. The memory device **412** provided in the layer **420A** preferably uses an OS memory as a memory element. As described above, the off-state current of an OS transistor hardly increases even in a high-temperature environment. The OS memory is less likely to be affected by heat generation of the transmission/reception device **421**, the signal processing device **414**, and the like included in the layer **410A**, and written data can be retained for a long time. Thus, a semiconductor device can achieve stable operation and high reliability even in a high-temperature environment.

21

A circuit included in the layer **410A** can be formed using a MOS-FET (Metal Oxide Semiconductor-Field Effect Transistor), an LDMOS-FET (Laterally Diffused Metal Oxide Semiconductor-Field Effect Transistor), a bipolar transistor, or the like.

FIG. **14** illustrates a cross-sectional structure example of a MOS-FET and an LDMOS-FET provided in the layer **410A**. A transistor **451** is an n-channel MOS-FET and a transistor **452** is a p-channel MOS-FET. A CMOS (Complementary MOS)-FET can be formed using an n-channel MOS-FET and a p-channel MOS-FET. A transistor **453** is an n-channel LDMOS-FET.

FIG. **14** illustrates an example in which a silicon substrate provided with a buried oxide film **481** (BOX: Buried Oxide) is used in the layer **410A**. The layer **410A** illustrated in FIG. **14** is formed using an SOI (Silicon On Insulator) substrate using an n-type semiconductor substrate.

The transistor **451**, the transistor **452**, and the transistor **453** are electrically separated from each other by the buried oxide film **481**, an element isolation region **483**, and an insulating layer **482**. The element isolation region **483** may be formed by a LOCOS (Local Oxidation of Silicon) method, for example. The insulating layer **482** may be formed by a DTI (Deep Trench Isolation) method.

The transistor **451** includes a high concentration n-type impurity region **461a**, a high concentration n-type impurity region **461b**, an insulating layer **462**, and an electrode **463**. A channel formation region of the transistor **451** is formed in part of a p-well **464**. The transistor **452** includes a high concentration p-type impurity region **465a**, a high concentration p-type impurity region **465b**, an insulating layer **466**, and an electrode **467**. A channel formation region of the transistor **452** is formed in part of an n-well **468**.

The transistor **453** includes a high concentration n-type impurity region **471a**, a high concentration n-type impurity region **471b**, an insulating layer **472**, and an electrode **473**. The insulating layer **472** and the electrode **473** each include a region overlapping with part of the element isolation region **483**. A channel formation region of the transistor **453** is formed in part of a p-well **474** and in part of the n-well **468**. A high concentration p-type impurity region **478** is provided in adjacent to the high concentration n-type impurity region **471a**.

An LDMOS-FET has a structure in which an avalanche breakdown is less likely to occur even when a high voltage is applied. Thus, an LDMOS-FET can be favorably used for the power amplifier **911** in the wireless transceiver **900**, for example.

An insulating layer, a conductive layer, or the like may be provided above the transistor **451**, the transistor **452**, and the transistor **453**.

Modification Example 2

FIG. **15A** illustrates a perspective view of a semiconductor device **400B**. The semiconductor device **400B** is a modification example of the semiconductor device **400A**. Accordingly, in order to avoid repeated description, differences of the semiconductor device **400B** from the semiconductor device **400A** are mainly described. The semiconductor device **400B** includes a layer **410B** instead of the layer **410A** in the semiconductor device **400A**.

FIG. **15B** is a perspective view for illustrating a structure of the semiconductor device **400B**. The layer **410B** includes a plurality of layers **410A**. For example, the layer **410B** includes the layers **410A** arranged in a matrix of four rows and four columns. In FIG. **15A** and/or FIG. **15B**, the layer

22

410A placed in a first row and a first column is denoted by a layer **410A**[**1,1**], the layer **410A** placed in a fourth row and the first column is denoted by a layer **410A**[**4,1**], the layer **410A** placed in the first row and a fourth column is denoted by a layer **410A**[**1,4**], and the layer **410A** placed in the fourth row and the fourth column is denoted by a layer **410A**[**4,4**].

In FIG. **15B**, the antenna **432** placed in the first row and the first column is denoted by an antenna **432**[**1,1**], the antenna **432** placed in the fourth row and the first column is denoted by an antenna **432**[**4,1**], and the antenna **432** placed in the first row and the fourth column is denoted by an antenna **432**[**1,4**].

For example, the antenna **432**[**1,1**] is electrically connected to the layer **410A**[**1,1**] and the antenna **432**[**4,1**] is electrically connected to the layer **410A**[**4,1**]. One antenna **432** and one layer **410A** are electrically connected to each other, whereby the processing speed of a received signal can be increased.

Modification Example 3

FIG. **16A** illustrates a perspective view of a semiconductor device **400C**. The semiconductor device **400C** is a modification example of the semiconductor device **400A**. Accordingly, in order to avoid repeated description, differences of the semiconductor device **400C** from the semiconductor device **400A** are mainly described. In the semiconductor device **400C**, the layer **430** includes a layer **430a** and a layer **430b**. The layer **430a** and the layer **430b** can be provided to overlap with each other.

FIG. **16B** is a perspective view illustrating a structure of the semiconductor device **400C**. The layer **430a** includes the antenna array **431**. The antenna array **431** includes a plurality of antennas **432**. In the semiconductor device **400C**, the antennas **432** are arranged in a matrix of four rows and four columns.

The layer **430b** includes an antenna array **441**. The antenna array **441** includes a plurality of antennas **442**. In the semiconductor device **400C**, the antennas **442** are arranged in a matrix of two rows and two columns.

The pitch of the antennas **432** arranged in a matrix is preferably different from the pitch of the antennas **442** arranged in a matrix. In addition, the size of the antennas **432** is preferably different from the size of the antennas **442**. In the case where the layer **430a** is provided over the layer **430b**, the antennas **432** are preferably smaller than the antennas **442**.

When antennas whose sizes are different from each other are stacked at different pitches, an increase in the area occupied by the antennas can be inhibited and transmission and/or reception of radio waves having different frequencies can be achieved. The antennas **432** and antennas **442** correspond to the antennas **931** described in the other embodiments, for example.

The composition, structure, method, and the like described above in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments and the example.

Embodiment 4

In this embodiment, structures of transistors that can be used in the semiconductor device described in the above embodiments are described. As an example, a structure in which transistors having different electrical characteristics are stacked is described. With the structure, the degree of

freedom in design of the semiconductor device can be increased. Stacking transistors having different electrical characteristics can increase the degree of integration of the semiconductor device.

FIG. 17 illustrates part of a cross-sectional structure of a semiconductor device. A semiconductor device illustrated in FIG. 17 includes a transistor 550, a transistor 500, and a capacitor 600. FIG. 19A is a cross-sectional view of the transistor 500 in the channel length direction, FIG. 19B is a cross-sectional view of the transistor 500 in the channel width direction, and FIG. 19C is a cross-sectional view of the transistor 550 in the channel width direction. For example, the transistor 500 and the transistor 550 correspond to the transistor 112 and the transistor 102, respectively, described in the above embodiments. The capacitor 600 corresponds to the capacitor 113.

The transistor 500 is an OS transistor. The transistor 500 has an extremely low off-state current. Accordingly, data voltage or charge written to a storage node through the transistor 500 can be retained for a long time. In other words, power consumption of the semiconductor device can be reduced because a storage node has a low frequency of refresh operation or requires no refresh operation.

In FIG. 17, the transistor 500 is provided above the transistor 550, and the capacitor 600 is provided above the transistor 550 and the transistor 500.

The transistor 550 is provided over a substrate 311 and includes a conductor 316, an insulator 315, a semiconductor region 313 that is part of the substrate 311, and a low-resistance region 314a and a low-resistance region 314b each functioning as a source region or a drain region.

As illustrated in FIG. 19C, in the transistor 550, a top surface and a side surface in the channel width direction of the semiconductor region 313 are covered with the conductor 316 with the insulator 315 therebetween. Such a Fin-type transistor 550 can have an increased effective channel width, and thus have improved on-state characteristics. In addition, since contribution of an electric field of a gate electrode can be increased, the off-state characteristics of the transistor 550 can be improved.

Note that the transistor 550 can be either a p-channel transistor or an n-channel transistor.

A region of the semiconductor region 313 where a channel is formed, a region in the vicinity thereof, the low-resistance region 314a and the low-resistance region 314b each functioning as a source region or a drain region, and the like preferably contain a semiconductor such as a silicon-based semiconductor, and preferably contain single crystal silicon. Alternatively, the regions may be formed using a material containing Ge (germanium), SiGe (silicon germanium), GaAs (gallium arsenide), GaAlAs (gallium aluminum arsenide), or the like. A structure may be employed in which silicon whose effective mass is controlled by applying stress to the crystal lattice and changing the lattice spacing is used. Alternatively, the transistor 550 may be a HEMT with the use of GaAs and GaAlAs, or the like.

The low-resistance region 314a and the low-resistance region 314b contain an element which imparts n-type conductivity, such as arsenic or phosphorus, or an element which imparts p-type conductivity, such as boron, in addition to the semiconductor material used for the semiconductor region 313.

For the conductor 316 functioning as a gate electrode, a semiconductor material such as silicon containing the element which imparts n-type conductivity, such as arsenic or phosphorus, or the element which imparts p-type conduc-

tivity, such as boron, or a conductive material such as a metal material, an alloy material, or a metal oxide material can be used.

Note that since the work function of a conductor depends on the material of the conductor, the threshold voltage of the transistor can be adjusted by selecting the material of the conductor. Specifically, it is preferable to use a material such as titanium nitride or tantalum nitride for the conductor. Moreover, in order to ensure both conductivity and embeddability, it is preferable to use stacked layers of metal materials such as tungsten and aluminum for the conductor, and it is particularly preferable to use tungsten in terms of heat resistance.

The transistor 550 may be formed using an SOI (silicon on insulator) substrate, for example.

As the SOI substrate, the following substrate may be used: an SIMOX (Separation by Implanted Oxygen) substrate which is formed in such a manner that after an oxygen ion is implanted into a mirror-polished wafer, an oxide layer is formed at a certain depth from the surface and defects generated in a surface layer are eliminated by high-temperature annealing, or an SOI substrate formed by using a Smart-Cut method in which a semiconductor substrate is cleaved by utilizing growth of a minute void, which is formed by implantation of a hydrogen ion, by thermal treatment; an ELTRAN method (a registered trademark: Epitaxial Layer Transfer); or the like. A transistor formed using a single crystal substrate contains a single crystal semiconductor in a channel formation region.

Note that the transistor 550 illustrated in FIG. 17 is an example and the structure is not limited thereto; an appropriate transistor can be used in accordance with a circuit configuration or a driving method. For example, when the semiconductor device is a single-polarity circuit using only OS transistors (which represent transistors having the same polarity, e.g., only n-channel transistors), the transistor 550 has a structure similar to that of the transistor 500 as illustrated in FIG. 18. Note that the details of the transistor 500 are described later.

An insulator 320, an insulator 322, an insulator 324, and an insulator 326 are stacked sequentially to cover the transistor 550.

For the insulator 320, the insulator 322, the insulator 324, and the insulator 326, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, aluminum nitride, or the like can be used, for example.

Note that in this specification, silicon oxynitride refers to a material that contains oxygen at a higher proportion than nitrogen, and silicon nitride oxide refers to a material that contains nitrogen at a higher proportion than oxygen. Furthermore, in this specification, aluminum oxynitride refers to a material that contains oxygen at a higher proportion than nitrogen, and aluminum nitride oxide refers to a material that contains nitrogen at a higher proportion than oxygen.

The insulator 322 may have a function of a planarization film for eliminating a level difference caused by the transistor 550 or the like provided below the insulator 322. For example, a top surface of the insulator 322 may be planarized by planarization treatment using a chemical mechanical polishing (CMP) method or the like to increase planarity.

In addition, for the insulator 324, it is preferable to use a film having a barrier property that prevents diffusion of hydrogen or impurities from the substrate 311, the transistor 550, or the like into a region where the transistor 500 is provided.

For the film having a barrier property against hydrogen, silicon nitride formed using a CVD method can be used, for example. Here, diffusion of hydrogen into a semiconductor element including an oxide semiconductor, such as the transistor 500, degrades the characteristics of the semiconductor element in some cases. Therefore, a film that inhibits hydrogen diffusion is preferably provided between the transistor 500 and the transistor 550. The film that inhibits hydrogen diffusion is specifically a film from which a small amount of hydrogen is released.

The amount of released hydrogen can be analyzed by thermal desorption spectroscopy (TDS) or the like, for example. The amount of hydrogen released from the insulator 324 that is converted into hydrogen atoms per area of the insulator 324 is less than or equal to 10×10^{15} atoms/cm², preferably less than or equal to 5×10^{15} atoms/cm², in the TDS analysis in a film-surface temperature range of 50° C. to 500° C., for example.

Note that the permittivity of the insulator 326 is preferably lower than that of the insulator 324. For example, the relative permittivity of the insulator 326 is preferably lower than 4, further preferably lower than 3. The relative permittivity of the insulator 326 is, for example, preferably 0.7 times or less, further preferably 0.6 times or less the relative permittivity of the insulator 324. When a material with a low permittivity is used for an interlayer film, parasitic capacitance generated between wirings can be reduced.

In addition, a conductor 328, a conductor 330, and the like that are connected to the capacitor 600 or the transistor 500 are embedded in the insulator 320, the insulator 322, the insulator 324, and the insulator 326. Note that the conductor 328 and the conductor 330 each have a function of a plug or a wiring. Furthermore, a plurality of conductors functioning as plugs or wirings are collectively denoted by the same reference numeral in some cases. Moreover, in this specification and the like, a wiring and a plug connected to the wiring may be a single component. That is, there are cases where part of a conductor functions as a wiring and part of a conductor functions as a plug.

As a material for each of the plugs and wirings (the conductor 328, the conductor 330, and the like), a single layer or a stacked layer of a conductive material such as a metal material, an alloy material, a metal nitride material, or a metal oxide material can be used. It is preferable to use a high-melting-point material that has both heat resistance and conductivity, such as tungsten or molybdenum, and it is preferable to use tungsten. Alternatively, it is preferable to form the plugs and wirings with a low-resistance conductive material such as aluminum or copper. The use of a low-resistance conductive material can reduce wiring resistance.

A wiring layer may be provided over the insulator 326 and the conductor 330. For example, in FIG. 17, an insulator 350, an insulator 352, and an insulator 354 are stacked sequentially. Furthermore, a conductor 356 is formed in the insulator 350, the insulator 352, and the insulator 354. The conductor 356 has a function of a plug or a wiring that is connected to the transistor 550. Note that the conductor 356 can be provided using a material similar to those for the conductor 328 and the conductor 330.

For example, like the insulator 324, the insulator 350 is preferably formed using an insulator having a barrier property against hydrogen. Furthermore, the conductor 356 preferably contains a conductor having a barrier property against hydrogen. In particular, the conductor having a barrier property against hydrogen is formed in an opening portion of the insulator 350 having a barrier property against hydrogen. With this structure, the transistor 550 and the transistor

500 can be separated by a barrier layer, so that diffusion of hydrogen from the transistor 550 into the transistor 500 can be inhibited.

Note that for the conductor having a barrier property against hydrogen, tantalum nitride is preferably used, for example. In addition, by stacking tantalum nitride and tungsten, which has high conductivity, the diffusion of hydrogen from the transistor 550 can be inhibited while the conductivity as a wiring is kept. In that case, a structure in which a tantalum nitride layer having a barrier property against hydrogen is in contact with the insulator 350 having a barrier property against hydrogen is preferable.

A wiring layer may be provided over the insulator 354 and the conductor 356. For example, in FIG. 17, an insulator 360, an insulator 362, and an insulator 364 are stacked sequentially. Furthermore, a conductor 366 is formed in the insulator 360, the insulator 362, and the insulator 364. The conductor 366 has a function of a plug or a wiring. Note that the conductor 366 can be provided using a material similar to those for the conductor 328 and the conductor 330.

For example, like the insulator 324, the insulator 360 is preferably formed using an insulator having a barrier property against hydrogen. Furthermore, the conductor 366 preferably contains a conductor having a barrier property against hydrogen. In particular, the conductor having a barrier property against hydrogen is formed in an opening portion of the insulator 360 having a barrier property against hydrogen. With this structure, the transistor 550 and the transistor 500 can be separated by a barrier layer, so that diffusion of hydrogen from the transistor 550 into the transistor 500 can be inhibited.

A wiring layer may be provided over the insulator 364 and the conductor 366. For example, in FIG. 17, an insulator 370, an insulator 372, and an insulator 374 are stacked sequentially. Furthermore, a conductor 376 is formed in the insulator 370, the insulator 372, and the insulator 374. The conductor 376 has a function of a plug or a wiring. Note that the conductor 376 can be provided using a material similar to those for the conductor 328 and the conductor 330.

For example, like the insulator 324, the insulator 370 is preferably formed using an insulator having a barrier property against hydrogen. Furthermore, the conductor 376 preferably contains a conductor having a barrier property against hydrogen. In particular, the conductor having a barrier property against hydrogen is formed in an opening portion of the insulator 370 having a barrier property against hydrogen. With this structure, the transistor 550 and the transistor 500 can be separated by a barrier layer, so that diffusion of hydrogen from the transistor 550 into the transistor 500 can be inhibited.

A wiring layer may be provided over the insulator 374 and the conductor 376. For example, in FIG. 17, an insulator 380, an insulator 382, and an insulator 384 are stacked sequentially. Furthermore, a conductor 386 is formed in the insulator 380, the insulator 382, and the insulator 384. The conductor 386 has a function of a plug or a wiring. Note that the conductor 386 can be provided using a material similar to those for the conductor 328 and the conductor 330.

For example, like the insulator 324, the insulator 380 is preferably formed using an insulator having a barrier property against hydrogen. Furthermore, the conductor 386 preferably contains a conductor having a barrier property against hydrogen. In particular, the conductor having a barrier property against hydrogen is formed in an opening portion of the insulator 380 having a barrier property against hydrogen. With this structure, the transistor 550 and the transistor

500 can be separated by a barrier layer, so that diffusion of hydrogen from the transistor **550** into the transistor **500** can be inhibited.

Although the wiring layer including the conductor **356**, the wiring layer including the conductor **366**, the wiring layer including the conductor **376**, and the wiring layer including the conductor **386** are described above, the semiconductor device of this embodiment is not limited thereto. Three or less wiring layers that are similar to the wiring layer including the conductor **356** may be provided, or five or more wiring layers that are similar to the wiring layer including the conductor **356** may be provided.

An insulator **510**, an insulator **512**, an insulator **514**, and an insulator **516** are stacked sequentially over the insulator **384**. A substance having a barrier property against oxygen or hydrogen is preferably used for any of the insulator **510**, the insulator **512**, the insulator **514**, and the insulator **516**.

For example, for the insulator **510** and the insulator **514**, it is preferable to use a film having a barrier property against hydrogen or impurities diffused from the substrate **311**, a region where the transistor **550** is provided, or the like into the region where the transistor **500** is provided. Thus, a material similar to that for the insulator **324** can be used.

For the film having a barrier property against hydrogen, silicon nitride formed using a CVD method can be used, for example. Here, diffusion of hydrogen into a semiconductor element including an oxide semiconductor, such as the transistor **500**, degrades the characteristics of the semiconductor element in some cases. Therefore, a film that inhibits hydrogen diffusion is preferably provided between the transistor **500** and the transistor **550**.

In addition, for the film having a barrier property against hydrogen, a metal oxide such as aluminum oxide, hafnium oxide, or tantalum oxide is preferably used for the insulator **510** and the insulator **514**, for example.

In particular, aluminum oxide has an excellent blocking effect that prevents the passage of both oxygen and impurities such as hydrogen and moisture which are factors of change in electrical characteristics of the transistor. Accordingly, aluminum oxide can prevent mixing of impurities such as hydrogen and moisture into the transistor **500** in the manufacturing process and after the manufacturing of the transistor. In addition, release of oxygen from the oxide included in the transistor **500** can be inhibited. Therefore, aluminum oxide is suitably used for the protective film of the transistor **500**.

In addition, for the insulator **512** and the insulator **516**, a material similar to that for the insulator **320** can be used, for example. Furthermore, when a material with a comparatively low permittivity is used for these insulators, parasitic capacitance generated between wirings can be reduced. A silicon oxide film, a silicon oxynitride film, or the like can be used for the insulator **512** and the insulator **516**, for example.

Furthermore, a conductor **518**, a conductor included in the transistor **500** (e.g., a conductor **503**), and the like are embedded in the insulator **510**, the insulator **512**, the insulator **514**, and the insulator **516**. Note that the conductor **518** has a function of a plug or a wiring that is connected to the capacitor **600** or the transistor **550**. The conductor **518** can be provided using a material similar to those for the conductor **328** and the conductor **330**.

In particular, the conductor **518** in a region in contact with the insulator **510** and the insulator **514** is preferably a conductor having a barrier property against oxygen, hydrogen, and water. With this structure, the transistor **550** and the transistor **500** can be separated by a layer having a barrier

property against oxygen, hydrogen, and water; thus, diffusion of hydrogen from the transistor **550** into the transistor **500** can be inhibited.

The transistor **500** is provided above the insulator **516**.

As illustrated in FIG. 19A and FIG. 19B, the transistor **500** includes the conductor **503** positioned to be embedded in the insulator **514** and the insulator **516**; an insulator **520** positioned over the insulator **516** and the conductor **503**; an insulator **522** positioned over the insulator **520**; an insulator **524** positioned over the insulator **522**; an oxide **530a** positioned over the insulator **524**; an oxide **530b** positioned over the oxide **530a**; a conductor **542a** and a conductor **542b** positioned apart from each other over the oxide **530b**; an insulator **580** that is positioned over the conductor **542a** and the conductor **542b** and is provided with an opening formed to overlap with a region between the conductor **542a** and the conductor **542b**; an insulator **545** positioned on a bottom surface and a side surface of the opening; and a conductor **560** positioned on a formation surface of the insulator **545**.

In addition, as illustrated in FIG. 19A and FIG. 19B, an insulator **544** is preferably positioned between the insulator **580** and the oxide **530a**, the oxide **530b**, the conductor **542a**, and the conductor **542b**. Furthermore, as illustrated in FIG. 19A and FIG. 19B, the conductor **560** preferably includes a conductor **560a** provided inside the insulator **545** and a conductor **560b** provided to be embedded inside the conductor **560a**. Moreover, as illustrated in FIG. 19A and FIG. 19B, an insulator **574** is preferably positioned over the insulator **580**, the conductor **560**, and the insulator **545**.

Note that in this specification and the like, the oxide **530a** and the oxide **530b** are sometimes collectively referred to as an oxide **530**.

Note that although a structure of the transistor **500** in which two layers of the oxide **530a** and the oxide **530b** are stacked in a region where a channel is formed and its vicinity is illustrated, the present invention is not limited thereto. For example, it is possible to employ a structure in which a single layer of the oxide **530b** or a stacked-layer structure of three or more layers is provided.

Furthermore, although the conductor **560** is illustrated to have a stacked-layer structure of two layers in the transistor **500**, the present invention is not limited thereto. For example, the conductor **560** may have a single-layer structure or a stacked-layer structure of three or more layers. Note that the transistors **500** illustrated in FIG. 17, FIG. 18, and FIG. 19A are examples, and the structures are not limited thereto; an appropriate transistor can be used in accordance with a circuit configuration, a driving method, or the like.

Here, the conductor **560** functions as a gate electrode of the transistor, and the conductor **542a** and the conductor **542b** each function as a source electrode or a drain electrode. As described above, the conductor **560** is formed to be embedded in the opening of the insulator **580** and the region between the conductor **542a** and the conductor **542b**. The positions of the conductor **560**, the conductor **542a**, and the conductor **542b** with respect to the opening of the insulator **580** are selected in a self-aligned manner. That is, in the transistor **500**, the gate electrode can be positioned between the source electrode and the drain electrode in a self-aligned manner. Thus, the conductor **560** can be formed without an alignment margin, resulting in a reduction in the area occupied by the transistor **500**. Accordingly, miniaturization and high integration of the semiconductor device can be achieved.

In addition, since the conductor **560** is formed in the region between the conductor **542a** and the conductor **542b** in a self-aligned manner, the conductor **560** does not have a

region overlapping with the conductor **542a** or the conductor **542b**. Thus, parasitic capacitance formed between the conductor **560** and each of the conductor **542a** and the conductor **542b** can be reduced. As a result, the switching speed of the transistor **500** can be improved, and the transistor **500** can have high frequency characteristics.

The conductor **560** sometimes functions as a first gate (also referred to as a top gate) electrode. In addition, the conductor **503** sometimes functions as a second gate (also referred to as a bottom gate) electrode. In that case, the threshold voltage of the transistor **500** can be controlled by changing a potential applied to the conductor **503** independently of a potential applied to the conductor **560**. In particular, when a negative potential is applied to the conductor **503**, the threshold voltage of the transistor **500** can be further increased, and the off-state current can be reduced. Thus, a drain current at the time when a potential applied to the conductor **560** is 0 V can be lower in the case where a negative potential is applied to the conductor **503** than in the case where a negative potential is not applied to the conductor **503**.

The conductor **503** is positioned to overlap with the oxide **530** and the conductor **560**. Thus, in the case where potentials are applied to the conductor **560** and the conductor **503**, an electric field generated from the conductor **560** and an electric field generated from the conductor **503** are connected, so that a channel formation region formed in the oxide **530** can be covered.

In this specification and the like, a transistor structure in which a channel formation region is electrically surrounded by electric fields of a pair of gate electrodes (a first gate electrode and a second gate electrode) is referred to as a surrounded channel (S-channel) structure. The S-channel structure disclosed in this specification and the like is different from a Fin-type structure and a planar structure. With the S-channel structure, resistance to a short-channel effect can be enhanced, that is, a transistor in which a short-channel effect is less likely to occur can be provided.

In addition, the conductor **503** has a structure similar to that of the conductor **518**; a conductor **503a** is formed in contact with an inner wall of an opening in the insulator **514** and the insulator **516**, and a conductor **503b** is formed on the inner side. Note that although the transistor **500** having a structure in which the conductor **503a** and the conductor **503b** are stacked is illustrated, the present invention is not limited thereto. For example, the conductor **503** may be provided as a single layer or to have a stacked-layer structure of three or more layers.

Here, for the conductor **503a**, a conductive material that has a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, and a copper atom (through which the impurities are unlikely to pass) is preferably used. Alternatively, it is preferable to use a conductive material that has a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like) (through which oxygen is unlikely to pass). Note that in this specification, a function of inhibiting diffusion of impurities or oxygen means a function of inhibiting diffusion of any one or all of the impurities and oxygen.

For example, when the conductor **503a** has a function of inhibiting diffusion of oxygen, a reduction in conductivity of the conductor **503b** due to oxidation can be inhibited.

In addition, in the case where the conductor **503** also functions as a wiring, a conductive material with high conductivity that contains tungsten, copper, or aluminum as its main component is preferably used for the conductor

503b. Note that although the conductor **503** has a stacked layer of the conductor **503a** and the conductor **503b** in this embodiment, the conductor **503** may have a single-layer structure.

The insulator **520**, the insulator **522**, and the insulator **524** have a function of a second gate insulating film.

Here, as the insulator **524** that is in contact with the oxide **530**, an insulator that contains oxygen more than oxygen in the stoichiometric composition is preferably used. Such oxygen is easily released from the insulator by heating. In this specification and the like, oxygen released by heating is sometimes referred to as "excess oxygen". That is, a region containing excess oxygen (also referred to as an "excess-oxygen region") is preferably formed in the insulator **524**. When such an insulator containing excess oxygen is provided in contact with the oxide **530**, oxygen vacancies (V_O) in the oxide **530** can be reduced and the reliability of the transistor **500** can be improved. When hydrogen enters the oxygen vacancies in the oxide **530**, such defects (hereinafter, referred to as V_OH in some cases) serve as donors and generate electrons serving as carriers in some cases. In other cases, bonding of part of hydrogen to oxygen bonded to a metal atom generates electrons serving as carriers. Thus, a transistor including an oxide semiconductor that contains a large amount of hydrogen is likely to have normally-on characteristics. Moreover, hydrogen in an oxide semiconductor is easily transferred by a stress such as heat or an electric field; thus, a large amount of hydrogen in an oxide semiconductor might reduce the reliability of the transistor. In one embodiment of the present invention, V_OH in the oxide **530** is preferably reduced as much as possible so that the oxide **530** becomes a highly purified intrinsic or substantially highly purified intrinsic oxide. It is important to remove impurities such as moisture or hydrogen in an oxide semiconductor (sometimes referred to as "dehydration" or "dehydrogenation treatment") and to compensate for oxygen vacancies by supplying oxygen to the oxide semiconductor (sometimes referred to as "oxygen adding treatment") in order to obtain an oxide semiconductor whose V_OH is sufficiently reduced. When an oxide semiconductor with sufficiently reduced impurities such as V_OH is used for a channel formation region of a transistor, the transistor can have stable electrical characteristics.

As the insulator including an excess-oxygen region, specifically, an oxide material that releases part of oxygen by heating is preferably used. An oxide that releases oxygen by heating is an oxide film in which the amount of released oxygen converted into oxygen atoms is greater than or equal to 1.0×10^{18} atoms/cm³, preferably greater than or equal to 1.0×10^{19} atoms/cm³, further preferably greater than or equal to 2.0×10^{19} atoms/cm³ or greater than or equal to 3.0×10^{20} atoms/cm³ in TDS (Thermal Desorption Spectroscopy) analysis. Note that the temperature of the film surface in the TDS analysis is preferably within the range of 100° C. to 700° C., or 100° C. to 400° C.

One or more of heat treatment, microwave treatment, and RF treatment may be performed in a state in which the insulator including the excess-oxygen region and the oxide **530** are in contact with each other. By the treatment, water or hydrogen in the oxide **530** can be removed. For example, in the oxide **530**, dehydrogenation can be performed when a reaction in which a bond of V_OH is cut occurs, i.e., a reaction of " $V_OH \rightarrow V_O + H$ " occurs. Part of hydrogen generated at this time is bonded to oxygen to be H_2O , and removed from the oxide **530** or an insulator near the oxide **530** in some cases. Some hydrogen may be getter into the conductor **542a** and/or the conductor **542b** in some cases.

For the microwave treatment, for example, an apparatus including a power supply that generates high-density plasma or an apparatus including a power supply that applies RF to the substrate side is suitably used. For example, the use of an oxygen-containing gas and high-density plasma enables high-density oxygen radicals to be generated, and application of the RF to the substrate side allows the oxygen radicals generated by the high-density plasma to be efficiently introduced into the oxide 530 or an insulator in the vicinity of the oxide 530. The pressure in the microwave treatment is higher than or equal to 133 Pa, preferably higher than or equal to 200 Pa, further preferably higher than or equal to 400 Pa. As a gas introduced into an apparatus for performing the microwave treatment, for example, oxygen and argon are used and the oxygen flow rate ($O_2/(O_2+Ar)$) is lower than or equal to 50%, preferably higher than or equal to 10% and lower than or equal to 30%.

In a manufacturing process of the transistor 500, heat treatment is preferably performed with the surface of the oxide 530 exposed. The heat treatment is performed at higher than or equal to 100° C. and lower than or equal to 450° C., preferably higher than or equal to 350° C. and lower than or equal to 400° C., for example. Note that the heat treatment is performed in a nitrogen gas or inert gas atmosphere, or an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more. For example, the heat treatment is preferably performed in an oxygen atmosphere. Accordingly, oxygen can be supplied to the oxide 530 to reduce oxygen vacancies (V_O). The heat treatment may be performed under reduced pressure. Alternatively, the heat treatment may be performed in such a manner that heat treatment is performed in a nitrogen gas or inert gas atmosphere, and then another heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more in order to compensate for released oxygen. Alternatively, the heat treatment may be performed in such a manner that heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more, and then another heat treatment is successively performed in a nitrogen gas or inert gas atmosphere.

Note that the oxygen adding treatment performed on the oxide 530 can promote a reaction in which oxygen vacancies in the oxide 530 are filled with supplied oxygen, i.e., a reaction of " $V_O+O\rightarrow\text{null}$ ". Furthermore, hydrogen remaining in the oxide 530 reacts with supplied oxygen, so that the hydrogen can be removed as H_2O (dehydration). This can inhibit recombination of hydrogen remaining in the oxide 530 with oxygen vacancies and formation of V_OH .

In addition, in the case where the insulator 524 includes an excess-oxygen region, it is preferable that the insulator 522 have a function of inhibiting diffusion of oxygen (e.g., an oxygen atom, an oxygen molecule, or the like) (through which oxygen is unlikely to pass).

When the insulator 522 has a function of inhibiting diffusion of oxygen or impurities, oxygen contained in the oxide 530 is not diffused into the insulator 520 side, which is preferable. Furthermore, the conductor 503 can be inhibited from reacting with oxygen contained in the insulator 524 or the oxide 530.

For the insulator 522, a single layer or stacked layers of an insulator containing what is called a high-k material such as aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), tantalum oxide, zirconium oxide, lead zirconate titanate (PZT), strontium titanate ($SrTiO_3$), or $(Ba,Sr)TiO_3$ (BST) are preferably used, for example. As miniaturization and high integration

of transistors progress, a problem such as leakage current might arise because of a thinner gate insulating film. When a high-k material is used for an insulator functioning as the gate insulating film, a gate potential during transistor operation can be reduced while the physical thickness is maintained.

It is particularly preferable to use an insulator containing an oxide of one or both of aluminum and hafnium, which is an insulating material having a function of inhibiting diffusion of impurities, oxygen, and the like (through which oxygen is unlikely to pass). Aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), or the like is preferably used as the insulator containing an oxide of one or both of aluminum and hafnium. In the case where the insulator 522 is formed using such a material, the insulator 522 functions as a layer that inhibits release of oxygen from the oxide 530 and mixing of impurities such as hydrogen from the periphery of the transistor 500 into the oxide 530.

Alternatively, aluminum oxide, bismuth oxide, germanium oxide, niobium oxide, silicon oxide, titanium oxide, tungsten oxide, yttrium oxide, or zirconium oxide may be added to these insulators, for example. Alternatively, these insulators may be subjected to nitriding treatment. The insulator over which silicon oxide, silicon oxynitride, or silicon nitride is stacked may be used.

In addition, it is preferable that the insulator 520 be thermally stable. For example, silicon oxide and silicon oxynitride are suitable because they are thermally stable. Furthermore, the combination of an insulator that is a high-k material and silicon oxide or silicon oxynitride enables the insulator 520 to have a stacked-layer structure that has thermal stability and a high dielectric constant.

Note that in the transistor 500 in FIG. 19A and FIG. 19B, the insulator 520, the insulator 522, and the insulator 524 are illustrated as the second gate insulating film having a stacked-layer structure of three layers; however, the second gate insulating film may be a single layer or may have a stacked-layer structure of two layers or four or more layers. In such cases, without limitation to a stacked-layer structure formed of the same material, a stacked-layer structure formed of different materials may be employed.

In the transistor 500, a metal oxide functioning as an oxide semiconductor is preferably used as the oxide 530 including a channel formation region. For example, as the oxide 530, a metal oxide such as an In-M-Zn oxide (the element M is one or more kinds selected from aluminum, gallium, yttrium, copper, vanadium, beryllium, boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like) is preferably used.

The metal oxide functioning as an oxide semiconductor may be formed by a sputtering method or an ALD (Atomic Layer Deposition) method. Note that the metal oxide functioning as an oxide semiconductor is described in detail in another embodiment.

Furthermore, the metal oxide functioning as the channel formation region in the oxide 530 has a band gap of more than or equal to 2 eV, preferably more than or equal to 2.5 eV. With the use of a metal oxide having such a wide bandgap, the off-state current of the transistor can be reduced.

When the oxide 530 includes the oxide 530a under the oxide 530b, it is possible to inhibit diffusion of impurities into the oxide 530b from the components formed below the oxide 530a.

Note that the oxide **530** preferably has a stacked-layer structure of a plurality of oxide layers that differ in the atomic ratio of metal atoms. Specifically, the atomic ratio of the element M to the constituent elements in the metal oxide used as the oxide **530a** is preferably higher than the atomic ratio of the element M to the constituent elements in the metal oxide used as the oxide **530b**. In addition, the atomic ratio of the element M to In in the metal oxide used as the oxide **530a** is preferably higher than the atomic ratio of the element M to In in the metal oxide used as the oxide **530b**. Furthermore, the atomic ratio of In to the element M in the metal oxide used as the oxide **530b** is preferably higher than the atomic ratio of In to the element M in the metal oxide used as the oxide **530a**.

In addition, the energy of the conduction band minimum of the oxide **530a** is preferably higher than the energy of the conduction band minimum of the oxide **530b**. In other words, the electron affinity of the oxide **530a** is preferably smaller than the electron affinity of the oxide **530b**.

Here, the energy level of the conduction band minimum gradually changes at a junction portion of the oxide **530a** and the oxide **530b**. In other words, the energy level of the conduction band minimum at the junction portion of the oxide **530a** and the oxide **530b** continuously changes or is continuously connected. To obtain this, the density of defect state in a mixed layer formed at an interface between the oxide **530a** and the oxide **530b** is preferably made low.

Specifically, when the oxide **530a** and the oxide **530b** contain a common element (as a main component) in addition to oxygen, a mixed layer with a low density of defect states can be formed. For example, in the case where the oxide **530b** is an In—Ga—Zn oxide, an In—Ga—Zn oxide, a Ga—Zn oxide, gallium oxide, or the like is preferably used as the oxide **530a**.

At this time, the oxide **530b** serves as a main carrier path. When the oxide **530a** has the above structure, the density of defect states at the interface between the oxide **530a** and the oxide **530b** can be made low. Thus, the influence of interface scattering on carrier conduction is small, and the transistor **500** can have a high on-state current.

The conductor **542a** and the conductor **542b** functioning as the source electrode and the drain electrode are provided over the oxide **530b**. For the conductor **542a** and conductor **542b**, it is preferable to use a metal element selected from aluminum, chromium, copper, silver, gold, platinum, tantalum, nickel, titanium, molybdenum, tungsten, hafnium, vanadium, niobium, manganese, magnesium, zirconium, beryllium, indium, ruthenium, iridium, strontium, and lanthanum; an alloy containing any of the above metal elements; an alloy containing a combination of the above metal elements; or the like. For example, it is preferable to use tantalum nitride, titanium nitride, tungsten, a nitride containing titanium and aluminum, a nitride containing tantalum and aluminum, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, an oxide containing lanthanum and nickel, or the like. In addition, tantalum nitride, titanium nitride, a nitride containing titanium and aluminum, a nitride containing tantalum and aluminum, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, and an oxide containing lanthanum and nickel are preferable because they are oxidation-resistant conductive materials or materials that retain their conductivity even after absorbing oxygen. Furthermore, a metal nitride film of tantalum nitride or the like is preferable because it has a barrier property against hydrogen or oxygen.

In addition, although the conductor **542a** and the conductor **542b** each having a single-layer structure are illustrated in FIG. **19A**, a stacked-layer structure of two or more layers may be employed. For example, it is preferable to stack a tantalum nitride film and a tungsten film. Alternatively, a titanium film and an aluminum film may be stacked. Alternatively, a two-layer structure where an aluminum film is stacked over a tungsten film, a two-layer structure where a copper film is stacked over a copper-magnesium-aluminum alloy film, a two-layer structure where a copper film is stacked over a titanium film, or a two-layer structure where a copper film is stacked over a tungsten film may be employed.

Other examples include a three-layer structure where a titanium film or a titanium nitride film is formed, an aluminum film or a copper film is stacked over the titanium film or the titanium nitride film, and a titanium film or a titanium nitride film is formed over the aluminum film or the copper film; and a three-layer structure where a molybdenum film or a molybdenum nitride film is formed, an aluminum film or a copper film is stacked over the molybdenum film or the molybdenum nitride film, and a molybdenum film or a molybdenum nitride film is formed over the aluminum film or the copper film. Note that a transparent conductive material containing indium oxide, tin oxide, or zinc oxide may be used.

In addition, as illustrated in FIG. **19A**, a region **543a** and a region **543b** are sometimes formed as low-resistance regions at an interface between the oxide **530** and the conductor **542a** (the conductor **542b**) and in the vicinity of the interface. In that case, the region **543a** functions as one of a source region and a drain region, and the region **543b** functions as the other of the source region and the drain region. Furthermore, the channel formation region is formed in a region between the region **543a** and the region **543b**.

When the conductor **542a** (the conductor **542b**) is provided to be in contact with the oxide **530**, the oxygen concentration in the region **543a** (the region **543b**) sometimes decreases. In addition, a metal compound layer that contains the metal contained in the conductor **542a** (the conductor **542b**) and the component of the oxide **530** is sometimes formed in the region **543a** (the region **543b**). In such a case, the carrier density of the region **543a** (the region **543b**) increases, and the region **543a** (the region **543b**) becomes a low-resistance region.

The insulator **544** is provided to cover the conductor **542a** and the conductor **542b** and inhibits oxidation of the conductor **542a** and the conductor **542b**. At this time, the insulator **544** may be provided to cover a side surface of the oxide **530** and to be in contact with the insulator **524**.

A metal oxide containing one kind or two or more kinds selected from hafnium, aluminum, gallium, yttrium, zirconium, tungsten, titanium, tantalum, nickel, germanium, neodymium, lanthanum, magnesium, and the like can be used as the insulator **544**. Alternatively, silicon nitride oxide, silicon nitride, or the like can be used for the insulator **544**.

It is particularly preferable to use an insulator containing an oxide of one or both of aluminum and hafnium, such as aluminum oxide, hafnium oxide, or an oxide containing aluminum and hafnium (hafnium aluminate), as the insulator **544**. In particular, hafnium aluminate has higher heat resistance than a hafnium oxide film. Therefore, hafnium aluminate is preferable because it is less likely to be crystallized by heat treatment in a later step. Note that the insulator **544** is not an essential component when the conductor **542a** and the conductor **542b** are oxidation-resistant materials or do not significantly lose their conductivity even after absorbing

oxygen. Design is appropriately set in consideration of required transistor characteristics.

When the insulator **544** is included, diffusion of impurities such as water and hydrogen contained in the insulator **580** into the oxide **530b** through the insulator **545** can be inhibited. Furthermore, oxidation of the conductor **560** due to excess oxygen contained in the insulator **580** can be inhibited.

The insulator **545** functions as a first gate insulating film. Like the insulator **524**, the insulator **545** is preferably formed using an insulator that contains excess oxygen and releases oxygen by heating.

Specifically, silicon oxide containing excess oxygen, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, or porous silicon oxide can be used. In particular, silicon oxide and silicon oxynitride are preferable because they are thermally stable.

When an insulator containing excess oxygen is provided as the insulator **545**, oxygen can be effectively supplied from the insulator **545** to the channel formation region of the oxide **530b**. Furthermore, as in the insulator **524**, the concentration of impurities such as water or hydrogen in the insulator **545** is preferably reduced. The thickness of the insulator **545** is preferably greater than or equal to 1 nm and less than or equal to 20 nm.

Furthermore, to efficiently supply excess oxygen contained in the insulator **545** to the oxide **530**, a metal oxide may be provided between the insulator **545** and the conductor **560**. The metal oxide preferably inhibits diffusion of oxygen from the insulator **545** into the conductor **560**. Providing the metal oxide that inhibits diffusion of oxygen inhibits diffusion of excess oxygen from the insulator **545** into the conductor **560**. That is, reduction in the amount of excess oxygen supplied to the oxide **530** can be inhibited. Moreover, oxidation of the conductor **560** due to excess oxygen can be inhibited. For the metal oxide, a material that can be used for the insulator **544** is used.

Note that the insulator **545** may have a stacked-layer structure like the second gate insulating film. As miniaturization and high integration of transistors progress, a problem such as leakage current might arise because of a thinner gate insulating film. For that reason, when the insulator functioning as the gate insulating film has a stacked-layer structure of a high-k material and a thermally stable material, a gate potential during transistor operation can be reduced while the physical thickness is maintained. Furthermore, the stacked-layer structure can be thermally stable and have a high relative permittivity.

Although the conductor **560** that functions as the first gate electrode and has a two-layer structure is illustrated in FIG. **19A** and FIG. **19B**, a single-layer structure or a stacked-layer structure of three or more layers may be employed.

For the conductor **560a**, it is preferable to use a conductive material having a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule (N_2O , NO , NO_2 , and the like), and a copper atom. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like). When the conductor **560a** has a function of inhibiting diffusion of oxygen, it is possible to inhibit a reduction in conductivity of the conductor **560b** due to oxidation caused by oxygen contained in the insulator **545**. As a conductive material having a function of inhibiting

diffusion of oxygen, for example, tantalum, tantalum nitride, ruthenium, ruthenium oxide, or the like is preferably used. For the conductor **560a**, the oxide semiconductor that can be used as the oxide **530** can be used. In that case, when the conductor **560b** is deposited using a sputtering method, the conductor **560a** can have a reduced value of electrical resistance to be a conductor. Such a conductor can be referred to as an OC (Oxide Conductor) electrode.

In addition, a conductive material containing tungsten, copper, or aluminum as its main component is preferably used for the conductor **560b**. Furthermore, the conductor **560b** also functions as a wiring and thus a conductor having high conductivity is preferably used as the conductor **560b**. For example, a conductive material containing tungsten, copper, or aluminum as its main component can be used. The conductor **560b** may have a stacked-layer structure, for example, a stacked-layer structure of any of the above conductive materials and titanium or titanium nitride.

The insulator **580** is provided over the conductor **542a** and the conductor **542b** with the insulator **544** therebetween. The insulator **580** preferably includes an excess-oxygen region. For example, the insulator **580** preferably contains silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, resin, or the like. In particular, silicon oxide and silicon oxynitride are preferable because they are thermally stable. In particular, silicon oxide and porous silicon oxide are preferable because an excess-oxygen region can be easily formed in a later step.

The insulator **580** preferably includes an excess-oxygen region. When the insulator **580** that releases oxygen by heating is provided, oxygen in the insulator **580** can be efficiently supplied to the oxide **530**. Note that the concentration of impurities such as water or hydrogen in the insulator **580** is preferably reduced.

The opening of the insulator **580** is formed to overlap with the region between the conductor **542a** and the conductor **542b**. Accordingly, the conductor **560** is formed to be embedded in the opening of the insulator **580** and the region between the conductor **542a** and the conductor **542b**.

The gate length needs to be short for miniaturization of the semiconductor device, but it is necessary to prevent a reduction in conductivity of the conductor **560**. When the conductor **560** is made thick to achieve this, the conductor **560** might have a shape with a high aspect ratio. In this embodiment, the conductor **560** is provided to be embedded in the opening of the insulator **580**; thus, even when the conductor **560** has a shape with a high aspect ratio, the conductor **560** can be formed without collapsing during the process.

The insulator **574** is preferably provided in contact with a top surface of the insulator **580**, a top surface of the conductor **560**, and a top surface of the insulator **545**. When the insulator **574** is deposited using a sputtering method, excess-oxygen regions can be provided in the insulator **545** and the insulator **580**. Accordingly, oxygen can be supplied from the excess-oxygen regions to the oxide **530**.

For example, a metal oxide containing one kind or two or more kinds selected from hafnium, aluminum, gallium, yttrium, zirconium, tungsten, titanium, tantalum, nickel, germanium, magnesium, and the like can be used as the insulator **574**.

In particular, aluminum oxide has a high barrier property, and even a thin aluminum oxide film having a thickness of greater than or equal to 0.5 nm and less than or equal to 3.0

nm can inhibit diffusion of hydrogen and nitrogen. Accordingly, aluminum oxide deposited by a sputtering method serves as an oxygen supply source and can also have a function of a barrier film against impurities such as hydrogen.

In addition, an insulator **581** functioning as an interlayer film is preferably provided over the insulator **574**. As in the insulator **524** or the like, the concentration of impurities such as water or hydrogen in the insulator **581** is preferably reduced.

Furthermore, a conductor **540a** and a conductor **540b** are positioned in openings formed in the insulator **581**, the insulator **574**, the insulator **580**, and the insulator **544**. The conductor **540a** and the conductor **540b** are provided to face each other with the conductor **560** therebetween. The structures of the conductor **540a** and the conductor **540b** are similar to a structure of a conductor **546** and a conductor **548** that are described later.

An insulator **582** is provided over the insulator **581**. A substance having a barrier property against oxygen or hydrogen is preferably used for the insulator **582**. Therefore, a material similar to that for the insulator **514** can be used for the insulator **582**. For the insulator **582**, a metal oxide such as aluminum oxide, hafnium oxide, or tantalum oxide is preferably used, for example.

In particular, aluminum oxide has an excellent blocking effect that prevents the passage of both oxygen and impurities such as hydrogen and moisture which are factors of change in electrical characteristics of the transistor. Accordingly, aluminum oxide can prevent mixing of impurities such as hydrogen and moisture into the transistor **500** in the manufacturing process and after the manufacturing of the transistor. In addition, release of oxygen from the oxide included in the transistor **500** can be inhibited. Therefore, aluminum oxide is suitably used for the protective film of the transistor **500**.

In addition, an insulator **586** is provided over the insulator **582**. For the insulator **586**, a material similar to that for the insulator **320** can be used. Furthermore, when a material with a comparatively low permittivity is used for these insulators, parasitic capacitance generated between wirings can be reduced. A silicon oxide film, a silicon oxynitride film, or the like can be used for the insulator **586**, for example.

Furthermore, the conductor **546**, the conductor **548**, and the like are embedded in the insulator **520**, the insulator **522**, the insulator **524**, the insulator **544**, the insulator **580**, the insulator **574**, the insulator **581**, the insulator **582**, and the insulator **586**.

The conductor **546** and the conductor **548** have functions of plugs or wirings that are connected to the capacitor **600**, the transistor **500**, or the transistor **550**. The conductor **546** and the conductor **548** can be provided using materials similar to those for the conductor **328** and the conductor **330**.

After the transistor **500** is formed, an opening may be formed to surround the transistor **500** and an insulator having a high barrier property against hydrogen or water may be formed to cover the opening. Surrounding the transistor **500** with the insulator having a high barrier property can prevent entry of moisture and hydrogen from the outside. Alternatively, a plurality of transistors **500** may be collectively surrounded by the insulator having a high barrier property against hydrogen or water. When an opening is formed to surround the transistor **500**, for example, the formation of an opening reaching the insulator **522** or the insulator **514** and the formation of the insulator having a high barrier property in contact with the insulator **522** or the

insulator **514** are suitable because these formation steps can also serve as part of the manufacturing steps of the transistor **500**. The insulator having a high barrier property against hydrogen or water is formed using a material similar to that for the insulator **522** or the insulator **514**, for example.

Next, the capacitor **600** is provided above the transistor **500**. The capacitor **600** includes a conductor **610**, a conductor **620**, and an insulator **630**.

In addition, a conductor **612** may be provided over the conductor **546** and the conductor **548**. The conductor **612** has a function of a plug or a wiring that is connected to the transistor **500**. The conductor **610** has a function of an electrode of the capacitor **600**. Note that the conductor **612** and the conductor **610** can be formed at the same time.

For the conductor **612** and the conductor **610**, a metal film containing an element selected from molybdenum, titanium, tantalum, tungsten, aluminum, copper, chromium, neodymium, and scandium; a metal nitride film containing the above element as its component (a tantalum nitride film, a titanium nitride film, a molybdenum nitride film, or a tungsten nitride film); or the like can be used. Alternatively, it is possible to use a conductive material such as indium tin oxide, indium oxide containing tungsten oxide, indium zinc oxide containing tungsten oxide, indium oxide containing titanium oxide, indium tin oxide containing titanium oxide, indium zinc oxide, or indium tin oxide to which silicon oxide is added.

Although the conductor **612** and the conductor **610** each having a single-layer structure are shown in this embodiment, the structure is not limited thereto; a stacked-layer structure of two or more layers may be employed. For example, between a conductor having a barrier property and a conductor having high conductivity, a conductor that is highly adhesive to the conductor having a barrier property and the conductor having high conductivity may be formed.

The conductor **620** is provided to overlap with the conductor **610** with the insulator **630** therebetween. Note that a conductive material such as a metal material, an alloy material, or a metal oxide material can be used for the conductor **620**. It is preferable to use a high-melting-point material that has both heat resistance and conductivity, such as tungsten or molybdenum, and it is particularly preferable to use tungsten. In addition, in the case where the conductor **620** is formed concurrently with another component such as a conductor, Cu (copper), Al (aluminum), or the like, which is a low-resistance metal material, is used.

An insulator **640** is provided over the conductor **620** and the insulator **630**. The insulator **640** can be provided using a material similar to that for the insulator **320**. In addition, the insulator **640** may function as a planarization film that covers an uneven shape therebelow.

With the use of this structure, a semiconductor device using a transistor including an oxide semiconductor can be miniaturized or highly integrated.

Examples of a substrate that can be used for the semiconductor device of one embodiment of the present invention include a glass substrate, a quartz substrate, a sapphire substrate, a ceramic substrate, a metal substrate (e.g., a stainless steel substrate, a substrate including stainless steel foil, a tungsten substrate, and a substrate including tungsten foil), a semiconductor substrate (e.g., a single crystal semiconductor substrate, a polycrystalline semiconductor substrate, and a compound semiconductor substrate), and an SOI (Silicon on Insulator) substrate. Alternatively, a plastic substrate having heat resistance to the processing temperature in this embodiment may be used. Examples of a glass substrate include a barium borosilicate glass substrate, an

aluminosilicate glass substrate, an aluminoborosilicate glass substrate, and a soda lime glass substrate. Alternatively, crystallized glass or the like can be used.

Alternatively, a flexible substrate, an attachment film, paper including a fibrous material, a base film, or the like can be used as the substrate. As examples of the flexible substrate, the attachment film, the base material film, and the like, the following can be given. Examples include plastics typified by polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polyether sulfone (PES), and polytetrafluoroethylene (PTFE). Another example is a synthetic resin such as acrylic. Other examples are polypropylene, polyester, polyvinyl fluoride, and polyvinyl chloride. Other examples are polyamide, polyimide, an aramid resin, an epoxy resin, an inorganic vapor deposition film, and paper. In particular, the use of a semiconductor substrate, a single crystal substrate, an SOI substrate, or the like enables the manufacture of small-sized transistors with a small variation in characteristics, size, shape, or the like and with high current capability. When a circuit is formed with such transistors, lower power consumption of the circuit or higher integration of the circuit can be achieved.

A flexible substrate may be used as the substrate, and a transistor, a resistor, a capacitor, and/or the like may be formed directly over the flexible substrate. Alternatively, a separation layer may be provided between the substrate and the transistor, the resistor, the capacitor, and/or the like. After part or the whole of a semiconductor device is completed over the separation layer, the separation layer can be used for separation from the substrate and transfer to another substrate. In such a case, the transistor, the resistor, the capacitor, and/or the like can be transferred to a substrate having low heat resistance or a flexible substrate. As the separation layer, a stack of inorganic films, namely a tungsten film and a silicon oxide film, an organic resin film of polyimide or the like formed over a substrate, or a silicon film containing hydrogen can be used, for example.

That is, a semiconductor device may be formed over one substrate and then transferred to another substrate. Examples of a substrate to which a semiconductor device is transferred include, in addition to the above-described substrates over which transistors can be formed, a paper substrate, a cellophane substrate, an aramid film substrate, a polyimide film substrate, a stone substrate, a wood substrate, a cloth substrate (including a natural fiber (silk, cotton, or hemp), a synthetic fiber (nylon, polyurethane, or polyester), a regenerated fiber (acetate, cupro, rayon, or regenerated polyester), or the like), a leather substrate, and a rubber substrate. With the use of any of these substrates, a flexible semiconductor device or a highly durable semiconductor device can be manufactured, high heat resistance can be provided, or a reduction in weight or thickness can be achieved.

Providing a semiconductor device over a flexible substrate can suppress an increase in weight and can produce a non-breakable semiconductor device.

Modification Example 1 of Transistor

A transistor **500A** illustrated in FIG. 20A, FIG. 20B, and FIG. 20C is a modification example of the transistor **500** having the structure illustrated in FIG. 19A and FIG. 19B. FIG. 20A is a top view of the transistor **500A**, FIG. 20B is a cross-sectional view of the transistor **500A** in the channel length direction, and FIG. 20C is a cross-sectional view of the transistor **500A** in the channel width direction. Note that for clarity of the drawing, some components are not illustrated in the top view of FIG. 20A. The structure illustrated

in FIG. 20A, FIG. 20B, and FIG. 20C can also be used for other transistors such as the transistor **550** included in the semiconductor device of one embodiment of the present invention.

The transistor **500A** having the structure illustrated in FIG. 20A, FIG. 20B, and FIG. 20C is different from the transistor **500** having the structure illustrated in FIG. 19A and FIG. 19B in that an insulator **552**, an insulator **513**, and an insulator **404** are included. Furthermore, the transistor **500A** is different from the transistor **500** having the structure illustrated in FIG. 19A and FIG. 19B in that the insulator **552** is provided in contact with a side surface of the conductor **540a** and a side surface of the conductor **540b**. Moreover, the transistor **500A** is different from the transistor **500** having the structure illustrated in FIG. 19A and FIG. 19B in that the insulator **520** is not included.

In the transistor **500A** having the structure illustrated in FIG. 20A, FIG. 20B, and FIG. 20C, the insulator **513** is provided over the insulator **512**. The insulator **404** is provided over the insulator **574** and the insulator **513**.

In the transistor **500A** having the structure illustrated in FIG. 20A, FIG. 20B, and FIG. 20C, the insulator **514**, the insulator **516**, the insulator **522**, the insulator **524**, the insulator **544**, the insulator **580**, and the insulator **574** are patterned and covered with the insulator **404**. That is, the insulator **404** is in contact with a top surface of the insulator **574**, a side surface of the insulator **574**, a side surface of the insulator **580**, a side surface of the insulator **544**, a side surface of the insulator **524**, a side surface of the insulator **522**, a side surface of the insulator **516**, a side surface of the insulator **514**, and a top surface of the insulator **513**. Thus, the oxide **530** and the like are isolated from the outside by the insulator **404** and the insulator **513**.

The insulator **513** and the insulator **404** preferably have high capability of inhibiting diffusion of hydrogen (e.g., at least one of a hydrogen atom, a hydrogen molecule, and the like) or a water molecule. For example, for the insulator **513** and the insulator **404**, silicon nitride or silicon nitride oxide that is a material having a high hydrogen barrier property is preferably used. This can inhibit diffusion of hydrogen or the like into the oxide **530**, thereby suppressing the degradation of the characteristics of the transistor **500A**. Consequently, the reliability of the semiconductor device of one embodiment of the present invention can be increased.

The insulator **552** is provided in contact with the insulator **581**, the insulator **404**, the insulator **574**, the insulator **580**, and the insulator **544**. The insulator **552** preferably has a function of inhibiting diffusion of hydrogen or water molecules. For example, for the insulator **552**, an insulator such as silicon nitride, aluminum oxide, or silicon nitride oxide that is a material having a high hydrogen barrier property is preferably used. In particular, it is preferable to use silicon nitride as the insulator **552** because of its high hydrogen barrier property. The use of a material having a high hydrogen barrier property for the insulator **552** can inhibit diffusion of impurities such as water and hydrogen from the insulator **580** and the like into the oxide **530** through the conductor **540a** and the conductor **540b**. Furthermore, oxygen contained in the insulator **580** can be inhibited from being absorbed by the conductor **540a** and the conductor **540b**. As described above, the reliability of the semiconductor device of one embodiment of the present invention can be increased.

Modification Example 2 of Transistor

A structure example of a transistor **500B** is described with reference to FIG. 21A, FIG. 21B, and FIG. 21C. FIG. 21A

is a top view of the transistor **500B**. FIG. **21B** is a cross-sectional view of a portion indicated by dashed-dotted line L1-L2 in FIG. **21A**. FIG. **21C** is a cross-sectional view of a portion indicated by dashed-dotted line W1-W2 in FIG. **21A**. Note that for clarity of the drawing, some components are not illustrated in the top view of FIG. **21A**.

The transistor **500B** is a modification example of the transistor **500A** and can be replaced with the transistor **500**. Accordingly, in order to avoid repeated description, differences of the transistor **500B** from the transistor **500A** are mainly described.

The transistor **500B** is different from the transistor **500A** in that an oxide **530c** is included. Specifically, the transistor **500B** includes the insulator **580** that has an opening formed to overlap with a portion between the conductor **542a** and the conductor **542b**, the oxide **530c** positioned on a bottom surface and a side surface of the opening, the insulator **545** positioned on a formation surface of the oxide **530c**, and the conductor **560** positioned on the formation surface of the insulator **545**.

Note that in this specification and the like, the oxide **530a**, the oxide **530b**, and the oxide **530c** are sometimes collectively referred to as an oxide **530**. A metal oxide that can be used as the oxide **530a** or the oxide **530b** can be used as the oxide **530c**.

As described above, the energy of the conduction band minimum of the oxide **530a** is preferably higher than the energy of the conduction band minimum of the oxide **530b**. Like the oxide **530a**, the energy of the conduction band minimum of the oxide **530c** is also preferably higher than the energy of the conduction band minimum of the oxide **530b**. In other words, the electron affinity of each of the oxide **530a** and the oxide **530c** is preferably smaller than the electron affinity of the oxide **530b**. In this manner, the energy level of the conduction band minimum gently changes at a junction portion of the oxide **530a** and the oxide **530b** and a junction portion of the oxide **530b** and the oxide **530c**. A mixed layer with a low density of defect states can be formed at an interface between the oxide **530a** and the oxide **530b** and an interface between the oxide **530b** and the oxide **530c**.

The influence of interface scattering on carrier conduction in the transistor **500B** including the oxide **530c** is smaller than that in the transistor **500** and the transistor **500A**. Thus, the transistor **500B** can obtain a higher on-state current than the transistor **500** and the transistor **500A**.

Moreover, including the oxide **530c** over the oxide **530b** makes it possible to inhibit diffusion of impurities into the oxide **530b** from the components formed above the oxide **530c**.

Modification Example 3 of Transistor

A structure example of a transistor **500C** is described with reference to FIG. **22A**, FIG. **22B**, and FIG. **22C**. FIG. **22A** is a top view of the transistor **500C**. FIG. **22B** is a cross-sectional view of a portion indicated by dashed-dotted line L1-L2 in FIG. **22A**. FIG. **22C** is a cross-sectional view of a portion indicated by dashed-dotted line W1-W2 in FIG. **22A**. Note that for clarity of the drawing, some components are not illustrated in the top view of FIG. **22A**.

The transistor **500C** is a modification example of the transistor **500** and can be replaced with the transistor **500**. Accordingly, in order to avoid repeated description, differences of the transistor **500C** from the transistor **500** are mainly described.

The conductor **560** functioning as a first gate electrode includes the conductor **560a** and the conductor **560b** over the conductor **560a**. For the conductor **560a**, a conductive material that has a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, and a copper atom is preferably used. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like).

When the conductor **560a** has a function of inhibiting diffusion of oxygen, the range of choices for the material of the conductor **560b** can be extended. That is, the conductor **560a** inhibits oxidation of the conductor **560b**, thereby preventing the decrease in conductivity.

The insulator **544** is preferably provided to cover the top surface and a side surface of the conductor **560** and a side surface of the insulator **545**. For the insulator **544**, an insulating material having a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen is preferably used. For example, aluminum oxide or hafnium oxide is preferably used. Moreover, it is possible to use, for example, a metal oxide such as magnesium oxide, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, or tantalum oxide or silicon nitride oxide, silicon nitride, or the like.

The insulator **544** can inhibit oxidation of the conductor **560**. Moreover, the insulator **544** can inhibit diffusion of impurities such as water and hydrogen contained in the insulator **580** into the transistor **500C**.

The transistor **500C** has the conductor **560** overlapping part of the conductor **542a** and part of the conductor **542b**, and thus tends to have larger parasitic capacitance than the transistor **500**. Consequently, the transistor **500C** tends to have a lower operating frequency than the transistor **500**. However, the transistor **500C** does not require steps of providing an opening in the insulator **580** and the like and embedding the conductor **560**, the insulator **545**, and the like in the opening; hence, the productivity of the transistor **500C** is higher than that of the transistor **500**.

The composition, structure, method, and the like described in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

Embodiment 5

In this embodiment, an oxide semiconductor which is a kind of metal oxides is described.

The metal oxide preferably contains at least indium or zinc. In particular, indium and zinc are preferably contained. In addition, aluminum, gallium, yttrium, tin, or the like is preferably contained. Furthermore, one or more kinds selected from boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, cobalt, and the like may be contained.

<Classification of Crystal Structure>

First, the classification of the crystal structures of an oxide semiconductor is described with reference to FIG. **23A**. FIG. **23A** is a diagram showing classification of crystal structures of an oxide semiconductor, typically IGZO (a metal oxide containing In, Ga, and Zn).

As shown in FIG. **23A**, an oxide semiconductor is roughly classified into "Amorphous", "Crystalline", and "Crystal". The term "Amorphous" includes completely amorphous. The term "Crystalline" includes CAAC (c-axis-aligned crys-

talline), nc (nanocrystalline), and CAC (cloud-aligned composite). Note that the term "Crystalline" excludes single crystal, poly crystal, and completely amorphous. The term "Crystal" includes single crystal and poly crystal.

Note that the structures in the thick frame in FIG. 23A are in an intermediate state between "Amorphous" and "Crystalline", and belong to a new crystalline phase. That is, these structures are completely different from "Amorphous", which is energetically unstable, and "Crystal".

A crystal structure of a film or a substrate can be analyzed with an X-ray diffraction (XRD) spectrum. Here, FIG. 23B shows an XRD spectrum, which is obtained by GIXD (Grazing-Incidence XRD) measurement, of a CAAC-IGZO film classified into "Crystalline". Note that a GIXD method is also referred to as a thin film method or a Seemann-Bohlin method. The XRD spectrum that is shown in FIG. 23B and obtained by GIXD measurement is hereinafter simply referred to as an XRD spectrum. The CAAC-IGZO film in FIG. 23B has a composition in the vicinity of In:Ga:Zn=4:2:3 [atomic ratio]. The CAAC-IGZO film in FIG. 23B has a thickness of 500 nm.

As shown in FIG. 23B, a clear peak indicating crystallinity is detected in the XRD spectrum of the CAAC-IGZO film. Specifically, a peak indicating c-axis alignment is detected at 2θ of around 31° in the XRD spectrum of the CAAC-IGZO film. As shown in FIG. 23B, the peak at 2θ of around 31° is asymmetric with respect to the axis of the angle at which the peak intensity is detected.

A crystal structure of a film or a substrate can also be evaluated with a diffraction pattern obtained by a nanobeam electron diffraction (NBED) method (such a pattern is also referred to as a nanobeam electron diffraction pattern). FIG. 23C shows a diffraction pattern of the CAAC-IGZO film. FIG. 23C shows a diffraction pattern obtained with the NBED method in which an electron beam is incident in the direction parallel to the substrate. The composition of the CAAC-IGZO film in FIG. 23C is In:Ga:Zn=4:2:3 [atomic ratio] or the neighborhood thereof. In the nanobeam electron diffraction method, electron diffraction is performed with a probe diameter of 1 nm.

As shown in FIG. 23C, a plurality of spots indicating c-axis alignment are observed in the diffraction pattern of the CAAC-IGZO film.

<<Structure of Oxide Semiconductor>>

Oxide semiconductors might be classified in a manner different from that in FIG. 23A when classified in terms of the crystal structure. Oxide semiconductors are classified into a single crystal oxide semiconductor and a non-single-crystal oxide semiconductor, for example. Examples of the non-single-crystal oxide semiconductor include the above-described CAAC-OS and nc-OS. Other examples of the non-single-crystal oxide semiconductor include a polycrystalline oxide semiconductor, an amorphous-like oxide semiconductor (a-like OS), and an amorphous oxide semiconductor.

Here, the above-described CAAC-OS, nc-OS, and a-like OS are described in detail.
[CAAC-OS]

The CAAC-OS is an oxide semiconductor that has a plurality of crystal regions each of which has c-axis alignment in a particular direction. Note that the particular direction refers to the film thickness direction of a CAAC-OS film, the normal direction of the surface where the CAAC-OS film is formed, or the normal direction of the surface of the CAAC-OS film. The crystal region refers to a region having a periodic atomic arrangement. When an atomic arrangement is regarded as a lattice arrangement, the

crystal region also refers to a region with a uniform lattice arrangement. The CAAC-OS has a region where a plurality of crystal regions are connected in the a-b plane direction, and the region has distortion in some cases. Note that the distortion refers to a portion where the direction of a lattice arrangement changes between a region with a uniform lattice arrangement and another region with a uniform lattice arrangement in a region where a plurality of crystal regions are connected. That is, the CAAC-OS is an oxide semiconductor having c-axis alignment and having no clear alignment in the a-b plane direction.

Note that each of the plurality of crystal regions is formed of one or more fine crystals (crystals each of which has a maximum diameter of less than 10 nm). In the case where the crystal region is formed of one fine crystal, the maximum diameter of the crystal region is less than 10 nm. In the case where the crystal region is formed of a large number of fine crystals, the size of the crystal region may be approximately several tens of nanometers.

In the case of an In-M-Zn oxide (the element M is one or more kinds selected from aluminum, gallium, yttrium, tin, titanium, and the like), the CAAC-OS tends to have a layered crystal structure (also referred to as a stacked-layer structure) in which a layer containing indium (In) and oxygen (hereinafter, an In layer) and a layer containing the element M, zinc (Zn), and oxygen (hereinafter, an (M,Zn) layer) are stacked. Indium and the element M can be replaced with each other. Therefore, indium may be contained in the (M,Zn) layer. In addition, the element M may be contained in the In layer. Note that Zn may be contained in the In layer. Such a layered structure is observed as a lattice image in a high-resolution TEM image, for example.

When the CAAC-OS film is subjected to structural analysis by out-of-plane XRD measurement with an XRD apparatus using $\theta/2\theta$ scanning, for example, a peak indicating c-axis alignment is detected at 2θ of 31° or around 31° . Note that the position of the peak indicating c-axis alignment (the value of 2θ) may change depending on the kind, composition, or the like of the metal element contained in the CAAC-OS.

For example, a plurality of bright spots are observed in the electron diffraction pattern of the CAAC-OS film. Note that one spot and another spot are observed point-symmetrically with a spot of the incident electron beam passing through a sample (also referred to as a direct spot) as the symmetric center.

When the crystal region is observed from the particular direction, a lattice arrangement in the crystal region is basically a hexagonal lattice arrangement; however, a unit lattice is not always a regular hexagon and is a non-regular hexagon in some cases. A pentagonal lattice arrangement, a heptagonal lattice arrangement, and the like are included in the distortion in some cases. Note that a clear grain boundary cannot be observed even in the vicinity of the distortion in the CAAC-OS. That is, formation of a crystal grain boundary is inhibited by the distortion of lattice arrangement. This is probably because the CAAC-OS can tolerate distortion owing to a low density of arrangement of oxygen atoms in the a-b plane direction, an interatomic bond distance changed by substitution of a metal atom, and the like.

A crystal structure in which a clear grain boundary is observed is what is called polycrystal. It is highly probable that the grain boundary becomes a recombination center and captures carriers and thus decreases the on-state current and field-effect mobility of a transistor, for example. Thus, the CAAC-OS in which no clear grain boundary is observed is one of crystalline oxides having a crystal structure suitable

for a semiconductor layer of a transistor. Note that Zn is preferably contained to form the CAAC-OS. For example, an In—Zn oxide and an In—Ga—Zn oxide are suitable because they can inhibit generation of a grain boundary as compared with an In oxide.

The CAAC-OS is an oxide semiconductor with high crystallinity in which no clear grain boundary is observed. Thus, in the CAAC-OS, reduction in electron mobility due to the grain boundary is less likely to occur. Moreover, since the crystallinity of an oxide semiconductor might be decreased by entry of impurities, formation of defects, or the like, the CAAC-OS can be regarded as an oxide semiconductor that has small amounts of impurities and defects (e.g., oxygen vacancies). Thus, an oxide semiconductor including the CAAC-OS is physically stable. Therefore, the oxide semiconductor including the CAAC-OS is resistant to heat and has high reliability. In addition, the CAAC-OS is stable with respect to high temperature in the manufacturing process (what is called thermal budget). Accordingly, the use of the CAAC-OS for the OS transistor can extend the degree of freedom of the manufacturing process.

[nc-OS]

In the nc-OS, a microscopic region (e.g., a region with a size greater than or equal to 1 nm and less than or equal to 10 nm, in particular, a region with a size greater than or equal to 1 nm and less than or equal to 3 nm) has a periodic atomic arrangement. In other words, the nc-OS includes a fine crystal. Note that the size of the fine crystal is, for example, greater than or equal to 1 nm and less than or equal to 10 nm, particularly greater than or equal to 1 nm and less than or equal to 3 nm; thus, the fine crystal is also referred to as a nanocrystal. Furthermore, there is no regularity of crystal orientation between different nanocrystals in the nc-OS. Thus, the orientation in the whole film is not observed. Accordingly, the nc-OS cannot be distinguished from an a-like OS or an amorphous oxide semiconductor with some analysis methods. For example, when an nc-OS film is subjected to structural analysis using out-of-plane XRD measurement with an XRD apparatus using $\theta/2\theta$ scanning, a peak indicating crystallinity is not detected. Furthermore, a diffraction pattern like a halo pattern is observed when the nc-OS film is subjected to electron diffraction (also referred to as selected-area electron diffraction) using an electron beam with a probe diameter larger than the diameter of a nanocrystal (e.g., larger than or equal to 50 nm). Meanwhile, in some cases, a plurality of spots in a ring-like region with a direct spot as the center are observed in the obtained electron diffraction pattern when the nc-OS film is subjected to electron diffraction (also referred to as nanobeam electron diffraction) using an electron beam with a probe diameter nearly equal to or smaller than the diameter of a nanocrystal (e.g., 1 nm or larger and 30 nm or smaller).

[a-Like OS]

The a-like OS is an oxide semiconductor having a structure between those of the nc-OS and the amorphous oxide semiconductor. The a-like OS includes a void or a low-density region. That is, the a-like OS has low crystallinity as compared with the nc-OS and the CAAC-OS. Moreover, the a-like OS has higher hydrogen concentration in the film than the nc-OS and the CAAC-OS.

<<Structure of Oxide Semiconductor>>

Next, the above-described CAC-OS is described in detail. Note that the CAC-OS relates to the material composition. [CAC-OS]

The CAC-OS refers to one composition of a material in which elements constituting a metal oxide are unevenly

distributed with a size greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 3 nm, or a similar size, for example. Note that a state in which one or more metal elements are unevenly distributed and regions including the metal element(s) are mixed with a size greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 3 nm, or a similar size in a metal oxide is hereinafter referred to as a mosaic pattern or a patch-like pattern.

In addition, the CAC-OS has a composition in which materials are separated into a first region and a second region to form a mosaic pattern, and the first regions are distributed in the film (this composition is hereinafter also referred to as a cloud-like composition). That is, the CAC-OS is a composite metal oxide having a composition in which the first regions and the second regions are mixed.

Note that the atomic ratios of In, Ga, and Zn to the metal elements contained in the CAC-OS in an In—Ga—Zn oxide are denoted with [In], [Ga], and [Zn], respectively. For example, the first region in the CAC-OS in the In-Ga-Zn oxide has [In] higher than [In] in the composition of the CAC-OS film. Moreover, the second region has [Ga] higher than [Ga] in the composition of the CAC-OS film. For example, the first region has higher [In] and lower [Ga] than the second region. Moreover, the second region has higher [Ga] and lower [In] than the first region.

Specifically, the first region includes indium oxide, indium zinc oxide, or the like as its main component. The second region includes gallium oxide, gallium zinc oxide, or the like as its main component. That is, the first region can be referred to as a region containing In as its main component. The second region can be referred to as a region containing Ga as its main component.

Note that a clear boundary between the first region and the second region cannot be observed in some cases.

For example, energy dispersive X-ray spectroscopy (EDX) is used to obtain EDX mapping, and according to the EDX mapping, the CAC-OS in the In—Ga—Zn oxide has a structure in which the region containing In as its main component (the first region) and the region containing Ga as its main component (the second region) are unevenly distributed and mixed.

In the case where the CAC-OS is used for a transistor, a switching function (on/off switching function) can be given to the CAC-OS owing to the complementary action of the conductivity derived from the first region and the insulating property derived from the second region. A CAC-OS has a conducting function in part of the material and has an insulating function in another part of the material; as a whole, the CAC-OS has a function of a semiconductor. Separation of the conducting function and the insulating function can maximize each function. Accordingly, when the CAC-OS is used for a transistor, a high on-state current (I_{on}), a high field-effect mobility (μ) and favorable switching operation can be achieved.

An oxide semiconductor has various structures with different properties. Two or more kinds among the amorphous oxide semiconductor, the polycrystalline oxide semiconductor, the a-like OS, the CAC-OS, the nc-OS, and the CAAC-OS may be included in an oxide semiconductor of one embodiment of the present invention.

<Transistor Including Oxide Semiconductor>

Next, the case where the above oxide semiconductor is used for a transistor is described.

When the above oxide semiconductor is used for a transistor, a transistor with high field-effect mobility can be achieved. In addition, a transistor having high reliability can be achieved.

An oxide semiconductor with a low carrier concentration is preferably used for the transistor. For example, the carrier concentration of an oxide semiconductor is lower than or equal to $1 \times 10^{17} \text{ cm}^{-3}$, preferably lower than or equal to $1 \times 10^{15} \text{ cm}^{-3}$, further preferably lower than or equal to $1 \times 10^{13} \text{ cm}^{-3}$, still further preferably lower than or equal to $1 \times 10^{11} \text{ cm}^{-3}$, yet further preferably lower than $1 \times 10^{10} \text{ cm}^{-3}$, and higher than or equal to $1 \times 10^{-9} \text{ cm}^{-3}$. In order to reduce the carrier concentration of an oxide semiconductor film, the impurity concentration in the oxide semiconductor film is reduced so that the density of defect states can be reduced. In this specification and the like, a state with a low impurity concentration and a low density of defect states is referred to as a highly purified intrinsic or substantially highly purified intrinsic state. Note that an oxide semiconductor having a low carrier concentration may be referred to as a highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor.

A highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has a low density of defect states and thus has a low density of trap states in some cases.

Electric charge trapped by the trap states in the oxide semiconductor takes a long time to disappear and might behave like fixed electric charge. Thus, a transistor whose channel formation region is formed in an oxide semiconductor with a high density of trap states has unstable electrical characteristics in some cases.

Accordingly, in order to obtain stable electrical characteristics of a transistor, reducing the impurity concentration in an oxide semiconductor is effective. In order to reduce the impurity concentration in the oxide semiconductor, it is preferable that the impurity concentration in an adjacent film be also reduced. Examples of impurities include hydrogen, nitrogen, an alkali metal, an alkaline earth metal, iron, nickel, and silicon.

<Impurity>

Here, the influence of each impurity in the oxide semiconductor is described.

When silicon or carbon, which is one of Group 14 elements, is contained in the oxide semiconductor, defect states are formed in the oxide semiconductor. Thus, the concentration of silicon or carbon in the oxide semiconductor and the concentration of silicon or carbon in the vicinity of an interface with the oxide semiconductor (the concentration obtained by secondary ion mass spectrometry (SIMS)) are each set lower than or equal to $2 \times 10^{18} \text{ atoms/cm}^3$, preferably lower than or equal to $2 \times 10^{17} \text{ atoms/cm}^3$.

When the oxide semiconductor contains an alkali metal or an alkaline earth metal, defect states are formed and carriers are generated in some cases. Thus, a transistor using an oxide semiconductor that contains an alkali metal or an alkaline earth metal is likely to have normally-on characteristics. Thus, the concentration of an alkali metal or an alkaline earth metal in the oxide semiconductor, which is obtained using SIMS, is lower than or equal to $1 \times 10^{18} \text{ atoms/cm}^3$, preferably lower than or equal to $2 \times 10^{16} \text{ atoms/cm}^3$.

Furthermore, when the oxide semiconductor contains nitrogen, the oxide semiconductor easily becomes n-type by generation of electrons serving as carriers and an increase in carrier concentration. As a result, a transistor using an oxide semiconductor containing nitrogen as a semiconductor is

likely to have normally-on characteristics. When nitrogen is contained in the oxide semiconductor, a trap state is sometimes formed. This might make the electrical characteristics of the transistor unstable. Therefore, the concentration of nitrogen in the oxide semiconductor, which is obtained using SIMS, is set lower than $5 \times 10^{19} \text{ atoms/cm}^3$, preferably lower than or equal to $5 \times 10^{18} \text{ atoms/cm}^3$, further preferably lower than or equal to $1 \times 10^{18} \text{ atoms/cm}^3$, still further preferably lower than or equal to $5 \times 10^{17} \text{ atoms/cm}^3$.

Hydrogen contained in the oxide semiconductor reacts with oxygen bonded to a metal atom to be water, and thus forms an oxygen vacancy in some cases. Entry of hydrogen into the oxygen vacancy generates an electron serving as a carrier in some cases. Furthermore, bonding of part of hydrogen to oxygen bonded to a metal atom causes generation of an electron serving as a carrier in some cases. Thus, a transistor using an oxide semiconductor containing hydrogen is likely to have normally-on characteristics. Accordingly, hydrogen in the oxide semiconductor is preferably reduced as much as possible. Specifically, the hydrogen concentration in the oxide semiconductor, which is obtained using SIMS, is set lower than $1 \times 10^{20} \text{ atoms/cm}^3$, preferably lower than $1 \times 10^{19} \text{ atoms/cm}^3$, further preferably lower than $5 \times 10^{18} \text{ atoms/cm}^3$, still further preferably lower than $1 \times 10^{18} \text{ atoms/cm}^3$.

When an oxide semiconductor with sufficiently reduced impurities is used for the channel formation region of the transistor, stable electrical characteristics can be given.

The composition, structure, method, and the like described in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

Embodiment 6

In this embodiment, application examples of the above-described semiconductor device are described.

[Semiconductor Wafer and Chip]

FIG. 24A is a top view of a substrate **711** before dicing treatment is performed. As the substrate **711**, a semiconductor substrate (also referred to as a "semiconductor wafer") can be used, for example. A plurality of circuit regions **712** are provided over the substrate **711**. A semiconductor device, a CPU, an RF tag, an image sensor, or the like of one embodiment of the present invention can be provided in the circuit region **712**.

The plurality of circuit regions **712** are each surrounded by a separation region **713**. Separation lines (also referred to as "dicing lines") **714** are set at a position overlapping with the separation regions **713**. The substrate **711** is cut along the separation lines **714**, whereby chips **715** including the circuit regions **712** can be cut out from the substrate **711**. FIG. 24B illustrates an enlarged view of the chip **715**.

Furthermore, a conductive layer or a semiconductor layer may be provided in the separation regions **713**. Providing a conductive layer or a semiconductor layer in the separation regions **713** relieves ESD that might be caused in a dicing step, preventing a decrease in the yield in the dicing step. Furthermore, a dicing step is generally performed while letting pure water whose specific resistance is decreased by dissolution of a carbonic acid gas or the like flow to a cut portion, in order to cool down a substrate, remove swarf, and prevent electrification, for example. Providing a conductive layer or a semiconductor layer in the separation regions **713** allows a reduction in the usage of the pure water. Therefore,

the manufacturing cost of the semiconductor device can be reduced. Furthermore, the productivity of the semiconductor device can be increased.

For a semiconductor layer provided in the separation regions **713**, a material having a bandgap greater than or equal to 2.5 eV and less than or equal to 4.2 eV is preferably used and a material having a bandgap greater than or equal to 2.7 eV and less than or equal to 3.5 eV is further preferable. The use of such a material allows accumulated charges to be released slowly; thus, rapid move of charges due to ESD can be suppressed and electrostatic breakdown is less likely to occur.

[Electronic Component]

An example where the chip **715** is used for an electronic component is described with reference to FIG. **25A** and FIG. **25B**. Note that the electronic component is also referred to as a semiconductor package or an IC package. For electronic components, there are various standards and names corresponding to a terminal extraction direction and a terminal shape.

The electronic component is completed when the semiconductor device described in the above embodiment is combined with components other than the semiconductor device in an assembly process (post-process).

The post-process is described with reference to a flow chart shown in FIG. **25A**. After an element substrate including the semiconductor device described in the above embodiment is completed in a pre-process, a “back surface grinding step” is performed to grind a back surface (a surface where the semiconductor device and the like are not formed) of the element substrate (Step **S721**). When the element substrate is thinned by grinding, warpage or the like of the element substrate is reduced, so that the size of the electronic component can be reduced.

Next, a “dicing step” for dividing the element substrate into a plurality of chips (the chips **715**) is performed (Step **S722**). Then, a “die bonding step” is performed to pick up the divided chips separately and bond them to a lead frame (Step **S723**). To bond a chip and a lead frame in the die bonding step, a method such as resin bonding or tape-automated bonding is selected as appropriate depending on products. Note that the chip may be bonded to an interposer substrate instead of the lead frame.

Next, a “wire bonding step” for electrically connecting a lead of the lead frame and an electrode on the chip through a metal fine line (wire) is performed (Step **S724**). A silver line or a gold line can be used as the metal fine line. Furthermore, ball bonding or wedge bonding can be used as the wire bonding.

The wire-bonded chip is subjected to a “sealing step (molding step)” of sealing the chip with an epoxy resin or the like (Step **S725**). Through the sealing step, the inside of the electronic component is filled with a resin, so that a circuit portion incorporated in the chip and a wire for connecting the chip to the lead can be protected from external mechanical force, and deterioration of characteristics (decrease in reliability) due to moisture or dust can be reduced.

Subsequently, a “lead plating step” for plating the lead of the lead frame is performed (Step **S726**). This plating treatment prevents rust of the lead and enables more reliable soldering at the time of mounting the electronic component on a printed board in a later step. Then, a “formation step” for cutting and processing the lead is performed (Step **S727**).

Next, a “marking step” for printing (marking) a surface of the package is performed (Step **S728**). Then, after a “testing step” (Step **S729**) for checking whether an external shape is

good and whether there is a malfunction, for example, the electronic component is completed.

FIG. **25B** illustrates a schematic perspective view of the completed electronic component. FIG. **25B** illustrates a schematic perspective view of a QFP (Quad Flat Package) as an example of the electronic component. An electronic component **750** illustrated in FIG. **25B** includes a lead **755** and a semiconductor device **753**. As the semiconductor device **753**, the semiconductor device described in the above embodiment or the like can be used.

The electronic component **750** illustrated in FIG. **25B** is mounted on a printed circuit board **752**, for example. A plurality of such electronic components **750** are combined and electrically connected to each other on the printed circuit board **752**; thus, a board on which the electronic components are mounted (a circuit board **754**) is completed. The completed circuit board **754** is used for an electronic device or the like.

[Electronic Device]

Next, examples of electronic devices including the semiconductor device of one embodiment of the present invention or the above-described electronic component are described.

Examples of electronic devices including the semiconductor device of one embodiment of the present invention or the electronic component include display devices of televisions, monitors, and the like; lighting devices; desktop personal computers; laptop personal computers; word processors; image reproduction devices that reproduce still images and moving images stored in recording media such as DVDs (digital versatile discs); portable CD players; radios; tape recorders; headphone stereos; stereos; table clocks; wall clocks; cordless phone handsets; transceivers; mobile phones; car phones; portable game machines; tablet terminals; large-sized game machines such as pachinko machines; calculators; portable information terminals; electronic notebooks; e-book readers; electronic translators; audio input devices; video cameras; digital still cameras; electric shavers; high-frequency heating appliances such as microwave ovens; electric rice cookers; electric washing machines; electric vacuum cleaners; water heaters; electric fans; hair dryers; air-conditioning systems such as air conditioners, humidifiers, and dehumidifiers; dishwashers; dish dryers; clothes dryers; futon dryers; electric refrigerators; electric freezers; electric refrigerator-freezers; freezers for preserving DNA; flashlights; tools such as chain saws; smoke detectors; and medical equipment such as dialyzers. Furthermore, industrial equipment such as guide lights, traffic lights, belt conveyors, elevators, escalators, industrial robots, power storage systems, and power storage devices for leveling the amount of power supply and smart grid can be given.

In addition, moving objects and the like driven by electric motors using power from the power storage devices are also included in the category of electronic devices. Examples of the moving objects include electric vehicles (EVs), hybrid electric vehicles (HEVs) that include both an internal-combustion engine and a motor, plug-in hybrid electric vehicles (PHEVs), tracked vehicles in which caterpillar tracks are substituted for wheels of these vehicles, motorized bicycles including motor-assisted bicycles, motorcycles, electric wheelchairs, golf carts, boats, ships, submarines, helicopters, aircraft, rockets, artificial satellites, space probes, planetary probes, and spacecraft.

51

The semiconductor device of one embodiment of the present invention or the electronic component can be used for a communication device or the like in any of the electronic devices.

The electronic device may include a sensor (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, a chemical substance, sound, time, hardness, electric field, current, voltage, power, radiation, flow rate, humidity, gradient, oscillation, a smell, or infrared rays) and the like.

The electronic device can have a variety of functions. For example, the electronic device can have a function of displaying a variety of data (a still image, a moving image, a text image, and the like) on the display portion, a touch panel function, a function of displaying a calendar, date, time, and the like, a function of executing a variety of software (programs), a wireless communication function, and a function of reading out a program or data stored in a recording medium.

FIG. 26 and FIG. 27A to FIG. 27F illustrate examples of electronic devices. In FIG. 26, a display device **8000** is an example of an electronic device including a semiconductor device **8004** of one embodiment of the present invention. Specifically, the display device **8000** corresponds to a display device for TV broadcast reception and includes a housing **8001**, a display portion **8002**, speaker portions **8003**, the semiconductor device **8004**, a power storage device **8005**, and the like. The semiconductor device **8004** of one embodiment of the present invention is provided in the housing **8001**. The semiconductor device **8004** can retain control data, a control program, or the like. The semiconductor device **8004** has a communication function, and the display device **8000** can function as an IoT device. The display device **8000** can receive power from a commercial power supply or can use power stored in the power storage device **8005**.

A display device such as a liquid crystal display device, a light-emitting display device in which a light-emitting element such as an organic EL element is provided in each pixel, an electrophoresis display device, a DMD (Digital Micromirror Device), a PDP (Plasma Display Panel), or an FED (Field Emission Display) can be used for the display portion **8002**.

Note that the display device includes, in its category, all of information display devices for personal computers, advertisement display, and the like besides for TV broadcast reception.

In FIG. 26, an installation lighting device **8100** is an example of an electronic device including a semiconductor device **8103** of one embodiment of the present invention. Specifically, the lighting device **8100** includes a housing **8101**, a light source **8102**, the semiconductor device **8103**, a power storage device **8105**, and the like. Although FIG. 26 illustrates the case where the semiconductor device **8103** is provided in a ceiling **8104** on which the housing **8101** and the light source **8102** are installed, the semiconductor device **8103** may be provided in the housing **8101**. The semiconductor device **8103** can retain data such as emission luminance of the light source **8102**, a control program, or the like. The semiconductor device **8103** has a communication function, and the lighting device **8100** can function as an IoT device. The display device **8100** can receive power from a commercial power supply or can use power stored in the power storage device.

Note that although the installation lighting device **8100** provided in the ceiling **8104** is illustrated in FIG. 26 as an

52

example, the semiconductor device of one embodiment of the present invention can be used in an installation lighting device provided in, for example, a wall **8405**, a floor **8406**, a window **8407**, or the like other than the ceiling **8104**.

Alternatively, the semiconductor device of one embodiment of the present invention can be used in a tabletop lighting device or the like.

As the light source **8102**, an artificial light source that emits light artificially by using power can be used. Specific examples of the artificial light source include an incandescent lamp, a discharge lamp such as a fluorescent lamp, and light-emitting elements such as an LED and an organic EL element.

In FIG. 26, an air conditioner including an indoor unit **8200** and an outdoor unit **8204** is an example of an electronic device including a semiconductor device **8203** of one embodiment of the present invention. Specifically, the indoor unit **8200** includes a housing **8201**, an air outlet **8202**, the semiconductor device **8203**, a power storage device **8205**, and the like. Although FIG. 26 illustrates the case where the semiconductor device **8203** is provided in the indoor unit **8200**, the semiconductor device **8203** may be provided in the outdoor unit **8204**. Alternatively, the semiconductor devices **8203** may be provided in both the indoor unit **8200** and the outdoor unit **8204**. The semiconductor device **8203** can retain control data, a control program, or the like of the air conditioner. The semiconductor device **8203** has a communication function, and the air conditioner can function as an IoT device. The air conditioner can receive power from a commercial power supply or can use power stored in the power storage device **8205**.

Note that although the split-type air conditioner including the indoor unit and the outdoor unit is illustrated in FIG. 26 as an example, the semiconductor device of one embodiment of the present invention can also be used in an air conditioner in which the functions of an indoor unit and an outdoor unit are integrated in one housing.

In FIG. 26, an electric refrigerator-freezer **8300** is an example of an electronic device including a semiconductor device **8304** of one embodiment of the present invention. Specifically, the electric refrigerator-freezer **8300** includes a housing **8301**, a refrigerator door **8302**, a freezer door **8303**, the semiconductor device **8304**, a power storage device **8305**, and the like. The power storage device **8305** is provided in the housing **8301** in FIG. 26. The semiconductor device **8304** can retain control data, a control program, or the like of the electric refrigerator-freezer **8300**. The semiconductor device **8304** has a communication function, and the electric refrigerator-freezer **8300** can function as an IoT device. The electric refrigerator-freezer **8300** can receive power from a commercial power supply or can use power stored in the power storage device **8305**.

FIG. 27A illustrates an example of a wrist-watch-type portable information terminal. A portable information terminal **6100** includes a housing **6101**, a display portion **6102**, a band **6103**, operation buttons **6105**, and the like. The portable information terminal **6100** further includes a secondary battery and the semiconductor device of one embodiment of the present invention or the electronic component. The portable information terminal **6100** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

FIG. 27B illustrates an example of a mobile phone. A portable information terminal **6200** includes a display portion **6202** incorporated in a housing **6201**, operation buttons **6203**, a speaker **6204**, a microphone **6205**, and the like.

The portable information terminal **6200** further includes a fingerprint sensor **6209** in a region overlapping with the display portion **6202**. The fingerprint sensor **6209** may be an organic optical sensor. Since a fingerprint differs between individuals, the fingerprint sensor **6209** can perform personal authentication when acquiring fingerprint patterns. As a light source for acquiring fingerprint patterns with the fingerprint sensor **6209**, light emitted from the display portion **6202** can be used.

The portable information terminal **6200** further includes a secondary battery and the semiconductor device of one embodiment of the present invention or the electronic component. The portable information terminal **6200** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

FIG. 27C illustrates an example of a cleaning robot. A cleaning robot **6300** includes a display portion **6302** placed on the top surface of a housing **6301**, a plurality of cameras **6303** placed on the side surface of the housing **6301**, a brush **6304**, operation buttons **6305**, a variety of sensors, and the like. Although not illustrated, the cleaning robot **6300** is provided with a tire, an inlet, and the like. The cleaning robot **6300** can run autonomously, detect dust **6310**, and vacuum the dust through the inlet provided on the bottom surface.

For example, the cleaning robot **6300** can analyze images taken by the cameras **6303** to judge whether there are obstacles such as a wall, furniture, or a step. In the case where an object that is likely to be caught in the brush **6304**, such as a wire, is detected by image analysis, the rotation of the brush **6304** can be stopped. The cleaning robot **6300** includes a secondary battery and the semiconductor device of one embodiment of the present invention or the electronic component. The cleaning robot **6300** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

FIG. 27D illustrates an example of a robot. A robot **6400** illustrated in FIG. 27D includes an arithmetic device **6409**, an illuminance sensor **6401**, a microphone **6402**, an upper camera **6403**, a speaker **6404**, a display portion **6405**, a lower camera **6406**, an obstacle sensor **6407**, and a moving mechanism **6408**.

The microphone **6402** has a function of detecting a speaking voice of a user, an environmental sound, and the like. The speaker **6404** also has a function of outputting sound. The robot **6400** can communicate with a user using the microphone **6402** and the speaker **6404**.

The display portion **6405** has a function of displaying various kinds of information. The robot **6400** can display user's desired information on the display portion **6405**. A touch panel may be incorporated in the display portion **6405**. Moreover, the display portion **6405** may be a detachable information terminal, in which case charging and data communication can be performed when the display portion **6405** is set at the home position of the robot **6400**.

The upper camera **6403** and the lower camera **6406** each have a function of taking an image of the surroundings of the robot **6400**. The obstacle sensor **6407** can detect the presence of an obstacle in the direction where the robot **6400** advances with the moving mechanism **6408**. The robot **6400** can move safely by recognizing the surroundings with the upper camera **6403**, the lower camera **6406**, and the obstacle sensor **6407**. The light-emitting device of one embodiment of the present invention can be used for the display portion **6405**.

The robot **6400** includes a secondary battery and the semiconductor device of one embodiment of the present invention or the electronic component. The robot **6400** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

FIG. 27E illustrates an example of a flying object. A flying object **6500** illustrated in FIG. 27E includes propellers **6501**, a camera **6502**, a battery **6503**, and the like and has a function of flying autonomously.

For example, image data taken by the camera **6502** is stored in an electronic component **6504**. The electronic component **6504** can analyze the image data to detect whether there is an obstacle in the way of the movement. Moreover, the electronic component **6504** can estimate the remaining battery level from a change in the power storage capacity of the battery **6503**. The flying object **6500** further includes the semiconductor device of one embodiment of the present invention or the electronic component. The flying object **6500** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

FIG. 27F illustrates an example of an automobile. An automobile **7160** includes an engine, tires, a brake, a steering gear, a camera, and the like. The automobile **7160** includes the semiconductor device of one embodiment of the present invention or the electronic component. The automobile **7160** including the semiconductor device of one embodiment of the present invention or the electronic component can function as an IoT device.

The composition, structure, method, and the like described in this example can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments and the example.

Embodiment 7

A normally-off CPU (also referred to as "Noff-CPU") can be achieved by using the OS transistor described in this specification and the like. Note that the Noff-CPU is an integrated circuit including a normally-off transistor, which is in a non-conduction state (also referred to as an off state) even when a gate voltage is 0 V.

In the Noff-CPU, power supply to a circuit that does not need to operate can be stopped so that the circuit can be brought into a standby state. The circuit brought into the standby state because of the stop of power supply does not consume power. Thus, the power usage of the Noff-CPU can be minimized. Moreover, the Noff-CPU can retain data necessary for operation, such as setting conditions, for a long time even when power supply is stopped. The return from the standby state requires only restart of power supply to the circuit and does not require rewriting of setting conditions or the like. In other words, high-speed return from the standby state is possible. As described here, the power consumption of the Noff-CPU can be reduced without a significant decrease in operation speed.

The Noff-CPU can be suitably used for a small-scale system such as an IoT end device (also referred to as an "endpoint microcomputer") **803** in the IoT (Internet of Things) field, for example.

FIG. 28 illustrates a hierarchical structure of an IoT network and tendencies of required specifications. FIG. 28 illustrates power consumption **804** and processing performance **805** as the required specifications. The hierarchical structure of the IoT network is roughly divided into a cloud

55

field **801** at the upper level and an embedded field **802** at the lower level. The cloud field **801** includes a server, for example. The embedded field **802** includes a machine, an industrial robot, an in-vehicle device, and a home appliance, for example.

Higher processing performance is required rather than lower power consumption towards the top of the hierarchical structure. Thus, a high-performance CPU, a high-performance GPU, a large-scale SoC (System on a Chip), and the like are used in the cloud field **801**. Furthermore, lower power consumption is required rather than higher processing performance towards the bottom of the hierarchical structure, and the number of devices is explosively increased. The semiconductor device of one embodiment of the present invention can be suitably used for a communication device in the IoT end device that needs to have low power consumption.

Note that an “endpoint” refers to an end region of the embedded field **802**. Examples of a device used in the endpoint include microcomputers used in a factory, a home appliance, infrastructure, agriculture, and the like.

FIG. 29 shows a conceptual diagram showing factory automation as an application example of the endpoint microcomputer. A factory **884** is connected to a cloud **883** through Internet connection (Internet). The cloud **883** is connected to a home **881** and an office **882** through the Internet connection. The Internet connection may be wired communication or wireless communication. In the case of wireless communication, for example, wireless communication based on a communication standard such as the fourth-generation mobile communications system (4G) or the fifth-generation mobile communications system (5G) is performed using the semiconductor device of one embodiment of the present invention for a communication device. The factory **884** may be connected to a factory **885** and a factory **886** through the Internet connection.

The factory **884** includes a master device (control device) **831**. The master device **831** is connected to the cloud **883** and has a function of transmitting and receiving data. The master device **831** is connected to a plurality of industrial robots **842** included in an IoT end device **841** through an M2M (Machine to Machine) interface **832**. As the M2M interface **832**, for example, industrial Ethernet (Ethernet is a registered trademark), which is a kind of wired communication, or local 5G, which is a kind of wireless communication, may be used.

A factory manager can check the operational status or the like from the home **881** or the office **882** connected to the factory **884** through the cloud **883**. In addition, the manager can check wrong items and part shortage, instruct a storage space, and measure takt time, for example.

In recent years, IoT has been globally introduced into factories; this situation is called “Smart Factory”. Smart Factory has been reported to enable not only simple examination and inspection by an endpoint microcomputer but also detection of failures and prediction of abnormality, for example.

The total power consumption of a small-scale system such as an endpoint microcomputer during operation is often small, which enhances the power reduction effect in a standby operation by the Noff-CPU. Meanwhile, the embedded field of IoT sometimes requires quick response; the use of the Noff-CPU achieves high-speed return from a standby operation.

The composition, structure, method, and the like described in this embodiment can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

56

Example

In this example, the simulation results of the cutoff frequency of an OS-FET that can be used for the present invention are described.

The cutoff frequency (f_T) of the OS-FET can be obtained by Formula 3 below.

10 [Formula 3]

$$f_T = \frac{g_m}{2\pi C_g} \quad (3)$$

Here, C_g and g_m denote the gate capacitance and transconductance of the OS-FET, respectively. The transconductance g_m at a particular drain voltage can be obtained by Formula 4 below.

20 [Formula 4]

$$g_m = \left(\frac{\partial I_d}{\partial V_g} \right)_{V_d} \quad (4)$$

In Formula 4 above, V_g , I_d , and V_d are the gate voltage, the drain current, and the drain voltage of the OS-FET, respectively.

For the calculation of the cutoff frequency, Atlas 3D, a device simulator from Silvaco Inc., was used. FIG. 30A to FIG. 30C illustrate a structure of the OS-FET used for the calculation. FIG. 30A is a schematic cross-sectional view of a center portion of the channel of the OS-FET in the channel length direction. FIG. 30B is a schematic cross-sectional view of the center portion of the channel of the OS-FET in the channel width direction. FIG. 30C is a schematic cross-sectional view of the source region or the drain region of the OS-FET in the channel width direction.

In FIG. 30A to FIG. 30C, the OS-FET includes BGE, BG11, BG12, OS1, OS2, a pair of SDs, TGI, and TGE. BGE functions as a back gate electrode and TGE functions as a gate electrode (also referred to as a top-gate electrode). OS1 and OS2 are a metal oxide having a stacked-layer structure. The pair of SDs each function as a source electrode and a drain electrode. BG11 and BG12 function as a gate insulating film having a stacked-layer structure that is provided between BGE and OS1, and TGI functions as a gate insulating film provided between OS2 and TGE.

As OS1, a metal oxide of In:Ga:Zn=1:3:4 [atomic ratio] was used. As OS2, a metal oxide of In:Ga:Zn=4:2:3 [atomic ratio] was used.

Furthermore, L in FIG. 30A, that is, the width of TGE denotes a channel length, and W in FIG. 30B, that is, the width of OS1 and OS2 denotes a channel width.

Next, Table 1 shows the calculation conditions.

TABLE 1

Software		Atlas3D from Silvaco Inc.	
Structure	L	30	nm
	W	30	nm
BGE	Work function	5.0	eV
	Thickness	20	nm
BG11	Relative permittivity	16.4	
	Thickness	20	nm
BG12	Relative permittivity	4.1	
	Thickness	30	nm

TABLE 1-continued

Software	Atlas3D from Silvaco Inc.		
TGI	Relative permittivity	4.1	nm
	Thickness	6	nm
TGE	Work function	4.7	eV
	Thickness	50	nm
SD	Work function	4.5	eV
	Thickness	20	nm
OS1	Electron affinity	4.5	eV
	Eg	3.4	eV
IGZO(134)	Relative permittivity	15	
	Electron mobility	1.5	cm ² /Vs
	Hole mobility	0.01	cm ² /Vs
	Nc	5E+18	/cm ³
	Nv	5E+18	/cm ³
	Thickness	5	nm
	Electron affinity	4.6	eV
	Eg	3.0	eV
	Relative permittivity	15	
	Electron mobility	8	cm ² /Vs
OS2	Hole mobility	0.01	cm ² /Vs
	Nc	5E+18	/cm ³
	Nv	5E+18	/cm ³
	Thickness	15	nm
	Nd in n + region	1E+20	/cm ³
IGZO(423)			

FIG. 31 shows the calculation results of the cutoff frequency of the OS-FET obtained with the above-described conditions. In FIG. 31, the horizontal axis represents the drain voltage (unit: V) of the OS-FET, and the vertical axis represents the cutoff frequency (unit: GHz). In the calculation, the gate voltage and the drain voltage have the same value.

From the results in FIG. 31, the cutoff frequency of the OS-FET was 38.6 GHz when the drain voltage was 1 V; the cutoff frequency was 71.5 GHz when the drain voltage was 2 V; the cutoff frequency was 104.4 GHz when the drain voltage was 3 V; the cutoff frequency was 132.8 GHz when the drain voltage was 4 V; and the cutoff frequency was 160.1 GHz when the drain voltage was 5 V. The calculation confirms that when the drain voltage is set higher than or equal to 3 V, a cutoff frequency of higher than or equal to 100 GHz can be obtained.

According to the above calculation results, the OS-FET can be favorably used as the transistor of one embodiment of the present invention.

The composition, structure, method, and the like described in this example can be used in combination as appropriate with the compositions, structures, methods, and the like described in the other embodiments, the example, and the like.

REFERENCE NUMERALS

100: semiconductor device, 101: transistor, 102: transistor, 103: functional element, 104: node, 105: coil, 106: capacitor, 107: coil, 108: coil, 109: parasitic capacitance, 111: memory element, 112: transistor, 113: capacitor, 114: node, 121: terminal, 122: terminal, 123: terminal, 124: terminal, 125: terminal, 131: operational amplifier

The invention claimed is:

1. A semiconductor device comprising a first transistor, a second transistor, a third transistor, a capacitor, and a functional element,

wherein one of a source and a drain of the first transistor is electrically connected to a first terminal,

wherein the other of the source and the drain of the first transistor is electrically connected to one of a source and a drain of the second transistor,

wherein a gate of the first transistor is electrically connected to a second terminal,

wherein the other of the source and the drain of the second transistor is electrically connected to a third terminal, wherein the other of the source and the drain of the second transistor is electrically connected to the functional element,

wherein one of a source and a drain of the third transistor is electrically connected to a fourth terminal,

wherein the other of the source and the drain of the third transistor is electrically connected to a gate of the second transistor,

wherein a gate of the third transistor is electrically connected to a fifth terminal,

wherein the capacitor is electrically connected to the other of the source and the drain of the third transistor,

wherein a semiconductor layer of the third transistor comprises an oxide semiconductor,

wherein the functional element comprises a resistor, a constant current source, or a parallel resonant circuit, and

wherein at least one of the first transistor to the third transistor is a multi-gate transistor.

2. The semiconductor device according to claim 1, wherein the second transistor comprises a back gate, and wherein the gate of the first transistor is electrically connected to the back gate.

3. The semiconductor device according to claim 1, wherein the oxide semiconductor comprises at least one of In and Zn.

4. The semiconductor device according to claim 1, wherein a semiconductor layer of the first transistor comprises an oxide semiconductor.

5. The semiconductor device according to claim 1, wherein a semiconductor layer of the second transistor comprises an oxide semiconductor.

6. The semiconductor device according to claim 1, being electrically connected to an antenna.

7. An electronic device comprising: the semiconductor device according to claim 1; and a speaker, a microphone, or a secondary battery.

8. The semiconductor device according to claim 1, further comprising:

a first coil electrically connected to the second terminal and the gate of the first transistor.

9. The semiconductor device according to claim 8, further comprising:

a second coil electrically connected to the first terminal and the one of the source and the drain of the first transistor.

10. The semiconductor device according to claim 1, further comprising:

an operation amplifier, wherein the operation amplifier is electrically connected to the other one of the source and the drain of the third transistor, the capacitor, and the gate of the second transistor.

11. A semiconductor device comprising a first transistor, a second transistor, a third transistor, a capacitor, and a functional element,

wherein one of a source and a drain of the first transistor is electrically connected to a first terminal,

wherein the other of the source and the drain of the first transistor is electrically connected to one of a source and a drain of the second transistor,

wherein a gate of the first transistor is electrically connected to a second terminal,

59

wherein the other of the source and the drain of the second transistor is electrically connected to a third terminal, wherein the other of the source and the drain of the second transistor is electrically connected to the functional element,

wherein one of a source and a drain of the third transistor is electrically connected to a fourth terminal,

wherein the other of the source and the drain of the third transistor is electrically connected to a gate of the second transistor,

wherein a gate of the third transistor is electrically connected to a fifth terminal,

wherein the capacitor is electrically connected to the other of the source and the drain of the third transistor,

wherein a semiconductor layer of the third transistor comprises an oxide semiconductor,

wherein the functional element comprises a resistor, a constant current source, or a parallel resonant circuit,

wherein the second transistor is a multi-gate transistor,

wherein a transconductance of the second transistor is higher than each of transconductances of the first transistor and the third transistor.

12. The semiconductor device according to claim 11, wherein the second transistor comprises a back gate, and wherein the gate of the first transistor is electrically connected to the back gate.

13. The semiconductor device according to claim 11, wherein the oxide semiconductor comprises at least one of In and Zn.

60

14. The semiconductor device according to claim 11, wherein a semiconductor layer of the first transistor comprises an oxide semiconductor.

15. The semiconductor device according to claim 11, wherein a semiconductor layer of the second transistor comprises an oxide semiconductor.

16. The semiconductor device according to claim 11, being electrically connected to an antenna.

17. An electronic device comprising:
the semiconductor device according to claim 11; and
a speaker, a microphone, or a secondary battery.

18. The semiconductor device according to claim 11, further comprising:
a first coil electrically connected to the second terminal and the gate of the first transistor.

19. The semiconductor device according to claim 18, further comprising:
a second coil electrically connected to the first terminal and the one of the source and the drain of the first transistor.

20. The semiconductor device according to claim 11, further comprising:
an operation amplifier,
wherein the operation amplifier is electrically connected to the other one of the source and the drain of the third transistor, the capacitor, and the gate of the second transistor.

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